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CliniCause: Constructing Clinically Grounded Observational Testbeds for Causal Analysis from Irregular ICU Time Series

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Kobi Kenzi

Supervised by: Dr. Dan Vilenchik

תקציר

רשומות רפואיות מיחידות טיפול נמרץ כוללות מדידות רבות הנאספות בזמנים לא אחידים ובתדירות המשתנה בין מטופלים ומשתנים. זמינותן החלקית של המדידות ודפוסי החסר עשויים לשקף הן את מצבו של המטופל והן את תהליך הטיפול. עבודה זו בוחנת מסגרת חישובית העוקבת במפורש אחר מקורות הראיות ומשלבת הפקת ידע קליני וסיבתי בסיוע מודל שפה גדול, הגדרה דטרמיניסטית של מצבי־פרוקסי, חיזוים מתוך סדרות זמן רפואיות לא־סדירות וניתוחים מתוקננים של תמותה באשפוז בהנחיית גרפים סיבתיים. מטרת העבודה היא לבחון את השילוב בין שלבי התהליך ואת המעברים ביניהם, ולא לספק תיקוף קליני, המלצת טיפול או מערכת המוכנה להטמעה.

המסגרת נבחנה בנפרד על נתוני PhysioNet 2012 ועל נתוני MIMIC-III. בשלב התכנון שימש מודל שפה גדול ככלי מסייע להצעת אונטולוגיות של מצבי־פרוקסי, משפחות של כללי החלטה, התייחסויות לדפוסי חסר וגרפים סיבתיים המותאמים לכל מאגר. ההצעות שנבחרו במסגרת הפרויקט קודדו לאחר מכן ככללים וכגרפים דטרמיניסטיים. ארבעה מודלים נלמדים לסדרות זמן — STraTS, GRU, GRU-D ו־TCN — אומנו לחזות את התוויות שהפיקו כללי ההחלטה. האיגוד ששימש בניתוחים הסיבתיים הסופיים כלל את ארבעת מקורות התוויות החזויות האלה ומקור אחד המבוסס על הכללים. מודל השפה לא השתתף כאומד בזמן הריצה. מצבי־הפרוקסי אינם אבחנות שאומתו בבדיקת תיקים רפואיים, והגרפים הסיבתיים לא נלמדו מנתוני המטופלים ולא עברו תיקוף קליני.

באומדני CausalForestDML הראשיים על העוקבות המקוריות התקבל אומדן ממוצע חיובי בכל תשעת מצבי־הפרוקסי שנבחנו ב־MIMIC-III. ב־PhysioNet התקבל אומדן חיובי בתשעה מתוך עשרת המצבים, ואילו אומדן ההלם היה שלילי. CausalForestDML ו־LinearDML הסכימו בכיוון בכל 19 ההשוואות בין מאגר למצב־פרוקסי. שלושת האומדים הסכימו בכיוון ב־18 מתוך 19 ההשוואות; CausalPFN נשמר כאומד חקרני בלבד בשל מגבלות בתיעוד המקור המתודולוגי ובתמיכה האבחונית הזמינה עבורו. ניתוחי ההתאמה שימשו כתמיכה תיאורית באמצעות השוואת תוצאות בזוגות מותאמים, ולא כהוכחה להשפעת התערבות. תוצאות שני המאגרים דווחו בנפרד ולא אוגמו.

תרומתה המרכזית של העבודה היא באינטגרציה שקופה של תהליך המשתרע מהצעות תכנון בסיוע מודל שפה, דרך כללים וגרפים המקודדים בקוד מקור, ועד לחיזוי ולניתוח תצפיתי. האומדנים תלויים בתקפות מצבי־הפרוקסי, בנכונות הגרפים, בסדר הזמנים, באיכות ההתאמה לגורמים מערפלים שנמדדו, בחפיפה בין הקבוצות, בהנחות המודלים ובהיעדר ערבול בלתי נמדד. לפיכך העבודה אינה קובעת שמצבי־הפרוקסי גרמו לתמותה או ששינוים ישנה תוצאות קליניות. תיעוד הראיות משפר את העקיבות המספרית, אך אינו מוכיח שחזור חישובי מלא. נדרשים בהמשך תיקוף קליני, תכנונים סיבתיים חזקים יותר, תיקוף חיצוני והשלמת הפרובננס.

Abstract

Intensive care records contain irregularly timed and incomplete measurements whose missingness may reflect both patient condition and clinical practice. This thesis evaluates a transparent computational workflow linking large-language-model-assisted design, deterministic construction of clinically inspired proxy states, prediction from irregular medical time series, and directed-acyclic-graph-guided observational analyses of in-hospital mortality. Its goal is methodological integration and traceability, not clinical validation, treatment recommendation, or deployment.

Evaluations were conducted separately in the PhysioNet 2012 and MIMIC-III cohorts. At design time, a large language model assisted in proposing dataset-specific proxy-state ontologies, decision-rule families, missingness considerations, and causal graphs; selected proposals were encoded as deterministic rules and graphs. Four learned time-series models—STraTS, GRU, GRU-D, and TCN—were evaluated for predicting rule-derived labels. The archived causal aggregate combined predictions from those four models with one rule-derived source. The language model was not a runtime estimator. The proxy states are not chart-adjudicated diagnoses, and the graphs were neither learned from patient data nor clinically validated.

In primary original-cohort CausalForestDML analyses, mean model-estimated effects were positive for all nine MIMIC-III proxy states and for nine of ten PhysioNet states; PhysioNet shock was negative. CausalForestDML and the secondary LinearDML comparator agreed in direction in all 19 dataset–proxy-state comparisons. All three estimators agreed in 18 of 19; CausalPFN remained exploratory because its methodological provenance and diagnostic support were more limited. Matching supplied descriptive matched-record outcome comparisons, not evidence of intervention effects. Dataset results were reported separately and not pooled.

The contribution is an integrated, inspectable workflow connecting language-model-assisted proposals to source-encoded rules and graphs, then tracing proxy constructs through prediction and observational analysis. Estimates remain conditional on proxy validity, graph and temporal-ordering correctness, adequate control of measured confounding, overlap, model assumptions, and the absence of important unmeasured confounding. They therefore do not establish that proxy states caused mortality or that changing them would improve outcomes. Clinical review, external validation, stronger causal designs, and fuller computational provenance remain necessary.

Keywords

English Keywords

Irregular clinical time series; intensive care; electronic health records; LLM-assisted knowledge elicitation; proxy states; programmatic labeling; weak supervision; multi-label prediction; directed acyclic graphs; observational causal inference; double machine learning; causal forests; sensitivity analysis; MIMIC-III; PhysioNet 2012.

מילות מפתח

סדרות זמן קליניות לא סדירות; טיפול נמרץ; רשומות בריאות אלקטרוניות; חילוץ ידע בסיוע LLM; מצבי פרוקסי; תיוגי תכנותי; פיקוח חלש; חיזוי רב-תוויתי; גרפים מכוונים חסרי מעגלים; הסקה סיבתית תצפיתית; למידת מכונה כפולה; יערות סיבתיים; ניתוח רגישות; MIMIC-III; PhysioNet 2012.

Contents

Abbreviations

Abbreviation	Meaning in this thesis
ICU	Intensive Care Unit.
EHR	Electronic Health Record.
DAG	Directed Acyclic Graph.
LLM	Large Language Model; used in this thesis for design-time knowledge elicitation, not runtime patient-level inference.
STraTS	Self-Supervised Transformer for Sparse and Irregularly Sampled Multivariate Clinical Time Series.
GRU	Gated Recurrent Unit.
GRU-D	Gated Recurrent Unit with Decay.
TCN	Temporal Convolutional Network.
DML	Double Machine Learning.
CATE	Conditional Average Treatment Effect; reported here only as a model-estimated quantity under the stated assumptions.
HTE	Heterogeneous Treatment Effect.
AUROC	Area Under the Receiver Operating Characteristic Curve.
AUPRC	Area Under the Precision–Recall Curve.
minRP	The maximum over thresholds of the minimum of precision and recall.
SOFA	Sequential Organ Failure Assessment.
ARDS	Acute Respiratory Distress Syndrome.
CausalPFN	Exploratory model name used in the project; its primary methodological source and expansion are unresolved.

Notation and Symbols

Symbol	Meaning in this thesis
i	Index of a normalized analysis record.
n_i	Number of observed measurement events for record i.
t_{ij}	Elapsed time in minutes for event j of record i.
v_{ij}	Observed variable name for event j of record i.
x_{ij}	Numeric value recorded for event j of record i.
$\mathcal{V}$	Set of observed measurement-variable names.
$\mathcal{O}_i$	Long-format measurement-event object for record i.
L_{ik}^{rule}	Rule-derived binary proxy label for record i and proxy state k.
$\hat{p}_{ik}^{(m)}$	Probability for proxy state k exported by model m for record i.
$\hat{L}_{ik}^{(m)}$	Binary predicted proxy label corresponding to $\hat{p}_{ik}^{(m)}$.
$\widetilde{L}_{ik}$	Aggregated predicted proxy-label representation; in the archived final causal runs, deterministic voting combined one rule-derived source with four aligned binary model-prediction sources.
A_i	Selected binary proxy-state exposure for record i.
Y_i	In-hospital mortality outcome for record i.
W_i	Observed adjustment variables selected from the project-specified DAG and available columns.
X_i	Effect-modifier features used to describe heterogeneity.
$Y_i(a)$	Conceptual potential outcome under exposure value a; its interpretation requires the assumptions stated in the methods.

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Chapter 1

Introduction

1.1 The Causal-Estimator Evaluation Problem

Causal inference concerns outcomes under alternative actions or exposure conditions, whereas each individual observational record reveals only the factual outcome under the condition that occurred. The counterfactual outcome needed to score an individual effect estimate is therefore unobserved. Unlike ordinary prediction, for which a held-out factual outcome can provide a direct target, observational data do not supply a direct causal answer key for evaluating an estimator. They also leave the true exposure-assignment process and the extent of unmeasured confounding unknown.

Evaluation consequently spans settings with different balances of control and observational fidelity. Fully synthetic studies specify covariates, exposure assignment, potential outcomes, confounding, support, and effect heterogeneity. Semi-synthetic studies retain selected empirical components while simulating assignments, outcomes, or both. Experimentally anchored and empirically anchored designs preserve additional observed structure while obtaining an answer through a controlled experiment, sampling rule, comparison construction, or generated mechanism. These settings permit causal estimation error to be measured against a specified target, but the evaluator also determines important aspects of the difficulty. Greater control makes a causal answer measurable, while greater observational fidelity makes the causal target less directly knowable. These are complementary evaluation properties rather than a simple hierarchy.

CliniCause addresses the observational end of this spectrum. It constructs resources for examining causal-estimator behavior under source-observed measurement, missingness, prevalence, support, care-process, cohort, and outcome patterns. Because those resources do not reveal the counterfactual truth, they cannot score causal estimation error against a known effect. Their role is instead to expose behavior and fragility under realistic observational conditions through several non-equivalent forms of evidence. Chapter 2 situates this role relative to controlled causal benchmarks.

1.2 Observational Fidelity in Irregular ICU Records

Intensive-care medicine produces a particularly demanding form of longitudinal data. Electronic health records contain laboratory values, vital signs, treatments, administrative fields, and outcomes, but these quantities are neither observed on a common clock nor collected with a common purpose. Measurements occur at nonuniform times; variables are sampled at markedly different frequencies; many values are absent at most time points; and the intensity of measurement may itself reflect clinical concern, workflow, or resource use. The resulting records are high dimensional and heterogeneous across patients, care trajectories, and institutions. Irregular medical time-series methods therefore face more than a standard sequence-modeling inconvenience: they must represent the timing, absence, and changing availability of observations while preserving a defensible analytical link to the question being asked [1, 2].

The conventional convenience of a fixed, complete measurement matrix can conceal these properties. Regular resampling, imputation, or summary aggregation may be useful engineering choices, but each makes choices about temporal resolution, which observations count, and how absences are represented. A bedside monitor, a laboratory order, and a static admission field need not convey the same temporal or clinical information. Moreover, the care process can affect both the values that are available and the chance that they are recorded. A representation that ignores timing or observation patterns may discard clinically relevant structure; a representation that exploits them can improve prediction while also encoding care-process signals. This tension makes irregular ICU data a methodological problem spanning statistical representation, workflow design, and interpretation rather than solely a model-selection problem.

For ICU analysis, missingness and irregularity matter at every layer of a study. They influence which variables a representation can retain, which patterns a prediction model can exploit, which records satisfy a cohort definition, and whether a rule can assign an analytical label. They also complicate causal adjustment: an absent measurement can represent no measurement, a measurement outside an available window, a source-data limitation, or a clinically informative decision not to measure. Handling masks, time gaps, or sparse events can improve a predictive representation, but it does not by itself remove the interpretive consequences of the observation process. The thesis consequently treats irregularity and informative measurement as motivations for careful representation and explicit limitations, rather than as local causal findings.

Two complementary needs follow. First, an analysis must accommodate irregular longitudinal measurements rather than force all observations into an unqualified regular grid. Architectures for sparse clinical series and missingness-aware recurrent models illustrate different ways to use values, time intervals, and observation patterns [3, 4]. Second, the analysis needs interpretable variables that can connect a rich event stream to later questions about prognosis and adjusted mortality contrasts. In this thesis, that intermediate layer consists of *clinically inspired proxy states*: rule-derived analytical constructs that summarize evidence for selected clinical patterns from available measurements.

This wording is deliberate. A rule-derived proxy label is not a chart-adjudicated diagnosis, ground truth, or a validated phenotype. A positive tag records that the project-specific source rules were satisfied for the available data; a negative tag does not establish absence of a clinical condition. Nevertheless, a transparent proxy-state layer can make the passage from irregular measurements to a structured downstream analysis inspectable. It supplies common targets for multi-label prediction and a documented set of proxy-state exposures for observational analyses, while retaining a visible link to the measurements and rules from which each construct was derived. Rule-based electronic phenotyping and weak-supervision research provide useful context for this design choice, but do not validate the local proxy definitions or their clinical meaning [5, 6].

Prediction and causal analysis then answer different questions. Predicting a rule-derived proxy label asks whether an irregular-time-series model can reproduce that label from the available inputs. Estimating an adjusted mortality contrast for a proxy-state exposure asks a separate observational question about modeled differences in outcome under explicit assumptions. Strong performance on the first task does not establish the clinical validity of the label, causal validity of the second analysis, or actionability of the exposure. The latter task requires a stated causal model, project-specified directed acyclic graphs (DAGs), exposure-specific adjustment logic, attention to support and overlap, sensitivity and permutation diagnostics, and traceable evidence. Even with these elements, adjusted estimates remain conditional on the adequacy of the proxy measurement, graph, temporal ordering, observed covariates, and model specification [7, 8, 9].

The framework is instantiated separately on PhysioNet 2012 and MIMIC-III, two public ICU time-series resources that provide complementary settings for this problem [10, 11]. Both contain longitudinal ICU measurements and in-hospital mortality information, yet they differ in era, scale, available variables, data contracts, measurement processes, proxy-state definitions, and project graphs. These differences make cross-dataset evaluation useful for examining methodological behavior across distinct observational settings, but they preclude treating similarly named constructs as automatically equivalent or pooling numerical estimates without further justification. The thesis therefore constructs two dataset-specific testbeds through a shared methodological sequence while preserving source-specific definitions and interpretation boundaries.

This problem setting also explains why the thesis moves from data contracts to proxy states, prediction, and only then to adjusted analysis. The same apparent clinical pattern may have different observable correlates when variable availability, sampling frequency, or coding conventions change. If a downstream analysis hides those transformations, a reader cannot determine whether a difference arose from the source data, the rule-derived label, the prediction export, or the effect-estimation model. Making these handoffs explicit is an engineering discipline with scientific consequences: it permits the analysis to be audited, identifies where limitations enter, and prevents an apparently seamless output from being interpreted as a seamless clinical fact.

1.3 Representation-Centered Observational Testbeds

The central move in CliniCause is to shift controlled researcher intervention away from simulating the patient-level data-generating process and toward specifying an inspectable *representation layer*. That layer comprises clinically motivated proxy constructs, deterministic operational rules and temporal cutoffs, analytical cohorts, covariates and outcomes, project-specified DAGs, exposure-specific adjustment assumptions, and provenance. Patient records, measurement and missingness patterns, empirical exposure support, care-process traces, and in-hospital mortality remain source-observed. CliniCause deterministically instantiates proxy labels from the selected rules; it does not simulate patient records, treatment or exposure-assignment mechanisms, or mortality outcomes. “Source-observed” does not imply that these data are error-free or causally sufficient.

The design and instantiation roles are deliberately separated. The LLM received schema-level information—available column names and the mortality outcome—together with instructions to use relevant literature. It did not receive patient records, empirical distributions, exposure prevalences, estimated effects, or execution access. Its output consisted of literature-grounded design proposals for proxy constructs, operational rules, missingness considerations, and dataset-specific graphs. Project-selected rules and graphs were subsequently encoded as deterministic source. The LLM was therefore a design assistant, not a clinical or causal expert, an executed patient-level component, or a source of ground truth.

This representation is instantiated independently on MIMIC-III and PhysioNet 2012. The resulting resources have a common analytical role but are not pooled replicas of one object. Similarly named proxy constructs can differ in measurement availability, operational definition, prevalence, missingness, support, graph structure, adjustment, and population. Each resource is a *testbed* when used to study causal-analysis behavior; the term does not imply a benchmark with known causal truth.

Figure 1.1 summarizes the separation among design, deterministic instantiation, and evaluation.

1.4 Thesis Gap and Objective

Prior work offers important components for this setting. Irregular-time-series research provides representations and predictive methods for sparse, asynchronously observed medical data. Electronic phenotyping provides strategies for constructing analytical variables when direct expert adjudication is unavailable [5]. Data programming formalizes programmatic heuristic labeling [12], while Snorkel provides related weak-supervision and downstream discriminative-training context [6]. Evaluations of medical LLM capabilities provide a bounded precedent for design-time clinical-knowledge elicitation [13]; work on LLM-derived causal-graph priors separately motivates their use as proposals rather than validated graphs [14]. Causal-inference research supplies DAG-based reasoning, double machine learning, heterogeneous-effect estimation, and diagnostics for observational studies [15, 16, 17]. ICU-focused reviews further emphasize the dif-

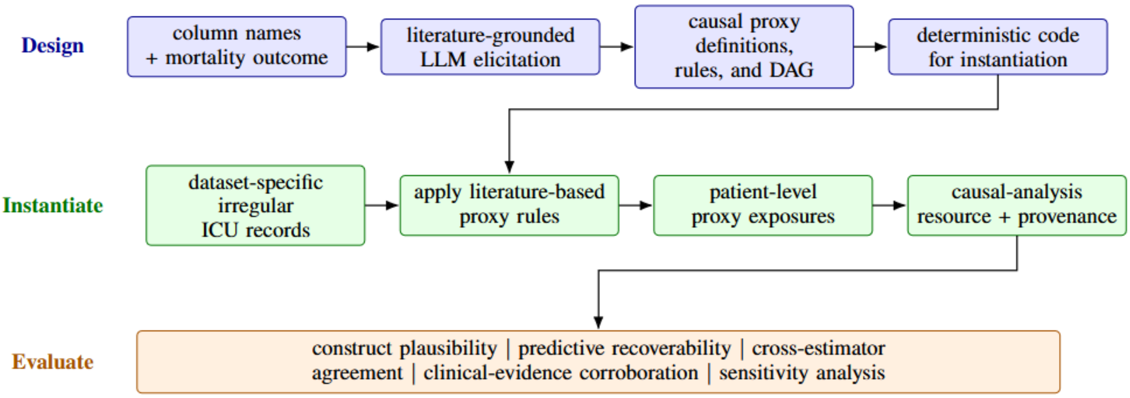

Figure 1.1: The Design, Instantiate, and Evaluate stages of CliniCause. Schema and literature information support LLM-assisted representation proposals, which project-selected deterministic code instantiates at patient level on source-observed ICU records and outcomes. The resulting dataset-specific resources are characterized through converging evidence without a known causal answer key; the LLM receives schema information rather than patient records.

ficulty of defining an actionable question, managing confounding, and distinguishing predictive patterns from treatment-effect claims in observational clinical records [18, 19].

The gap addressed here is not merely the absence of an integrated pipeline. Controlled benchmarks provide measurable causal accuracy under specified mechanisms, while routine observational data provide realistic processes without a known causal answer. A reusable observational evaluation object must make the intervening representation decisions explicit: LLM-generated proposals must be distinguished from selected source-encoded designs; source-specific preprocessing must yield well-defined records; rule-derived labels must be distinguishable from model predictions and algorithmic aggregates; prediction exports must be normalized before downstream use; and causal analyses must state how their exposures, covariates, graphs, diagnostics, populations, and provenance relate. Without these interfaces, a resource can appear coherent while its design assumptions, labels, analysis rows, adjustment variables, or interpretation change silently between stages.

The objective of this thesis is therefore *to construct and characterize clinically grounded observational causal-analysis testbeds from source-observed irregular ICU records through an evidence-tracked, representation-centered framework, while making their proxy definitions, data contracts, causal assumptions, analytical interfaces, diagnostics, provenance, and limitations explicit*. "Evidence-tracked" here means that the thesis distinguishes LLM design proposals, selected source-coded behavior, checked numerical artifacts, and unresolved provenance or validation requirements. The testbeds and their enabling methodology constitute an observational evaluation contribution; they are not claims of clinical validation, deployment readiness, causal identification, or full clean-checkout computational reproducibility.

The framework has two distinct layers. At design time, a structured LLM-assisted protocol elicited proposed proxy-state ontologies, decision-rule families, missingness considerations, and dataset-specific DAGs; project-selected designs were then encoded in deterministic tagger and graph source files. During analytical execution, the sequence is dataset-specific preprocessing; deterministic generation of rule-derived proxy labels; multi-label proxy-state prediction; prediction normalization and algorithmic aggregation; exposure-specific adjustment under the project-specified DAGs; matching and heterogeneous-effect estimation; and robustness and provenance evaluation. The LLM is not invoked in this executed patient-level pipeline. The proxy states are not validated clinical diagnoses; the DAGs are not learned from data; no unconditional causal effect is claimed; and many illness-state proxies do not define well-specified interventions. The workflow does not provide clinical recommendations, bedside decision support, or prospective evidence of benefit.

The integrated workflow is the enabling machinery that makes the testbeds inspectable rather than assumed. Dataset-specific preprocessing preserves the distinction between source records and analysis-ready objects. Rule-based construction makes the origin of each proxy label visible. Prediction normalization distinguishes probabilities from thresholded labels, and algorithmic aggregation distinguishes a deterministic vote from clinical consensus. Project-specified DAGs and exposure-specific adjustment sets make the causal assumptions operationally visible, while matching and model-estimated CATE summaries provide different forms of observational evidence. Finally, robustness and provenance checks separate a checked numerical statement from stronger claims about causal identification, clinical meaning, or rerun reproducibility. The framework's purpose is not to erase these distinctions, but to retain them throughout construction and characterization of the resources.

This framing has practical consequences for how the results should be read. A checked prediction metric concerns the archived evaluation of a model against rule-derived labels, not an independently verified clinical endpoint. A matching output is a descriptive matched-pair outcome difference, not automatically a treatment effect. A mean model-estimated CATE is an aggregate of fitted conditional estimates under the project assumptions, not an unconditional population quantity. Sensitivity, permutation, and support diagnostics contribute evidence about particular failure modes, but none can substitute for a justified intervention definition, an adequate causal graph, or clinical review. By placing these distinctions in the framework itself, the thesis avoids treating provenance, statistical modeling, and clinical interpretation as afterthoughts to an otherwise complete pipeline.

LLM-assisted clinical and causal knowledge elicitation is therefore a substantive design-time stage of CliniCause. The structured protocol produced proposals for the proxy-state ontology, binary decision rules, missingness reasoning, and dataset-specific DAGs; the project selected designs and encoded them in active deterministic source code. This role does not make the LLM a validated medical expert, establish that every proposal was implemented unchanged, or validate the proxy constructs or graphs. Nor was the LLM an executed patient-level estimator: it did not preprocess records, generate runtime predictions, select adjustment variables dynamically, or estimate causal effects. The distinction between design provenance and executable authority makes the representation layer inspectable without treating generated proposals as clinical or causal evidence.

1.5 Research Questions and Contributions

The main research question is:

> How can clinically grounded observational causal-analysis testbeds be constructed from source-observed irregular ICU records—without simulating patient records, treatment or exposure-assignment mechanisms, or outcomes—and what converging evidence can characterize their usefulness and limitations for evaluating causal estimators?

The words "observational" and "characterize" are central. The thesis investigates how a linked workflow can construct and qualify retrospective evaluation resources; it does not claim to score estimates against causal truth, prove that proxy-state exposures cause mortality, or show that manipulating a proxy would improve outcomes. The main question is addressed through five secondary research questions:

SRQ-1: How can clinically and literature-grounded representation designs be instantiated deterministically and separately in MIMIC-III and PhysioNet while retaining source-specific measurement and missingness processes?

SRQ-2: Which data contracts, normalization, aggregation, DAG, adjustment, and provenance mechanisms make the constructed resources explicit, auditable, and reusable?

SRQ-3: How recoverable are the deterministic proxy labels from irregular time series, and what mortality-relevant predictive information is retained by the constructed proxy representations?

SRQ-4: What converging evidence—including estimator sign, rank, and magnitude agreement, clinical-literature comparison, matching and support, permutation disruption, and omitted-variable sensitivity—characterizes each resource?

SRQ-5: Which limitations prevent the resources from being interpreted as clinical validation, known causal truth, or proof of estimator correctness?

The primary contribution is a representation-centered framework and two concrete, dataset-specific observational causal-analysis testbeds constructed from source-observed MIMIC-III and PhysioNet 2012 records. The framework preserves observed records and outcomes while making its proxy definitions, operational rules, cohort choices, DAGs, adjustment assumptions, and provenance explicit and replaceable. Because no known causal answer is available, it characterizes the resources through converging, non-equivalent evidence rather than estimator error against ground truth.

The enabling methods include schema-bounded LLM-assisted elicitation and deterministic source code for proxy rules and DAGs. They also include dataset-specific data contracts, irregular-time-series proxy prediction, deterministic aggregation, exposure-specific DAG adjustment, matching, heterogeneous-effect estimation, diagnostics, and provenance. The compared estimators are CausalForestDML, LinearDML, and CausalPFN. CausalForestDML remains the primary model-estimated causal-analysis estimator, LinearDML is a secondary comparator, and CausalPFN is retained as an exploratory complementary estimator because it lacks the complete diagnostic support available for the DML estimators.

The empirical contribution is a checked cross-dataset comparison of proxy recoverability using four learned prediction models, original-cohort analyses for all 19 dataset–exposure combinations, cross-estimator directional comparison, separate outcome-downsampling robustness analyses, and cross-dataset characterization without numerical pooling. CausalForestDML and LinearDML agree in direction for 19 of 19 combinations; all three estimators agree for 18 of 19, with PhysioNet shock retained as the exception. Matching and support diagnostics, mortality-relevant prediction, clinical-literature comparison, permutation disruption, omitted-variable sensitivity, population perturbation, and provenance provide additional evidence classes whose numerical findings and qualifications are reported in their designated chapters. The thesis also contributes checked numerical tables, a results manifest, checksums, a source packet, a decision register, corrected result figures, and explicit tracking of unresolved provenance. These materials support numerical traceability, but they do not establish that the analysis can be rerun from a clean checkout or that every historical producing setting has been recovered.

These contributions are cumulative rather than interchangeable. Construction of the two observational testbeds is primary; the integrated workflow and shared proxy/data-contract layers make their handoffs inspectable; empirical comparisons and estimator triangulation characterize the resources; and evidence records preserve the boundary between numerical traceability and full computational reproducibility. Agreement can show that a pattern is not unique to one fitted estimator under one representation, but it does not establish estimator accuracy, graph correctness, causal identification, or clinical validity. None of these layers upgrades a rule-derived label to a diagnosis or a graph-guided contrast to a well-defined intervention effect.

As empirical orientation, the four-model comparison was dataset dependent: STraTS led the four archived MIMIC-III test metrics and GRU-D led the corresponding PhysioNet metrics. The original-cohort directional agreements summarized above motivate the later analysis, while the PhysioNet shock exception remains visible. These patterns do not establish statistical superiority, causal validity, positivity, or clinical utility; matching and outcome-downsampled results remain qualified robustness evidence.

1.6 Thesis Organization

The remainder of the thesis follows construction and characterization of the two testbeds from context to interpretation. Background and study design establish the evaluation spectrum, resource contracts, questions, and assumptions. Chapters 4–8 construct and characterize the resources through source-specific contracts, proxy instantiation and prediction, DAG-guided adjustment and estimation, diagnostics, and provenance. The final chapters report the checked evidence, interpret its limitations, and state the conclusions and future-work priorities.

Chapter 2

Background and Related Work

This chapter places the thesis at the intersection of causal-method evaluation and five methodological traditions. Causal-evaluation research clarifies what is gained and lost as an evaluation design moves from controlled causal mechanisms toward source-observed processes. Critical-care data research supplies the two ICU data sources and exposes the consequences of irregular observation. Sequence-modeling research offers several ways to represent sparse measurements for prediction. Electronic-phenotyping and weak-supervision research explains how useful analytical labels can be constructed when direct adjudication is unavailable. Research on LLM medical knowledge and causal-graph priors motivates a carefully bounded design-time elicitation role. Causal-inference research provides a language for specifying questions, encoding assumptions, selecting adjustment variables, estimating heterogeneous contrasts, and diagnosing limited support or sensitivity to omitted variables. Each tradition addresses a necessary part of the problem, but none by itself defines the complete observational evaluation object constructed here. The chapter therefore emphasizes the connections and boundaries among them; project-specific implementation details begin in Chapter 3.

2.1 Causal-Estimator Evaluation: Control and Observational Fidelity

Evaluating a causal estimator differs from evaluating a factual-outcome predictor because the individual counterfactual outcome is not observed. Evaluation designs therefore differ in how they obtain an answer key and in how much of the observed system they preserve. The resulting spectrum is not a ranking in which one setting dominates the others: control supports direct error measurement, whereas observational fidelity preserves difficulties that a specified mechanism may omit.

2.1.1 Fully synthetic evaluation

In a fully synthetic setting, the evaluator specifies the covariates, exposure-assignment mechanism, potential outcomes, confounding structure, empirical-support regime, and treatment-effect heterogeneity. The true causal contrast is consequently known, and assumptions such as ignorability or positivity can be varied deliberately. This control makes estimator error measurable under the chosen mechanisms. It also makes the task's difficulty conditional on choices made by the evaluator, including the response surface, assignment model, overlap, and form of heterogeneity.

2.1.2 Semi-synthetic evaluation

Semi-synthetic designs retain empirical components, commonly a covariate distribution, while simulating exposure assignments, outcomes, or both. IHDP-style evaluation and ACIC-based studies illustrate how observed covariates can be combined with constructed outcome or assignment mechanisms [20, 21]. Influence-function-based evaluation has likewise compared causal models across semi-synthetic competition datasets with shared empirical features and distinct simulated mechanisms [22]. These resources retain more empirical structure than fully synthetic data and still provide a constructed causal answer, but performance remains specific to the imposed mechanisms.

2.1.3 Experimentally anchored evaluation

Experimental or randomized-trial information can anchor causal-method evaluation in real measurements and outcomes. One approach induces an observational study from experimental data through a controlled sampling mechanism, so that estimates from the constructed observational sample can be compared with the experimental reference [23]. The experimental anchor strengthens the target comparison, while researcher-controlled sampling, population selection, and harmonization remain part of the evaluation design. Such designs therefore occupy a different point on the control–fidelity spectrum rather than eliminating construction choices.

2.1.4 Empirically anchored generated evaluation

Empirically anchored generation can learn additional distributional structure from observed data and then generate datasets under user-specified effects and confounding. Credence exemplifies this strategy for validating causal-inference methods [24]. Compared with a simple parametric simulation, this approach can preserve richer empirical structure, yet the causal mechanisms and answer key remain generated under evaluator-specified conditions.

2.1.5 Source-observed observational testbeds

A source-observed observational testbed occupies the complementary end of the spectrum. It retains measurement and missingness patterns, exposure prevalence, empirical support, care processes, cohort selection, and outcomes as recorded in routine data. It therefore preserves observational conditions that a controlled design may simplify or generate. The corresponding limitation is fundamental: no counterfactual answer key is available, so the testbed cannot measure causal estimation error against truth. Its appropriate role is to study estimator behavior, agreement, discrepancies, support, and sensitivity under realistic but causally unresolved conditions.

CliniCause is positioned in this latter category. It does not replace fully synthetic, semi-synthetic, experimentally anchored, or empirically generated benchmarks. Controlled benchmarks provide measurable causal accuracy under specified mechanisms; the CliniCause testbeds provide diagnostic realism under source-observed mechanisms whose causal truth remains unknown. Their construction must consequently expose the representation choices and assumptions that controlled generation would otherwise encode in a known data-generating process.

2.2 ICU EHR Datasets and Irregular Sampling

2.2.1 Critical-care EHR time series

ICU electronic health records combine physiological measurements, laboratory results, administered treatments, clinical observations, and administrative information. These sources do not share a natural sampling clock. Vital signs may be recorded repeatedly, laboratory tests are ordered at variable intervals, and static admission fields may appear only once. Observation times are therefore nonuniform both within a variable and across variables, while different patients can have markedly different measurement density and duration. Reviews of irregular medical time series distinguish irregular spacing within a sequence from heterogeneous sampling rates across variables, and emphasize that either feature complicates models built for complete, fixed-dimensional sequences [1].

Missingness in this setting is not a single phenomenon. A value may be absent because a test was not ordered, a device was not connected, a source table did not contain the measurement, the observation fell outside the analysis window, or preprocessing rejected it. Clinical concern and care processes can influence whether and when a variable is measured, so the observation pattern may contain predictive information [2, 4]. It would nevertheless be too strong to treat every absence as informative or clinically meaningful. Observation patterns can mix physiology, workflow, resource availability, and data-quality mechanisms; exploiting them predictively does not identify which mechanism generated them.

Several further properties make ICU EHR data methodologically demanding. Illness trajectories are heterogeneous, measurement values are high dimensional, physiological and administrative records coexist, and identifiers may refer to a person, admission, ICU stay, or challenge record rather than to the same unit. Repeated stays can create dependence if they are mistaken for independent people. Cohort entry, time zero, and the interval over which measurements are summarized can also change the scientific meaning of a record. These are not merely data-cleaning details: they determine what information is available to a model and whether variables can be interpreted as preceding an outcome or exposure.

Regular resampling and imputation can create a convenient matrix, but they necessarily choose a time resolution and a rule for unobserved cells. Sparse-event representations instead preserve observed values and times, while missingness-aware recurrent models attach masks and elapsed-time information to a regularized sequence. Neither strategy is universally preferable. The relevant choice depends on the source data, measurement density, task, and the distinction between modeling the value process and modeling the observation process. This thesis therefore treats irregular sampling as both a representation problem and an interpretation boundary.

2.2.2 PhysioNet/Computing in Cardiology Challenge 2012

The PhysioNet/Computing in Cardiology Challenge 2012 was organized around patient-specific prediction of in-hospital mortality from early ICU information. It combined admission descriptors with time-stamped vital-sign and laboratory series and provided a structured training-and-evaluation setting for mortality classifiers and risk estimators [10]. Its challenge format makes it a useful compact benchmark-style source: the prediction target, observation window, and evaluation setting were defined for a shared research task rather than left to every investigator to derive independently.

That role must be separated from this thesis's local processing. The source paper documents the challenge data and rules, but it does not establish that a later repository reproduces every inclusion rule, missing-value convention, split, or score. Nor does a challenge record automatically correspond to the same unit as an ICU stay in another database. The thesis uses PhysioNet 2012 as one source on which to exercise a common workflow; exact local data contracts and preprocessing decisions are described in Chapter 4, without assuming full reproduction of the original competition.

2.2.3 MIMIC-III and benchmark research

MIMIC-III is a deidentified, single-center critical-care database integrating bedside observations, laboratory data, medications, procedures, coded information, notes, and outcomes over an extended data era. It was made available to qualified researchers subject to training and a data-use agreement, enabling reproducible critical-care research while retaining controlled-access obligations [11]. Unlike a fixed prediction challenge, MIMIC-III is a relational clinical database from which researchers must define the study unit, cohort, variables, time windows, and outcomes.

Benchmark work on MIMIC-III illustrates how a database can be converted into standardized clinical time-series tasks and reusable baselines [25]. The benchmark is a particular extraction and task definition, not the database itself. Consequently, citing the benchmark supports the value of reproducible task construction but does not imply that this thesis uses its cohorts or tasks unchanged. Local preprocessing may draw on a benchmark style while defining different proxy-label targets and downstream analysis objects.

2.2.4 Dataset differences relevant to this thesis

PhysioNet 2012 functions as a bounded mortality-challenge resource with admission descriptors and time-stamped early-ICU measurements, whereas MIMIC-III is a broader relational database requiring investigator-defined extraction and linkage. Their complementary roles test whether one analytical sequence can operate across distinct scales, eras, variables, identifiers, and measurement processes. Neither role implies that the local pipeline reproduces the original challenge or benchmark unchanged.

Table 2.1 summarizes these roles and their principal interpretation boundary.

Table 2.1: Conceptual roles of the ICU datasets used in this thesis.

Dataset	Role in the study	Interpretation boundary
PhysioNet 2012	Bounded mortality-challenge resource and compact benchmark-style source.	The local pipeline need not reproduce every original challenge rule, split, or preprocessing convention.
MIMIC-III	Broad relational critical-care database requiring cohort extraction and linkage.	The local extraction is one site-, era-, and definition-specific view of the database.

The purpose of using both sources is methodological portability, not population pooling. Dataset-specific cohorts, preprocessing, proxy rules, and graphs are retained because similarly named variables or proxy states can differ in units, acquisition, aggregation, missingness, and construct definition. Cross-dataset differences may therefore reflect the data source, care process, representation, or model interface rather than biology alone.

2.3 Irregular Time-Series Representation Models

2.3.1 Representation challenges and design choices

Predictive models for irregular clinical time series differ first in what they consider an input. One family regularizes observations onto a sequence of time bins and supplies imputed values, masks, or time gaps. Another works directly with observed events, representing measurement identity, value, and time without filling every unobserved cell. Within these representations, temporal dependence can be modeled by recurrent state updates, temporal convolution, or attention. These choices encode distinct assumptions about which temporal relationships should be easy to learn and how the observation process should enter the model [1].

No representation removes the need to define the task and preprocessing contract. Dense sequences require a bin width, aggregation rule, and treatment of missing cells. Sparse events require a variable vocabulary, continuous or discrete encodings of time and value, and a policy for very long event streams. Positional or temporal encodings provide order information to attention mechanisms, while masks can prevent padded or future positions from entering a computation. Self-supervised pretraining can exploit unlabeled records, but its utility depends on the pretext task and the relation between pretraining and supervised populations. The four learned models reviewed below are the predictive families compared by the thesis; their published forms are conceptual precedents rather than proof that every local adaptation is identical.

2.3.2 Recurrent baselines: GRU and GRU-D

The gated recurrent unit (GRU) updates a hidden state sequentially using reset and update gates. Conceptually, the update gate balances retention of the previous state against incorporation of a candidate state, while the reset gate controls how strongly earlier information contributes to that candidate. This gating provides a flexible general sequence baseline with fewer gating components than a conventional long short-term memory unit [26]. A standard GRU can learn from a suitably prepared clinical sequence, but it does not inherently know how much time elapsed between observations or why a variable is absent. Its handling of irregularity therefore depends on the input representation supplied to it.

GRU-D modifies recurrent modeling for multivariate series with missing values. It uses observation masks and elapsed-time information, and learns decay mechanisms that move missing inputs toward empirical reference values and attenuate hidden states as gaps increase [4]. This formulation makes the missingness pattern and time since observation explicit rather than treating imputation as an invisible preprocessing step. It is one principled response to irregular sampling, not a universal solution: learned decay imposes its own structure, the observation process may differ across sites, and predictive use of missingness does not correct causal bias from measurement decisions.

2.3.3 Temporal convolution: TCN

Temporal convolutional networks apply one-dimensional convolution in temporal order. Causal convolution prevents future positions from contributing to an earlier output, dilation expands the receptive field without requiring a kernel at every lag, and residual blocks stabilize deeper models. Because convolutions for many positions can be computed in parallel, a TCN offers a different computational and inductive bias from a recurrent state update [27]. The canonical TCN study is a general sequence-modeling comparison rather than an ICU-specific method paper. In an irregular ICU setting, the model still requires an appropriate regular sequence representation, so resampling, aggregation, masks, and time features remain part of its effective design.

2.3.4 Sparse-event transformer modeling: STraTS

STraTS is designed for sparse and irregular multivariate clinical series. It treats an observed measurement as a triplet of time, variable, and value; continuous embeddings of time and value are combined with a variable embedding, and transformer-style attention contextualizes the observed events. This avoids constructing a fully imputed variable-by-time matrix and preserves fine-grained observation times. The method also introduces a self-supervised forecasting objective for pretraining representations before supervised prediction [3].

This design is well matched to sparse event streams, but it does not make preprocessing irrelevant. Event selection, normalization, truncation, static features, and the variable vocabulary still determine what the transformer sees. Moreover, the published method and the thesis's supervised multi-label adaptation are distinct objects. The paper supports the triplet representation, attention mechanism, and pretraining rationale; Chapter 6 defines the local target heads, training, and export contract.

2.3.5 Relationship among the four included model families

The comparison in Table 2.2 is best understood as a comparison of inductive biases rather than a ladder of sophistication. GRU supplies a general recurrent baseline; GRU-D makes missingness and elapsed time explicit within recurrence; TCN emphasizes multiscale local-to-long-range temporal filters; and STraTS applies attention to observed event triplets with a pretraining option. Explicit missingness handling is central to GRU-D, implicit in the absence and timing of STraTS events, and otherwise dependent on the prepared inputs. Likewise, pretraining is a defined part of STraTS but not an assumed property of every model.

Table 2.2: Predictive model families included in the thesis comparison.

Model	Core temporal mechanism	Input and irregularity handling	Role in this thesis	Primary reference
STraTS	Transformer-style contextualization and attention pooling.	Sparse observed time–variable–value triplets; timing and absence are retained without a fully imputed grid; self-supervised pretraining is supported.	Sparse-event focal model with a later local multi-label adaptation.	[3]
GRU	Gated recurrent hidden-state updates.	Requires a prepared sequence; irregular time and missingness are not inherently resolved by the ordinary recurrent unit.	General recurrent baseline.	[26]
GRU-D	Recurrent updates with trainable input and hidden-state decay.	Uses masks and elapsed-time features to model missingness explicitly.	Missingness-aware recurrent baseline.	[4]
TCN	Dilated temporally ordered convolutions with residual blocks.	Operates on a regular sequence representation whose binning, masks, and aggregation must be defined.	Convolutional sequence baseline.	[27]

Architecture suitability can vary with measurement density, missingness, preprocessing, label construction, sample size, and the supervised task. A model can also exploit site-specific observation patterns rather than transferable physiology. For these reasons, related work motivates empirical comparison but not a universal ranking. The thesis-specific implementations and results are reserved for Chapters 6 and 9.

2.4 Proxy Phenotyping and Weak Supervision

2.4.1 Electronic phenotyping and rule-derived labels

Electronic phenotyping identifies patients or records with characteristics of interest from routinely collected EHR data. Phenotypes may combine diagnosis or procedure codes, laboratory and vital measurements, medications, text, temporal criteria, deterministic rules, statistical models, or machine learning. The field has consequently developed from manually specified rule sets toward hybrid and learned approaches, while retaining the need to validate the construct for its intended use [5].

Rule-based phenotypes remain attractive when transparency and inspectability matter. An investigator can trace a positive label to named variables, thresholds, and time conditions; the same rule can be rerun; and domain reviewers can identify implausible clauses. A rule can also provide a practical label when manual adjudication is unavailable. Concrete respiratory-failure phenotyping work illustrates how operational cohort definitions can be constructed from EHR evidence while requiring validation against the intended clinical concept [28].

Transparency is not equivalence to clinical truth. Thresholds can be sensitive to units and preprocessing, required variables may be missing, and a rule developed at one site may not transport to another. Temporal ambiguity can cause measurements before, during, and after a clinical episode to be collapsed into one label. Codes and treatments may reflect billing or care processes as well as physiology. Rule outputs can therefore be reproducible yet noisy, and a negative label can mean insufficient evidence rather than absence of the condition. Validation must be tied to the intended use: a cohort-retrieval rule, prediction target, and causal exposure may require different evidence.

2.4.2 Clinical-definition sources and approximation

Clinical consensus documents provide important conceptual grounding for organ dysfunction and acute syndromes, but their definitions are usually multicomponent and context dependent. Sepsis-3 links sepsis to infection-associated organ dysfunction and gives specific clinical operationalizations; KDIGO defines acute kidney injury using creatinine and urine-output changes; and the ISTH work proposes clinical and laboratory scoring for disseminated intravascular coagulation [29, 30, 31]. SOFA and the Berlin ARDS definition likewise supply authoritative terminology for organ dysfunction and respiratory syndromes [32, 33].

These sources should not be used to upgrade a partial rule into a formal diagnosis. A project rule may use measurements related to a definition, draw on one threshold family, or approximate part of a syndrome while omitting duration, baseline change, imaging, intervention, or contextual requirements. The appropriate claim is therefore that a proxy family is *clinically informed by* or *draws on concepts from* these sources. Whether each local rule captures its intended construct requires rule-level clinical review and, ideally, chart-adjudicated evaluation. The two definition sources without local PDFs are used here only for this limited conceptual role; no detailed local implementation claim depends on them.

2.4.3 Weak supervision and label aggregation

Weak supervision addresses settings in which accurate hand labels are scarce but imperfect programmatic sources are available. Data programming expresses domain heuristics, distant supervision, or other weak sources as labeling functions and models their noisy, potentially conflicting outputs to create training data for downstream discriminative models [12]. Snorkel implements this broader paradigm as a system with a learned generative label model that estimates source accuracies and dependencies without requiring every training example to be manually labeled [6]. The central lesson for this thesis is that label-source quality and correlation matter: adding sources does not guarantee that shared bias cancels.

A deterministic majority vote is not the same method. Voting assigns the class supported by a specified number of binary sources and treats their contributions according to the fixed vote rule. A learned weak-supervision label model instead estimates latent source-quality or dependence parameters and can weight conflicting sources differently. This thesis uses deterministic majority voting, not the Snorkel generative model. The output is algorithmic aggregation, not clinical consensus; correlated voters trained on the same rule-derived labels can preserve or amplify common error.

2.4.4 Implications for this thesis's proxy states

The thesis uses *clinically inspired proxy state*, *rule-derived proxy label*, *proxy phenotype*, and *weak clinical label* to distinguish analytical constructs from verified diagnoses. Repository identifiers beginning with `LAT_` are historical software names. They do not establish that a formally identified latent clinical variable exists. This distinction is essential because the same column interface is used across several stages without making the generating mechanisms equivalent.

Proxy-state validity can be separated into at least five questions. *Implementation validity* asks whether code executes the documented rule. *Schema consistency* asks whether identifiers and columns align across artifacts. *Predictive learnability* asks whether models can estimate rule-derived labels from time-series inputs. *Clinical construct validity* asks whether the labels correspond to the intended clinical concept. *Causal measurement validity* asks whether a proxy can serve as a sufficiently accurate and temporally coherent exposure for the stated causal question. Repository evidence supports the first three more directly than the last two.

The shared proxy layer has three connected roles in this thesis: it supplies supervised prediction targets, provides binary inputs for algorithmic aggregation, and defines downstream proxy-state exposure representations. A common interface makes handoffs auditable and permits the same construct name to be traced across stages. It also creates a compounded error path: rule design can mismeasure a construct, prediction can reproduce or add classification error, aggregation can preserve correlated error, and DAG-guided causal analysis can then condition on or contrast a mismeasured exposure. Predictive accuracy against rule labels demonstrates learnability of those labels; it is not clinical validation and does not establish causal measurement validity.

2.4.5 LLM-Assisted Clinical and Causal Knowledge Elicitation

Advanced LLMs can encode and express substantial medical knowledge under structured evaluation. In MultiMedQA evaluations, PaLM-family models produced medically relevant answers and reasoning, and instruction tuning improved several measured capabilities [13]. The same study's human evaluation exposed clinically important gaps, and Med-PaLM remained inferior to clinicians. This evidence supports the feasibility of using an LLM to generate design proposals; it does not establish reliability for a particular proxy-design task, correctness of every rule, suitability as an autonomous medical expert, or clinical validation of CliniCause outputs.

LLM semantic judgments have also been studied as prior information for causal-graph construction. Darvariu and colleagues evaluate prompt designs and soft LLM-derived priors for causal discovery, reporting the greatest benefit for edge orientation in selected common-sense benchmark settings while also finding that standalone LLM priors are not uniformly beneficial [14]. This is a methodological precedent for treating model output as candidate relations or orientation information, not evidence that the CliniCause DAGs are correct. The local graphs were not learned from patient-level data, the study did not implement the paper's discovery algorithm, and formal adjustment validity still depends on the correctness and completeness of the project-specified assumptions.

A separate precedent for turning elicited reasoning into executable labels is programmatic labeling. Data programming represents heuristics as inspectable labeling functions that can generate large weakly labeled data sets for discriminative learning [12]. The deterministic CliniCause decision trees play a conceptually related role by translating selected design proposals into source-coded rules that generate proxy-label targets from observed variables. CliniCause does not reproduce the data-programming generative model, its noise-aware loss, or every element of its multi-source framework.

Snorkel further illustrates a workflow in which noisy and correlated rule sources are combined through a learned generative label model before a downstream discriminative model is trained [6]. CliniCause instead uses its own deterministic decision-rule construction and implemented aggregation workflow. Its majority vote is a fixed algorithmic operation, not Snorkel's learned generative label model, and the citation supplies conceptual weak-supervision context rather than implementation equivalence.

Within this thesis, the LLM-assisted elicitation protocol is consequently treated as a substantive design-time component with bounded authority. It produced proposals for proxy-state ontologies, binary decision rules, missingness considerations, and dataset-specific DAGs; the project selected designs and encoded them in deterministic source files. The prompt artifacts document that process, whereas active tagger and graph scripts define implemented behavior. Neither the literature above nor the local prompt record establishes clinical validity, complete human review, or causal identification.

2.5 Causal Diagrams, Identification, HTE, Overlap, and Sensitivity

2.5.1 Observational causal questions and target-trial thinking

Causal analysis begins with a question, not an estimator. Target-trial thinking makes the desired comparison explicit by specifying eligibility, treatment or exposure strategies, assignment, time zero, follow-up, outcome, causal contrast, and analysis plan before observational data are analyzed [8]. This discipline exposes mismatches such as using post-baseline information to define eligibility, comparing prevalent with incident exposure, or allowing eligibility and follow-up to begin at different times. ICU EHR data make these issues acute because clinical state, treatment, measurement, and selection evolve rapidly and are documented through care processes.

The exposure must also correspond to a sufficiently well-defined intervention or contrast. Different ways of producing the same measured state can lead to different outcomes, so a contrast on a state such as body mass index—or, here, an illness-state proxy—may violate consistency when the underlying intervention is unspecified [9]. This concern is not repaired by choosing a flexible estimator. It requires clarity about what changes, when it changes, and which versions of the exposure are being compared.

Retrospective ICU studies additionally face confounding by indication, changing severity, informative measurement, treatment–confounder feedback, selection, and limitations in outcome measurement. A review of observational causal inference in intensive care emphasizes careful question formulation, assumptions, data preparation, and reporting across the analysis workflow [18]. Marginal structural models provide important background for time-varying treatment and confounder processes, but they are not the point/binary exposure estimator implemented in this thesis. Nor does this thesis claim a complete target-trial emulation.

2.5.2 DAGs, backdoor reasoning, and adjustment

A directed acyclic graph (DAG) represents assumed causal relations through nodes and directed edges. Paths into an exposure can encode confounding routes, while colliders require special care because conditioning on them can open otherwise blocked associations. Backdoor reasoning uses the graph to identify covariates that block noncausal paths without adjusting for inappropriate descendants or colliders [7]. A DAG thus makes assumptions inspectable and connects scientific reasoning to adjustment-set selection.

Graphical transparency must not be mistaken for empirical proof. A DAG does not establish that its arrows are correct, and a set that blocks paths in the drawn graph does not guarantee exchangeability in the clinical population. Required confounders may be unavailable in the observed data; measurement can be an imperfect proxy for a graph node; and temporal order must be justified scientifically rather than inferred from an arrow. In this thesis, project-specified DAGs guide observed-variable adjustment at a high level. Their dataset-specific nodes, mappings, and implementation are deferred to Chapter 7.

2.5.3 Double machine learning

Double machine learning (DML) separates the causal parameter of interest from flexible nuisance functions, such as outcome regression and exposure propensity. Machine-learning estimators can fit these functions, but naively substituting regularized predictions into a causal estimating equation can transfer regularization and overfitting bias to the target parameter. DML addresses this problem through Neyman-orthogonal scores, which reduce first-order sensitivity to nuisance-estimation error, and cross-fitting, which estimates nuisance functions on data separate from the observations used to evaluate the score [15].

In a residualization interpretation, covariate-explained components are removed from the outcome and exposure, and the residual association enters the orthogonal score. Cross-fitting rotates training and evaluation folds so observations can contribute efficiently without using the same fitted nuisance error twice. These devices broaden the nuisance models that can be used while retaining inference under stated regularity and rate conditions. They do not eliminate the substantive assumptions of causal inference. DML cannot correct an omitted confounder that was never measured, create overlap where none exists, turn a proxy into an accurately measured exposure, repair an incorrect DAG, or define an otherwise ambiguous intervention.

EconML provides software implementations for heterogeneous-effect estimators, including DML-related workflows, and is useful for documenting implementation context [34]. The software reference is distinct from the theoretical basis: library availability does not validate local nuisance models, cross-fitting behavior, or identification assumptions.

2.5.4 Causal forests and heterogeneous effects

Outcome prediction estimates how outcomes vary with covariates under the observed data distribution; effect estimation asks how an outcome contrast varies between exposure conditions under causal assumptions. A strong prognostic variable need not be an effect modifier, and a subgroup with high baseline risk need not receive greater benefit from an intervention. Keeping prognostic and treatment-effect heterogeneity separate is therefore fundamental.

Causal forests adapt random-forest ideas to heterogeneous treatment-effect estimation. Recursive partitioning seeks regions in covariate space with different treatment contrasts, and aggregation across trees stabilizes local estimates. Under conditions including unconfoundedness and appropriate forest construction, the method provides a nonparametric approach to conditional treatment effects and inference [16]. Generalized random forests extend the adaptive-neighborhood view of forests to parameters defined through local moment equations, providing a broader framework in which heterogeneous-effect estimation can be formulated [17].

These methods are attractive when effect modification is nonlinear or involves interactions, but flexibility does not validate discovered subgroups. Estimated conditional effects can be unstable in sparse regions, depend on nuisance models and tuning, and identify patterns that do not transport. Forest feature importance describes the fitted model rather than a biological mechanism. High-dimensional individual estimates are not automatically clinically actionable.

2.5.5 HTE in EHR and critical care

EHR research is motivated by the possibility that large, heterogeneous observational populations can inform conditional or individualized treatment-effect estimation. The same data introduce confounding, informative missingness, treatment-selection bias, and covariate shift, while individual counterfactual outcomes remain unobserved [19]. Validation is consequently harder than ordinary prediction: a held-out factual outcome cannot directly reveal the accuracy of an individual's unobserved alternative outcome.

Modern HTE reviews emphasize principled separation of hypothesis generation from confirmation, careful performance criteria, and the instability of data-driven subgroups [35, 36]. Critical-care populations add marked variation in baseline risk, disease mechanisms, and competing trajectories. Heterogeneity in observed outcomes or prognosis can be mistaken for heterogeneity of treatment effect, and average trial results can conceal clinically important variation while exploratory subgroup analyses can also produce spurious patterns [37]. These considerations motivate conditional-effect modeling in the thesis but do not imply that its proxy-state analyses validate individualized clinical effects.

The empirical study evaluates CausalPFN in an exploratory role, while its detailed theoretical positioning remains limited pending a verified primary method source.

2.5.6 Overlap and empirical support

Positivity requires that the exposure conditions being compared are possible within relevant covariate strata. Its empirical counterpart, overlap or common support, asks whether comparable exposed and unexposed observations exist in the analyzed sample. When propensity scores approach their boundaries, estimates rely on a small number of observations or extrapolation, producing instability and changing the population for which a comparison is credible. Work on limited overlap formalizes the variance and target-population consequences and motivates explicit attention to the region of adequate support [38].

Overlap is especially important for heterogeneous-effect estimation because local estimates divide the sample into increasingly specific covariate neighborhoods. A global mixture of exposed and unexposed records can coexist with local support failures. Matching rates, failures, and distances can provide indirect evidence about available comparisons, but they depend on the matching representation and algorithm. They do not replace propensity-score distributions, common-support plots, or covariate-balance assessment. Conversely, empirical overlap does not establish exchangeability or validate the DAG. This thesis therefore treats support diagnostics as necessary qualification and does not claim that positivity has been established.

2.5.7 Omitted-variable sensitivity and diagnostic workflows

Even after adjustment, observational estimates can be biased by variables that affect both exposure and outcome but are absent from the model. Omitted-variable-bias sensitivity analysis asks how strongly such a variable would need to relate to the residualized exposure and outcome to change a conclusion. Robustness values provide a compact strength-of-confounding summary, while comparison with observed covariates can make a hypothetical confounder scale more interpretable [39]. Extensions to causal machine learning formulate analogous bias bounds and inference for broader target parameters and flexible nuisance estimation [40].

Sensitivity results remain model- and scale-dependent. Benchmarking is persuasive only when the selected observed covariate is scientifically meaningful, and no observed benchmark necessarily bounds every unmeasured process. A robustness value does not prove that unmeasured confounding is absent. Sensitivity analysis also does not validate a DAG, create support, or repair exposure misclassification. Its role is to quantify how conclusions would respond to specified departures from an identifying assumption.

More broadly, transparent causal workflows separate modeling assumptions, identification, estimation, and attempts to refute or perturb the analysis. DoWhy articulates this model–identify–estimate–refute sequence and implements placebo and other robustness checks as software-supported diagnostics [41]. The value of such checks is procedural: they can expose sensitivity to broken relationships or alternative assumptions. They do not turn every perturbation into a formal randomization test, and the citation does not imply that every DoWhy refuter was used in this thesis.

2.5.8 Positioning of the present study

The literature supplies complementary evaluation designs and methodological components rather than the observational evaluation object constructed in this thesis. Models for irregular series address representation and prediction; phenotyping provides transparent routes from EHR evidence to analytical labels; LLM research motivates bounded knowledge elicitation and graph-prior proposals; data programming and weak supervision explain how domain heuristics can create imperfect training labels and how noisy sources may be combined; DAGs make identification assumptions explicit; DML and forest methods enable flexible adjusted and heterogeneous-effect estimation; and overlap and sensitivity methods expose fragility in support and unmeasured-confounding assumptions. CliniCause uses these components to construct two reusable observational resources in which representation decisions, patient-level instantiation, adjustment assumptions, estimator interfaces, diagnostics, and provenance are explicit. The integration contribution makes that construction auditable; it is not a claim to have invented each component or to have created causal ground truth.

Representation-centered construction and integration also compound risk. Unsupported LLM suggestions can enter selected rule or graph designs; measurement error in source events can affect rule-derived labels; model error can alter predicted labels; aggregation can preserve shared error; a project-specified DAG can be misspecified; local comparisons can lack support; and unmeasured confounding can remain after flexible estimation. Provenance gaps can make it difficult to determine which version of an artifact entered the next stage. LLM-assisted elicitation is a substantive design-time method in this framework, but it was not an executed estimator, a patient-data causal-discovery method, a source of authoritative clinical judgment, or a source of clinical validation. Chapters 3–8 therefore define the actual testbed objects, source-code implementation, assumptions, estimator interfaces, and diagnostic hierarchy. This progression preserves the central lesson of the related work: a credible observational testbed keeps every representation decision, handoff, evidence class, and limitation visible rather than absorbing them into a single model output.

Chapter 3

Problem Definition and Study Design

This chapter defines CliniCause as a framework for constructing two dataset-specific observational causal-analysis testbeds. Each resource packages source-observed ICU records and outcomes together with an explicit analytical representation, project-specified causal assumptions, estimator interfaces, diagnostics, and provenance. MIMIC-III and PhysioNet 2012 are instantiated separately: their shared contract identifies corresponding analytical roles, but it neither pools records nor makes similarly named proxies, graphs, adjustment sets, or estimands identical.

3.1 Dataset-Specific Observational Testbed Contract

For dataset d, the constructed resource links a source-specific record identity; source-observed longitudinal measurements; observation and missingness information; a deterministic proxy-state representation; model-generated proxy annotations where used; fixed aggregation outputs where used; source-observed in-hospital mortality; observed covariates; a project-specified DAG; exposure-specific adjustment sets; an analytical cohort definition; and provenance records that distinguish the evidence class of each component. The resource is an observational *testbed* when used to examine causal-estimator behavior. It is not a benchmark with a known counterfactual answer.

These components do not have a common evidential status. Record identifiers, recorded measurement events, their availability patterns, observed covariates, and mortality originate in the source data subject to extraction and data-quality limitations. Proxy definitions, cut-offs, temporal rules, cohort definitions, and covariate roles are representation-defined research choices. Rule-derived proxy labels are deterministic derived objects instantiated from those choices and the available observations. Learned proxy annotations are model-generated estimates of the rule-derived labels, while the archived majority-vote interface is a fixed derived analytical output. DAGs and exposure-specific adjustment sets are project-specified causal assumptions. Provenance and evidence-class records link these objects without implying that source observation, deterministic derivation, model estimation, and causal assumption provide equivalent evidence.

The controlled intervention in this construction therefore occurs at the representation layer. Researchers specify which observed evidence defines a proxy, which records form an analytical cohort, which variables serve as covariates and outcomes, and which graph and adjustment assumptions govern each exposure analysis. Deterministic source code then instantiates the selected proxy rules on patient records. CliniCause does not simulate those patient records, a treatment or exposure-assignment mechanism, or mortality outcomes, and deterministic proxy-label construction does not make the underlying clinical state known. The causal answer remains unknown because the data and outcomes are observational and the representation choices do not establish identification.

3.2 Units, Time Horizons, and Data Objects

This thesis studies retrospective intensive-care records represented as irregular clinical time series. The operational study unit is a time-series record or ICU stay, depending on the source dataset and preprocessing path. PhysioNet 2012 source files use record identifiers, whereas MIMIC-III source tables use ICU-stay and admission identifiers. The implemented pipelines normalize these source-specific identifiers to the canonical join key `ts_id`. This identifier should be read as the analysis-record identifier used by the software; it is not assumed to be a globally unique person identifier across repeated admissions or stays.

For a study unit i, the observed longitudinal data are represented as a finite collection of measurement events

$$\mathcal{O}_i = \{(i, t_{ij}, v_{ij}, x_{ij}) : j = 1, \ldots, n_i\},$$

where i is the normalized `ts_id`, t_{ij} is elapsed time in minutes from record start or ICU admission, $v_{ij} \in \mathcal{V}$ is the observed variable name, and x_{ij} is the numeric value recorded for that variable. In the causal repository this long-format object is stored in `ts` with the required columns `ts_id`, `minute`, `variable`, and `value`. Static or baseline quantities such as age, sex or gender coding, height, weight, and ICU-type indicators are represented within the same event schema as variables observed at or near the beginning of the record when the preprocessing implementation provides them.

The event representation is intentionally irregular. Observations occur at nonuniform times, different variables are measured with different frequencies, and most variables are absent at most possible times. A missing measurement is therefore not interpreted as a normal value. It may reflect a clinical decision not to measure, an unrecorded observation, a source-table limitation, or a preprocessing exclusion. The thesis treats informative measurement and missingness processes as important threats to interpretation, not as causal findings established by the repository.

The primary patient- or stay-level outcome used by the downstream causal repository is `in_hospital_mortality`. In the causal processed artifact, this outcome is stored in `oc`, the outcome table keyed by `ts_id`. The third object in that processed artifact, `ts_ids`, is the sorted collection of normalized identifiers represented in `ts`. Later predictive processing uses a distinct split-aware artifact and must not be conflated with this causal contract.

3.3 Design-Time Knowledge Construction and Executed Analysis

The framework distinguishes a design-time representation-construction layer from deterministic patient-level instantiation and subsequent analysis. At design time, the official dataset-description link and instructions to research relevant clinical, causal, and time-series literature were supplied to a structured LLM-assisted elicitation protocol. The protocol requested a dataset audit, proxy-state ontology, binary decision-rule families, reasoning about missingness and measurement processes, and a dataset-specific DAG. The LLM received schema-level information and the mortality outcome, not patient records, empirical distributions, exposure prevalences, estimated effects, or execution access. The resulting proposals were project-selected and encoded in deterministic tagger and graph source files. Available evidence does not establish that every proposal was implemented unchanged, a complete line-by-line prompt-output-to-code mapping, or a formal clinician-review chain.

The analytical layer begins only after those designs have been encoded. It preprocesses raw ICU data, applies deterministic decision-tree rules to generate rule-derived proxy labels, trains deep irregular-time-series models to predict those labels, normalizes predicted outputs, and forms the archived final aggregate from one rule-derived source and four predicted-label sources before DAG-guided matching, effect estimation, sensitivity analysis, and permutation diagnostics. No LLM is invoked for patient-level preprocessing, label generation, prediction, adjustment selection, or effect estimation during this executed layer. Thus prompts document design proposals, active source defines implementation, and every downstream inference remains conditional on proxy, graph, support, and model assumptions. The two datasets follow this factorization separately; source-specific measurement availability and rule semantics are retained rather than forced into a common pooled representation.

3.4 Prediction Task

The prediction task is a multi-label proxy-state prediction problem over irregular ICU time series. For each study unit i and proxy state k, let $L_{ik}^{rule} \in \{0, 1\}$ denote a rule-derived proxy label. These labels are clinically inspired analytical constructs generated from observed data by the deterministic source-code rules selected and encoded after LLM-assisted design elicitation. They are weak clinical labels or proxy phenotypes, not chart-adjudicated diagnoses, manually verified clinical states, or validated phenotypes unless separate validation evidence is supplied in a later chapter.

The supervised predictive workflow estimates a vector of proxy-state labels from the observed time-series representation. Its target matrix is loaded from a latent-label CSV keyed by `ts_id`; the supervised proxy-label targets are not taken from the mortality column in `oc`. For model m, the exported prediction object may contain probabilities $\hat{p}_{ik}^{(m)}$ and corresponding binary predicted labels $\hat{L}_{ik}^{(m)}$. These outputs are model estimates of rule-derived proxy labels rather than direct measurements of patient physiology.

Within construction and characterization of each testbed, five analytical tasks are distinguished. First, deterministic proxy-state construction derives binary rule-derived proxy labels from observed clinical measurements. Second, multi-label proxy-state prediction estimates those rule-derived labels from irregular time-series inputs and measures proxy recoverability rather than clinical accuracy. Third, the archived final causal runs aggregate one rule-derived table with four aligned binary model-prediction tables into a majority-vote proxy-label table. Fourth, mortality prediction or association analysis evaluates whether proxy-state representations contain prognostic information about `in_hospital_mortality`. Fifth, observational causal-effect estimation treats selected columns from that mixed rule-and-prediction aggregate as modeled exposures and estimates adjusted contrasts under explicit assumptions. These tasks create or characterize different resource components; none supplies known causal truth. Detailed proxy rules, model architectures, estimator algorithms, and numerical results are intentionally deferred to later chapters.

3.5 Causal Question, Exposures, Outcome, Estimands, Assumptions

The downstream causal-analysis setting is retrospective and observational. No intervention is assigned by this study. The repository uses software parameters named `treatment` for selected binary `LAT_*` columns, but many such columns represent illness-state or organ-dysfunction proxy states rather than well-defined administered therapies. In the thesis narrative these variables are therefore described as proxy-state exposures or modeled treatment variables unless a later section explicitly defines a manipulable intervention.

Let $A_i \in \{0, 1\}$ denote a selected proxy-state exposure, Y_i denote the binary outcome `in_hospital_mortality`, W_i denote observed adjustment variables selected from a project-specified DAG and available columns, and X_i denote effect-modifier features used to describe heterogeneity. The dataset-specific DAGs were developed through a project design process that included LLM-assisted clinical and causal elicitation and were then encoded in source code; the implemented graph scripts, not the prompt outputs alone, define the DAG artifacts used by the pipeline. Potential-outcome notation such as $Y_i(a)$ is useful as conceptual language, but causal interpretation requires more than the presence of a graph or an estimator. It depends on assumptions about the causal graph, consistency, exchangeability conditional on observed covariates, positivity or overlap, outcome and exposure measurement, and unmeasured confounding [7, 8, 9].

Accordingly, later estimates should be interpreted as adjusted effect estimates under project assumptions, not as unconditional clinical effects. The existence of a DAG, matching script, or CATE estimator does not prove identification. Proxy states may be useful for structured analysis while still carrying label noise, temporal ambiguity, and intervention-definition limitations.

The main research question is how clinically grounded observational causal-analysis testbeds can be constructed from source-observed irregular ICU records—without simulating patient records, treatment or exposure-assignment mechanisms, or outcomes—and what converging evidence can characterize their usefulness and limitations for evaluating causal estimators. The five secondary questions stated in Chapter 1 cover separate deterministic instantiation in the two datasets; data contracts, normalization, aggregation, DAG adjustment, and provenance; proxy recoverability and mortality-relevant information; estimator, clinical-comparison, matching/support, permutation, and sensitivity evidence; and the limitations that preclude clinical validation, known causal truth, or proof of estimator correctness. These are study questions and design commitments; their answers are reported in Chapters 9 and 10.

Chapter 4

Data and Preprocessing

Preprocessing instantiates the source-specific data contract for each CliniCause resource. It does not generate synthetic patients or mortality outcomes: it selects, normalizes, aligns, summarizes, masks, imputes, or transforms source-observed measurements and outcomes according to the implemented pipeline. These operations distinguish source-observed events from extraction choices, cohort restrictions, preprocessing transformations, analysis-ready representations, and the downstream derived labels introduced in Chapter 5.

The contract is scientifically consequential because it determines which records enter, which measurements are available, how time and absence are represented, which identifiers survive, and which later proxy and causal-analysis objects can be instantiated. MIMIC-III and PhysioNet 2012 remain separate resources throughout. Their source-observed measurement, missingness, care-process, and support patterns pass through different extraction and transformation paths, so cross-dataset heterogeneity is a defining feature of the two observational testbeds rather than a preprocessing defect. Nothing in this chapter authorizes pooling their records or treating similarly named fields as equivalent.

4.1 PhysioNet 2012 Pipeline

The PhysioNet/Computing in Cardiology Challenge 2012 dataset provides ICU time-series records and associated in-hospital mortality outcomes for a mortality-prediction challenge setting [10]. In this thesis it supplies the source-observed measurements and outcomes from which one dataset-specific resource is instantiated for proxy-state prediction and downstream causal analysis. The local repository does not track the raw challenge files or a verified processed-data pickle, so this section describes the implemented preprocessing contract rather than asserting a raw preprocessing cohort total. Chapter 9 separately reports the validated causal-analysis model-row count.

The causal-repository PhysioNet preprocessing implementation traverses the raw `set-a`, `set-b`, and `set-c` folders and reads one patient-record file at a time. It removes rows with missing parameter names, skips records with at most five retained measurement rows, and removes observations with negative values, which the source code treats as missingness indicators. It converts source times from HH:MM strings into elapsed minutes, renames RecordID to ts_id, Parameter to variable, and Value to value, and reads outcome files by mapping In-hospital_death to in_hospital_mortality. After concatenating the source sets, it drops duplicate events and converts the categorical ICUType measurement into one-hot-style variables ICUType_1 through ICUType_4.

The resulting causal processed artifact is serialized as three objects: ts, oc, and ts_ids. The ts object is a long event table with ts_id, minute, variable, and value. The oc object contains ts_id, length_of_stay, in_hospital_mortality, and a source subset indicator. The ts_ids object is derived from identifiers observed in ts. This artifact is the expected input to rule-based tagging, mortality-from-proxy analysis, matching, and CATE estimation in the causal repository.

4.2 MIMIC-III Pipeline

MIMIC-III is a critical-care electronic health-record database used here to instantiate the second, separate ICU resource [11]. The implemented workflow follows a clinical time-series benchmark style in that it transforms raw ICU events into stay-level time-series examples suitable for prediction, while adapting the output contract for this thesis pipeline [25]. The resulting analysis-ready representation is constructed from source observations rather than being a synthetic patient or outcome generator. The repository does not establish a tracked raw-data snapshot, extraction date, or final processed artifact hash.

The active causal-repository MIMIC preprocessing implementation reads broad categories of MIMIC-III source tables: ICU-stay timing from ICUSTAYS, demographics from PATIENTS, bedside measurements from CHARTEVENTS, laboratory measurements from LABEVENTS, fluid outputs from OUTPUTEVENTS, administered inputs from INPUTEVENTS_CV and INPUTEVENTS_MV, and mortality/timing information from ADMISSIONS. Large event tables are processed in chunks and written to temporary shards before feature rows are collected. The implementation filters to adult ICU stays, removes chart events marked with an error flag, requires valid chart times and numeric or textual values, applies item-ID mappings and range filters for selected variables, converts selected units, and assigns events without direct ICU-stay identifiers to an ICU interval when possible.

For the causal contract, event times are converted to elapsed minutes relative to ICU admission, events are restricted to the early ICU window used by the implementation, age and gender are added as static rows at minute zero, and duplicate (ts_id, minute, variable) events are collapsed by averaging. The helper contract canonicalizes stay identifiers, normalizing numeric-looking identifiers such as 12345.0 to 12345. It defines the canonical ts columns as ts_id, minute, variable, and value, defines the canonical oc columns as ts_id, length_of_stay, in_hospital_mortality, and subset, and intentionally excludes internal MIMIC identifiers such as HADM_ID and SUBJECT_ID from the exported outcome table.

Inspection found a source-level issue that should be resolved before relying on local MIMIC preprocessing execution: the active script reads `ADMISSIONS.csv` with `DEATHTIME` for early-death filtering, while the canonical outcome helper requires `HOSPITAL_EXPIRE_FLAG` to construct `in_hospital_mortality`. This does not change the documented target contract, but it remains a source-level reproducibility risk.

4.3 STraTS Split-Aware Artifacts Versus Causal Artifacts

The predictive STraTS repository uses a different processed-data contract from the causal repository. Its loader expects a five-object pickle consisting of `events`, `oc`, `train_ids`, `val_ids`, and `test_ids`, where the final three objects explicitly encode train, validation, and test identifiers. The loader accepts stay identifiers named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and validates the split arrays after the same normalization. This split-aware contract is therefore not interchangeable with the causal repository's unsplit `[ts, oc, ts_ids]` artifact.

In supervised STraTS runs, the latent-label CSV is joined to the selected split identifiers after ID normalization. The loader filters labels to the supervised IDs, raises an error when labels are missing for any supervised `ts_id`, raises an error for duplicate normalized labels, and uses all non-`ts_id` columns in the latent CSV as proxy-label targets. The mortality column in `oc` is not used as the supervised proxy-label target. Prediction export writes `ts_id`, one probability column per proxy state, and one thresholded binary column per proxy state; the inspected wrapper scripts export the `all` split for their prediction CSVs, although final archive-copy provenance remains incomplete.

The predictive loader also implements split-aware leakage controls. It keeps only variables observed in the training split, derives normalization statistics from the training split, computes static-feature normalization from training rows, and validates label alignment before training. For MIMIC-III supervised prediction it restricts events to the first 24 hours. The pretraining path removes test data, uses training-derived variable statistics, uses a 48-hour maximum window for PhysioNet 2012, and permits MIMIC-III observations up to five days while sampling 24-hour input windows. These controls reduce some sources of leakage, but they do not guarantee absence of leakage, nor do they resolve missing raw-data and split-provenance limitations.

The parent router can bridge contracts by reading a causal `[ts, oc, ts_ids]` artifact and constructing a STraTS-style `[ts, oc, train_ids, val_ids, test_ids]` payload through a seeded identifier shuffle. That bridge is an orchestration convenience and does not prove how every archived final STraTS artifact was generated.

Several provenance limitations remain. Raw PhysioNet and MIMIC-III data are not tracked in Git. The processed pickles used by final runs are external or absent from the audited repository. Some run records contain machine-specific absolute paths, and final cohort counts require verified processed artifacts or manifests. Clean-checkout reproduction therefore requires

Table 4.1: Processed artifact contracts used by the causal and STraTS repositories.

Property	Causal repository	STraTS repository
Processed artifact	`[ts, oc, ts_ids]`	`[events, oc, train_ids, val_ids, test_ids]`
Split representation	Unsplit ID collection	Explicit train/validation/test IDs
Main use	Tagging and causal analysis	Predictive training and export
Identifier convention	Canonical `ts_id`	Loader-normalized `ts_id`
Proxy-label source	Separate latent CSV downstream	Separate latent CSV joined to split IDs

authorized data access, artifact hashes, and a documented split-generation record.

Accordingly, the two preprocessing paths instantiate two resource contracts rather than one pooled clinical table. They preserve the distinction between source-observed records and mortality outcomes, extraction and cohort decisions, transformed analysis objects, and downstream representation-defined constructs. The observational data-generating and measurement processes remain unknown; explicit preprocessing makes the transformations applied to their source-observed traces inspectable without claiming that every raw event or missingness pattern survives unchanged.

Chapter 5

Clinical Proxy-State Construction

This chapter defines the representation layer and its deterministic instantiation within each testbed. The construction sequence is schema and literature review, LLM-assisted candidate representation, project selection, deterministic source encoding, patient-level proxy instantiation from available observations, optional model-generated proxy annotations, and the fixed downstream aggregate where used. The LLM proposed designs but did not execute on patient records; active tagger source, rather than prompt text, defines every implemented proxy rule. Project-selected ontologies, decision rules, missingness considerations, and DAG proposals were encoded in tagger and graph source files.

The taggers instantiate representation-defined analytical constructs from source-observed measurements. A positive label means that the encoded rule was satisfied by available evidence. A zero label means that the rule did not fire on that evidence; because a missing measurement can prevent a clause from firing, zero does not establish clinical absence. Model predictions are learned annotations of the rule-derived labels, and the aggregate is a deterministic analytical interface. None of these objects is a diagnosis, expert adjudication, clinical consensus, or causal truth. Interpretability comes from inspectable rules and stable `LAT_*` interfaces, not from chart-adjudicated reference labels.

The chapter retains the scientific meaning of each proxy family and the active aggregation rule. Exhaustive prompt inventories, artifact hashes, prompt-to-source matrices, and clause-level rule tables are maintained in the repository evidence package rather than repeated in the main narrative. Predictive architectures appear in Chapter 6, and numerical results in Chapter 9.

5.1 Rationale and Terminology

Electronic phenotyping uses structured measurements, codes, and rules to construct analytical variables when direct adjudication is unavailable [5]. Here a *proxy state* or *proxy phenotype* is a binary construct summarizing available evidence for a clinically meaningful condition or physiologic pattern. A *rule-derived proxy label* is assigned by deterministic source rules; a *predicted proxy label* is a supervised model estimate of that rule-derived label; and an *aggregated* *predicted proxy-label representation* is the deterministic mixed-source vote used downstream. A binary proxy-state tag is the realized 0/1 value for one `ts_id` and proxy-state column.

Repository names such as `latent`, `latent tag`, and `LAT_*` remain stable implementation vocabulary, but they do not show that a hidden biological state was discovered. The proxy layer converts selected designs into source-coded constructs, supplies multi-label prediction targets, and provides a common schema for aggregation, mortality analysis, matching, and CATE estimation. No chart-adjudicated gold standard was found; missing measurements can prevent clauses from firing; thresholds approximate rather than reproduce complete clinical definitions; and dataset-specific states are not automatically construct-equivalent. This project also does not implement the Snorkel generative label model [6].

5.1.1 LLM-Assisted Design of Proxy States, Rules, and DAGs

The structured elicitation protocol requested a dataset audit, proxy-state ontology, binary decision-rule families, missingness and measurement-process reasoning, and a dataset-specific DAG for PhysioNet 2012 and MIMIC-III. The final dataset prompts instantiate a common template with dataset-specific names and official description links. Their exported runs preserve the resulting design proposals, but the exact system instructions, decoding settings, hosted-model configuration, follow-up interactions, and export procedure are incompletely documented.

The outputs were proposals subsequently selected and encoded by the project, not clinical validation or executable authority. The LLM did not inspect patient records at runtime, generate final labels, aggregate voters, select adjustment variables dynamically, or estimate effects. Active tagger and graph source defines implemented behavior; prompts document design provenance. Deterministic code then evaluates the encoded clauses against the available observations for each patient-level record. The final outputs align at schema level with the eleven active PhysioNet states and ten active MIMIC states, but complete accepted/rejected decisions and line-level prompt-output-to-code correspondence have not been established.

The full prompt inventory, version comparison, hashes, execution-metadata boundary, and prompt-to-source assessment are retained in `thesis-writing/audit/llm_prompt_provenance_audit.md`. Exact implemented proxy clauses and their citation status are retained in the active taggers and `thesis-writing/logs/stage_4_9B_proxy_rule_citation_matrix.csv`. These repository records form the technical evidence package; the main text below summarizes the constructs and behaviors needed to understand the method.

Figure 5.1 illustrates this design-to-instantiation distinction for the MIMIC-III shock rule. Its observed-evidence clauses correspond to the active operational definition in Table 5.2; the figure does not define an additional rule.

5.2 PhysioNet Proxy-State Rules

The active PhysioNet tagger pivots the long event table by `ts_id` and minute, computes per-record minima, maxima, means, first values, and last values, and applies one decision func-

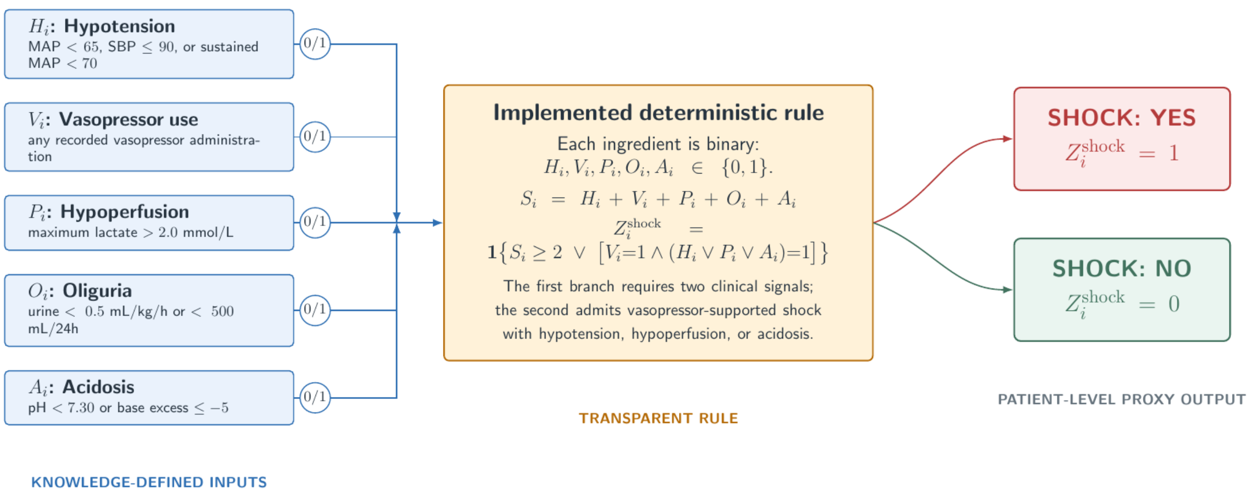

Figure 5.1: Illustrative deterministic instantiation of the project-selected MIMIC-III shock proxy. Observed-evidence clauses feed the encoded rule, and the patient-level label is derived from the available observations. A non-firing clause can reflect absent qualifying evidence; proxy value zero therefore does not prove absence of clinical shock. The figure illustrates the representation contract rather than validating the clinical construct.

tion per proxy column. Its eleven-state ontology comprises chronic baseline risk, global severity, shock, respiratory failure, renal dysfunction, hepatic dysfunction, coagulation/hematologic dysfunction, inflammation/sepsis burden, neurologic dysfunction, cardiac injury/strain, and metabolic derangement. The archived rule-derived CSV matches this schema and contains binary tags without duplicate normalized identifiers in the validated artifact; producing-command, input-hash, source-version, and threshold-mode provenance remain incomplete.

Numeric comparisons contribute positive evidence only when a non-missing value exists, so missing measurements are not treated as normal or abnormal. The active summarizer does not impose new time windows beyond the processed record. Urine clauses require a compatible precomputed urine summary, and oxygenation uses an available PaO2/FiO2 summary or an approximation from component summaries. Default source thresholds can be replaced by an explicit optimized-threshold file, but the archive does not establish that this option produced the final CSV.

Table 5.1: Operational PhysioNet 2012 rule-derived proxy-state definitions. Thresholds are active source defaults; the archive does not establish use of the optional optimized-threshold mode.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_CHRONIC_BASELINE_RISK`: baseline vulnerability or limited reserve.	First age; BMI calculated from first height and weight; minimum albumin; first ICU type.	Positive when at least 2 of age ≥ 75, BMI < 18.5 or ≥ 40, albumin < 3.0, and ICU type 1 or 3 are present.	Admission-wide summaries; BMI requires both inputs. The score and ICU-type clause are project specific.
`LAT_GLOBAL_SEVERITY`: multi-domain acute physiologic severity.	Seven domains: hemodynamic pressure; ventilation/oxygenation/respiratory rate; GCS; creatinine/BUN/urine; bilirubin/platelets; lactate/pH/bicarbonate; troponin/heart rate.	Positive for at least 3 abnormal domains, or at least 2 plus lactate ≥ 4.0, pH < 7.20, mechanical ventilation, GCS ≤ 8, or MAP < 60. Main domain cutoffs include MAP < 70, systolic pressure ≤ 100, PF < 300, SaO2 < 92, respiratory rate ≥ 22, GCS < 15, creatinine ≥ 2.0, BUN ≥ 40, urine < 500, bilirubin ≥ 2.0, platelets < 100, lactate > 2.0, and HR ≥ 130.	A project domain count, not SOFA; missing measurements suppress their domains.
`LAT_SHOCK`: circulatory shock or hemodynamic instability.	Minimum invasive/noninvasive MAP and systolic pressure; maximum lactate and HR; compatible urine summary; minimum pH and bicarbonate.	Positive when at least 2 of low MAP (< 70), low systolic pressure (≤ 90), lactate > 2.0, HR ≥ 110, urine < 500, or acidosis (pH < 7.30 or bicarbonate < 18) are present; low MAP with lactate ≥ 4.0 is also an explicit sufficient pair.	No vasopressor input in this dataset rule; urine evidence exists only when precomputed.
`LAT_RESPIRATORY_FAILURE`: hypoxemia, ventilatory failure, or ventilatory support.	Mechanical-ventilation flag; PF ratio; minimum SaO2 and PaO2; extreme respiratory rate; maximum PaCO2 with pH.	Positive for at least 2 of ventilation, PF < 300, SaO2 < 92 or PaO2 < 60, respiratory rate ≥ 22 or < 8, and PaCO2 ≥ 50 or PaCO2 ≥ 45 with pH < 7.30; ventilation with PF < 300 is an explicit sufficient pair.	PF may be approximated from component summaries; not adjudicated ARDS or respiratory failure.
`LAT_RENAL_DYSFUNCTION`: acute or clinically meaningful renal dysfunction.	First and maximum creatinine; maximum BUN; compatible urine summary; maximum potassium; minimum bicarbonate.	Positive for at least 2 of creatinine ≥ 2.0 or rise from first value ≥ 0.3, BUN ≥ 40, urine < 500, and potassium ≥ 5.5 or bicarbonate < 18; creatinine ≥ 3.5 is sufficient.	Not full KDIGO staging; timing is the available record and urine may be unavailable.
`LAT_HEPATIC_DYSFUNCTION`: hepatic injury or poor reserve.	Maximum bilirubin, AST, ALT, and ALP; minimum albumin and platelets.	Weighted score ≥ 2: bilirubin ≥ 2.0 contributes 2; AST or ALT ≥ 200, ALP ≥ 250, albumin < 2.5, and platelets < 100 each contribute 1.	Mixes injury, nutritional/reserve, and platelet evidence; project-specific construct.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_COAG_HEME_DYSFUNCTION`: coagulation or hematologic stress.	Minimum platelets and HCT; maximum HCT and WBC; first-to-minimum platelet change.	Weighted score ≥ 2: platelets < 100 contributes 2; platelets < 150, HCT < 25 or > 55, WBC < 4 or > 20, and platelet drop ≥ 50 each contribute 1.	Not a complete DIC or SOFA score; anemia and leukocyte extremes broaden the construct.
`LAT_INFLAMMATION_SEPSIS_BURDEN`: systemic inflammatory or sepsis-like burden.	Temperature and WBC extremes; maximum HR, respiratory rate, and lactate; minimum PaCO2 and platelets.	Positive for at least 3 of temperature > 38.3 or < 36, WBC > 12 or < 4, HR > 90, respiratory rate > 20 or PaCO2 < 32, lactate > 2.0, and platelets < 150; temperature or WBC abnormality together with lactate stress and score ≥ 2 is also sufficient.	No infection anchor; explicitly not confirmed sepsis or a Sepsis-3 implementation.
`LAT_NEUROLOGIC_DYSFUNCTION`: depressed consciousness or neurologic/metabolic stress.	Minimum GCS, sodium, glucose, SaO2, and pH; maximum sodium, glucose, and PaCO2.	Weighted score ≥ 2: GCS ≤ 12 contributes 2; GCS < 15, sodium < 130 or > 150, glucose < 70 or > 300, and PaCO2 ≥ 50, SaO2 < 90, or pH < 7.25 each contribute 1.	Sedation and ventilation are not separated from neurologic impairment.
`LAT_CARDIAC_INJURY_STRAIN`: myocardial injury, ischemia, or severe strain.	Maximum troponin I/T and HR; minimum HR, MAP, and systolic pressure; maximum lactate; first ICU type.	Weighted score ≥ 2: troponin I > 0.1 or troponin T > 0.01 contributes 2; ICU type 1 or 2, HR ≥ 130 or < 50, MAP < 70 or systolic pressure ≤ 90, and lactate > 2.0 each contribute 1.	Troponin alone can make the rule positive; assay/context harmonization is not encoded.
`LAT_METABOLIC_DERANGEMENT`: acid–base, lactate, electrolyte, or glucose derangement.	Minima/maxima of pH, bicarbonate, sodium, potassium, magnesium, glucose, and PaCO2; maximum lactate.	Positive for at least 2 abnormal domains: pH < 7.30 or > 7.50; bicarbonate < 18 or > 32; lactate > 2.0; sodium < 130 or > 150, potassium < 3.0 or > 5.5, or magnesium < 0.6 or > 1.2; glucose < 70 or > 250; PaCO2 < 32 or > 50. pH < 7.20, lactate ≥ 4.0, or potassium ≥ 6.0 is sufficient.	Project-specific composite; an isolated critical marker can determine the label.

The rules combine clinically recognizable evidence domains with project-specific Boolean or score thresholds. Global severity aggregates several organ domains; shock combines hypotension, perfusion, urine, and acid-base evidence; respiratory failure combines oxygenation, ventilation, and respiratory-rate evidence; renal dysfunction combines creatinine, BUN, urine, electrolyte, and acid-base evidence; and the remaining families combine their named physiologic domains. Table 5.1 states the active operational definitions. Sepsis-3, SOFA, KDIGO, ISTH DIC, ARDS, and respiratory-phenotyping sources provide limited conceptual grounding [29, 32, 30, 31, 33, 28]. The implementation is not a complete reproduction of those formal definitions, and the chronic, hepatic, cardiac, and metabolic clauses remain partly project specific.

5.3 MIMIC Proxy-State Rules and Validation Artifacts

The active MIMIC tagger defines ten states: chronic burden, inflammation/sepsis, global severity, shock, respiratory failure, renal dysfunction, combined hepatic/coagulation dysfunction, neurologic dysfunction, metabolic derangement, and cardiac strain. Relative to PhysioNet, it uses a chronic-burden construct, combines hepatic and coagulation evidence, and changes several available inputs and helper flags. These are construct and source-feature differences, not cosmetic renaming.

The tagger accepts three input modes. Summary-CSV mode consumes one row per stay. Canonical-pickle mode reconstructs rule features from `[ts, oc, ts_ids]`, including GCS components, first-24-hour urine output, rolling urine rates when weight is available, identifier normalization, and selected measurement aliases. Raw-concept mode aggregates admission, diagnosis, vital, laboratory, urine, vasopressor, ventilation, infection, and biomarker sources when supplied. Positive binary helpers and non-missing numeric comparisons provide evidence; absent numeric evidence does not make a rule positive. Input mode and extraction completeness therefore affect which clauses can fire.

MIMIC rules use the same broad score-or-sufficient-condition structure while exploiting richer care-process evidence. Examples include infection anchors for inflammation/sepsis, vasopressor and hypoperfusion evidence for shock, respiratory-support evidence for respiratory failure, dialysis evidence for renal dysfunction, and biomarker or rhythm evidence for cardiac strain. Table 5.2 gives the active dataset-specific definitions. The clinical sources above provide partial grounding, but the chronic-burden, combined hepatic/coagulation, metabolic, and cardiac constructs require qualified review and remain project-specific proxies.

Table 5.2: Operational MIMIC-III rule-derived proxy-state definitions. Available evidence depends on summary-CSV, canonical-pickle, or raw-concept input mode.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_CHRONIC_BURDEN`: chronic burden or baseline vulnerability.	Age or deidentified-old flag; comorbidity count or Elixhauser score; chronic-disease helpers; malignancy/immunosuppression; admission type.	Positive when at least 2 of age ≥ 75 or old-age flag, comorbidity count ≥ 2 or Elixhauser ≥ 5, chronic organ/ICD evidence, malignancy or immunosuppression, and emergency/urgent admission are present.	Helper availability is input-mode dependent; admission urgency is part of this project-specific baseline construct.
`LAT_INFLAMMATION_SEPSIS`: inflammation, suspected infection, or sepsis-like burden.	Temperature and WBC extremes; maximum lactate; culture/suspected-infection and antibiotic/suspected-infection helpers.	Score ≥ 2 across temperature ≥ 38.3 or < 36, WBC > 12 or < 4, lactate > 2.0, culture evidence, and antibiotic evidence; at least one culture, antibiotic, temperature, or WBC anchor is required.	Not formal Sepsis-3; culture and antibiotic helpers reflect care processes as well as disease evidence.
`LAT_GLOBAL_SEVERITY`: multi-domain acute physiologic severity.	Seven domains: pressure/vasopressors; oxygenation/ventilation; GCS/RASS; creatinine/urine; lactate/pH/bicarbonate; temperature/WBC; bilirubin/platelets/INR.	Positive for at least 3 abnormal domains. Main cutoffs are MAP < 70 or SBP ≤ 100; SpO2 < 92, PF ≤ 300, or ventilation; GCS < 15 or RASS $\leq -3/\geq 2$; creatinine ≥ 2.0 or 24-hour urine < 500; lactate > 2.0, pH < 7.30, or bicarbonate < 18; inflammatory thresholds; bilirubin ≥ 2.0, platelets < 100, or INR ≥ 1.5.	A project domain count, not SOFA; there is no single-marker sufficient condition.
`LAT_SHOCK`: shock or hemodynamic instability.	Minimum MAP/SBP and sustained-MAP helper; vasopressors; maximum lactate; 24-hour or rolling 6-hour urine; minimum pH and base excess.	Positive for score ≥ 2 across hypotension (MAP < 65, SBP ≤ 90, or sustained MAP < 70), vasopressors, lactate > 2.0, urine < 500 ml/24 h or < 0.5 ml/kg/h over 6 h, and pH < 7.30 or base excess ≤ -5; vasopressors plus hypoperfusion, acidosis, or hypotension is also explicit.	Support and urine helpers depend on extraction mode; not a formal shock diagnosis.

Proxy state and construct	Operational evidence	Decision form	Principal boundary
`LAT_RESPIRATORY_FAILURE`: gas-exchange or ventilatory failure/support.	Minimum SpO2 and PF ratio; maximum FiO2; mechanical/noninvasive ventilation; respiratory-rate extremes; maximum PaCO2 with pH.	Positive for score ≥ 2 across SpO2 < 92, PF ≤ 300, support (ventilation or FiO2 ≥ 0.5), RR ≥ 30 or ≤ 8, and PaCO2 ≥ 50 with pH < 7.30; support plus hypoxemia, low PF, or ventilatory failure is also explicit.	PF is derived from summary extrema; not adjudicated ARDS or respiratory failure.
`LAT_RENAL_DYSFUNCTION`: renal dysfunction.	Maximum and change-from-first creatinine; maximum BUN; 24-hour/rolling 6-hour urine; maximum potassium; minimum bicarbonate; dialysis/CRRT helper.	Positive for score ≥ 2 across creatinine ≥ 2.0 or delta ≥ 0.3, BUN ≥ 40, low urine, potassium ≥ 5.5 or bicarbonate < 18, and dialysis; dialysis/CRRT alone is sufficient.	Not full KDIGO staging; weight is required for normalized urine rate.
`LAT_HEPATIC_COAG_DYSFUNCTION`: combined hepatic and coagulation dysfunction.	Maximum bilirubin, AST, ALT, and INR; minimum platelets and albumin; prolonged-PT/PTT helpers.	Positive for score ≥ 2 across bilirubin ≥ 2.0, AST or ALT ≥ 120, platelets < 100, INR ≥ 1.5 or prolonged PT/PTT, and albumin < 2.5.	Combines liver injury, reserve, and coagulation evidence; not full SOFA or DIC scoring.
`LAT_NEUROLOGIC_DYSFUNCTION`: neurologic dysfunction with sedation caveat.	Minimum GCS; abnormal GCS-component helper; RASS extremes; pupil/focal-neurologic helper; sedation/intubation helper.	GCS ≤ 8 contributes 2; GCS ≤ 13, an abnormal GCS component, RASS ≤ -3 or ≥ 2, and pupil/focal abnormality each contribute 1; score ≥ 2 is positive. Source explicitly rejects an isolated one-point score when sedation/intubation is present.	Sedation is only partially handled; GCS components are mandatory in canonical-pickle construction.
`LAT_METABOLIC_DERANGEMENT`: acid–base, lactate, electrolyte, or glucose derangement.	Minima/maxima of pH, bicarbonate, potassium, sodium, and glucose; minimum base excess; maximum lactate.	Positive for at least 2 abnormal domains: pH < 7.30 or > 7.55; bicarbonate < 18 or > 35 or base excess ≤ -5; lactate > 2.0; potassium < 3.0 or ≥ 5.5; sodium < 130 or > 150; glucose < 70 or > 250.	Project-specific composite with no single-marker sufficient condition.
`LAT_CARDIAC_STRAIN`: cardiac strain or injury.	Troponin flags or fallback T/I values; CK-MB helper/value; arrhythmia or HR extremes; HR-plus-hypotension stress; cardiac unit/surgery context.	Positive for score ≥ 2 across troponin evidence (T ≥ 0.1 or I ≥ 0.4 fallback), CK-MB evidence (including a recorded value > 0), rhythm instability/HR > 150 or < 40, HR stress with hypotension, and cardiac-care context; the final condition additionally requires troponin-defined biomarker or rhythm evidence.	CK-MB contributes but cannot satisfy the anchor alone; assay units and clinical adjudication are not encoded.

The MIMIC output package contains the binary tag table, the merged features used by the rules, prevalence and co-occurrence summaries, unadjusted mortality-by-tag summaries, a JSON validation summary, and a decision-tree artifact. These files support implementation and descriptive validation only: prevalence is not clinical accuracy, risk ratios in the mortality table are not causal effects, and co-occurrence is not construct validation.

5.4 Predicted and Aggregated Proxy Labels

The same `LAT_*` interface can represent three generation mechanisms. Rule-derived labels come from the taggers, predicted labels from the four supervised time-series models in Chapter 6, and the final aggregate from deterministic voting over aligned binary files. Across all twelve archived final causal runs, the logs enumerate exactly five sources: one dataset-specific rule-derived table and predicted-label tables from GRU, GRU-D, STraTS, and TCN. The logs establish filenames and downstream handoff but do not co-archive the voter bytes, hashes, or checkpoint-to-export mappings.

Prediction exports contain `ts_id`, one probability column `<label>_prob` per target, and one corresponding binary column thresholded at 0.5. Voting consumes only the binary columns. The normalization helpers separate probability and binary fields, enforce the configured proxy order, and reject duplicate identifiers, nonbinary values, or incompatible schemas. Voter rows are aligned on the shared normalized identifier intersection, with the first sorted voter file defining output column and row order.

For record i, proxy state k, and M aligned voter files, let $L_{ik}^{(m)}$ be the binary value from voter m. The implemented rule is

$$\widetilde{L}_{ik} = \mathbb{I}\left(\sum_{m=1}^{M} L_{ik}^{(m)} \geq \left\lceil \frac{M}{2} \right\rceil\right).$$

It is equivalent to `2 * ones_count >= n_voters`, so an even-voter tie maps to 1. The orchestrator writes the resulting `ts_id`-keyed mixed-source table and passes it to mortality prediction, matching, CATE estimation, sensitivity analysis, and permutation stages.

This majority vote is a fixed analytical exposure interface, not expert consensus, five independent ground-truth labels, clinical truth, or a learned probabilistic label model. It consists of one rule-derived source and the four learned sources STraTS, GRU, GRU-D, and TCN. Those predicted voters were trained against the rule-derived targets, and the rule source also participates in the vote; correlated error can therefore persist or be amplified. Interpretation depends on stable construct definitions, aligned identifiers, binary-only inputs, and the exact five-source lineage. Aggregation is consequently subordinate to representation instantiation: it supplies one documented downstream interface without redefining or clinically validating the underlying proxy constructs.

Chapter 6

Predictive Modeling of Proxy States

This chapter evaluates *proxy recoverability*: whether deterministic rule-derived proxy labels correspond to learnable patterns in each testbed's irregular source-derived representation. Its supervision chain is LLM-assisted decision-rule design, project selection and deterministic source encoding, generation of rule-derived proxy labels, and deep-model prediction of those labels. The supervised target is the resulting rule-derived proxy-label matrix loaded from a separate CSV file keyed by `ts_id`; the inputs are irregular measurement histories plus static covariates prepared under the data contracts described in Chapter 4. The model outputs are learned annotations of analytical proxy labels, not source-observed exposures or adjudicated clinical states.

The final comparison contains four learned models: STraTS, GRU, GRU-D, and TCN. This chapter defines their representation, training, evaluation, and export mechanics; experiment selection and numerical comparisons are deferred to Chapters 8 and 9. Dataset-specific differences in the leading model characterize differences in the two resources and support dataset-specific model choice; they do not establish universal architectural superiority.

6.1 Self-Supervised STraTS Pretraining

STraTS represents an irregular time series as a set of observed measurement tokens rather than as a fully imputed grid. The implemented STraTS class constructs each token by adding three learned components: a continuous embedding of measurement time, a continuous embedding of the normalized measurement value, and an embedding of the variable identity. The token sequence is contextualized with a transformer-style block and then pooled with fusion attention to obtain a fixed-dimensional time-series representation [3]. Static covariates are combined with this representation before either the self-supervised forecasting head or the supervised proxy-label head is applied.

In the verified STraTS implementation, contextualized observation embeddings are pooled and combined with a learned embedding of static covariates. The recurrent, convolutional, and decay baselines in the final comparison do not have a supported self-supervised pretraining path in the current source.

The pretraining dataset loader reads the split-aware processed artifact and removes held-out test identifiers before constructing unsupervised examples. It builds the variable vocabulary from variables observed in the pretraining train split, derives value means and standard deviations from that train split, and saves the resulting variable list, normalization table, and maximum time horizon to `pt_saved_variables.pkl`. For PhysioNet 2012 the maximum time is 48 hours. For MIMIC-III the loader permits observations up to five days but samples at most a 24-hour input context and clamps implausibly old age values in the same way as the supervised loader.

Each self-supervised example is formed by sampling a forecast anchor from observed timestamps that are not the final timestamp and are at least 12 hours after record start. Observations up to the anchor form the historical context, truncated by the maximum observation count and, for MIMIC-III, by the 24-hour context limit. The target window extends two hours after the anchor. For each variable observed in that future window, the implementation keeps the last observed normalized value and sets the corresponding forecast mask entry to one; variables without future observations are masked out of the loss. Input times are rescaled to the interval $[-1, 1]$ using the dataset-specific maximum time.

The pretraining head maps the combined time-series and static representation to one forecast value per variable. Its objective is masked squared error over variables observed in the sampled future window, so unobserved future variables do not contribute to the loss. The pretraining evaluator materializes batches for each evaluation split, sampling three forecast batches per evaluation chunk when a split is first evaluated, and reports `loss_neg`, the negative masked loss, for checkpoint selection. The best pretraining checkpoint is written as `checkpoint_best.bin`; loss values themselves are not reported in this methods chapter.

Checkpoint-loaded supervised training is implemented by constructing the target supervised architecture, loading the supplied pretraining checkpoint, and copying overlapping tensors into the current model state before calling `load_state_dict`. When `--load_ckpt_path` is supplied, the supervised dataset also loads `pt_saved_variables.pkl` from the same directory to reuse the pretraining variable vocabulary, normalization statistics, and maximum-time metadata. This source path supports warm-started STraTS-family training, but the archived final exports still lack a complete manifest linking each exported CSV to the intended pretraining checkpoint, supervised checkpoint, split artifact, and archive-copy step.

6.2 Supervised Multi-Label Models

The supervised prediction task uses a deterministic rule-derived latent-label CSV as its target matrix. The loader accepts identifier columns named `ts_id`, `icustay_id`, or `ICUSTAY_ID`, canonicalizes them to `ts_id`, and checks consistency when more than one accepted identifier column is present. It then filters the label table to the supervised split identifiers, raises an error for missing labels, raises an error for duplicate normalized identifiers, sorts rows to match the internal supervised identifier order, and treats every non-`ts_id` column as a proxy-label target. The number and order of supervised targets are therefore determined at runtime by the CSV schema, but their semantic status remains rule-derived rather than clinical ground truth.

All supervised backends share a multi-label output contract. During training the model returns binary cross-entropy with logits against the full target vector, using per-target positive-class weights computed from the supervised train split. During evaluation or prediction export, the same model call without labels returns sigmoid probabilities. Static covariates are normalized from training rows and combined with each model's time-series representation. In scratch training the binary head maps the combined representation directly to proxy-label logits; checkpoint-loaded STraTS training uses an intermediate forecasting head so overlapping pretraining tensors can be reused.

The STraTS supervised backend uses the irregular observation triplets described in Section 6.1: time, value, and variable embeddings, transformer contextualization, fusion attention, and static-feature combination [3].

The GRU baseline uses a dense hourly grid rather than the raw irregular token set. For each hour and variable, the loader builds value, observation-mask, and elapsed-time-since-observation channels, mean-fills unobserved values from training data, normalizes the filled values, and concatenates the three channel groups before passing them to a gated recurrent unit [26]. The final recurrent hidden state is concatenated with the static-feature embedding and projected through the shared supervised head.

The GRU-D backend uses a model-specific irregular sequence representation containing observed values, missingness masks, elapsed-time tensors, sequence lengths, and static covariates [4]. The implemented cell updates last observed values, applies learned exponential decay to inputs and hidden states as a function of elapsed time, and includes missingness masks in the recurrent gates. The representation passed to the supervised head is the final valid recurrent state for each sequence combined with the static-feature embedding.

The TCN baseline uses the same dense hourly value, observation-mask, and elapsed-time representation as the GRU baseline, but processes the sequence with temporal convolutional residual blocks [27]. The implementation permutes the dense grid into channel-first form, applies a temporal convolutional network, takes the last temporal embedding, and concatenates it with the static-feature embedding before projection to proxy-label logits.

Together, these backends compare sparse-event attention, ordinary recurrence on a prepared grid, missingness-aware recurrence with learned decay, and temporal convolution. All four have archived prediction outputs for both datasets, while checkpoint-to-export, split, and archive-copy lineage remains incomplete. Table 2.2 summarizes their inductive biases without ranking them by sophistication.

The supervised evaluator computes metrics independently for each rule-derived target column in the requested split. Target columns with only one observed class in that split are skipped. For the remaining targets, it computes AUROC, area under the precision-recall curve, and the maximum value of the pointwise minimum of precision and recall. The reported split-level values are simple averages over the non-degenerate target columns, with loss and negative loss also recorded. During supervised training, checkpoint selection uses the validation value of `auprc + auroc`; this validation score should not be interpreted as a test score. These metrics quantify learnability or reproducibility of rule-derived targets, not clinical construct validity or correctness of the LLM-assisted decision-rule design. A model can reproduce both useful rule structure and systematic rule error.

Training mechanics are shared across the supervised backends at the top-level script. The script constructs a deterministic seed by adding the run index to the configured seed, uses AdamW for trainable parameters, clips gradient norms at 0.3, evaluates at the configured validation schedule, and saves `checkpoint_best.bin` when the validation selection metric improves. If an `eval_train` split is evaluated, it is the first 2000 training examples in the current supervised split object, which makes it a diagnostic subset rather than a separate external evaluation set.

6.3 Prediction Export and Normalization

Prediction export is requested with `--save_pred_csv_path`. The export routine can run after a training run or in an export-only invocation with no training steps scheduled. Immediately before export, the script reloads `checkpoint_best.bin` from the current output directory only if that file exists. The exported split is selected by `--predict_split`, whose accepted values are `train`, `val`, `test`, and `all`; in the `all` case the exporter iterates over all supervised identifiers in the train, validation, and test split object. The inspected wrapper scripts and archived export logs support `predict_split=all` for the final prediction CSVs. The logs also record `seed=2023`, and archived `checkpoint_best.bin` files can be hashed; nevertheless, the records do not supply split identifiers, a processed-artifact hash, or an independently verified checkpoint-to-export and archive-copy manifest.

For every exported row, the exporter removes labels from the batch, obtains probability outputs from the model, thresholds each probability at 0.5, and writes a combined CSV. The first column is `ts_id`. It is followed by one probability column named `<label>_prob` for each target and then one thresholded binary prediction column named exactly as the target label. These binary columns are model-predicted proxy labels, not rule-derived labels and not clinical adjudications.

The combined prediction CSV is not the same file shape required by downstream voter aggregation. The normalization helper validates `ts_id`, separates probability and binary fields, and writes binary-only voter files. The majority-vote contract then requires a shared identifier and binary proxy schema in the approved order. The archived final vote combines one rule-derived table with predicted-label tables from GRU, GRU-D, STraTS, and TCN; Chapter 5 defines its semantics. This is an algorithmic handoff, not expert consensus.

Probability exports and thresholded binary labels therefore serve distinct roles: probabilities retain model confidence for predictive evaluation, whereas binary exports enter the fixed aggregation interface. Prediction error can propagate through thresholding, aggregation, and the downstream observational analysis. Proxy recoverability shows only that the rule-derived constructs correspond to patterns learnable from the source-derived inputs; it does not establish clinical validity, diagnostic accuracy, causal validity, correctness of a proxy definition or DAG, or the usefulness of an intervention. Chapter 8 accordingly treats recoverability as one non-equivalent axis in the testbed's converging-evidence characterization.

Chapter 7

Causal Graph and Effect-Estimation Methodology

This chapter defines the causal and estimator interfaces packaged with each observational testbed. The project-specified DAGs expose representation-layer assumptions. Exposure-specific adjustment sets operationalize those graphs, while matching and heterogeneous-effect estimators characterize behavior conditional on the resulting interface. Transparency makes these assumptions inspectable and replaceable; it does not make the graphs correct or supply a known causal answer.

7.1 Dataset-Specific DAGs

The causal component of this project does not learn a causal graph from the observational data. It uses two LLM-assisted, project-specified directed acyclic graphs (DAGs), one for PhysioNet 2012 and one for MIMIC-III, as explicit representation-layer assumptions for selecting adjustment variables and organizing later effect-estimation outputs. A structured clinical and causal elicitation process generated graph proposals that were selected and encoded as active graph-construction source code. Work on LLM-derived soft priors provides a methodological precedent for using semantic judgments as candidate causal relations or orientation information [14], but it does not validate these local graphs or imply that its discovery algorithm was implemented here. The diagrams should therefore be read as source-encoded causal hypotheses that were not validated edge by edge and are not guaranteed complete, not as clinically validated or statistically discovered ground truth.

This separation is important because the main exposure variables in the causal stage are binary proxy states rather than randomized treatments. In the causal code, each selected proxy state is treated as a binary exposure A, the outcome is in-hospital mortality Y, and the graph is used to identify candidate pre-exposure background or latent proxy variables for adjustment. The thesis follows the graphical language of Pearl's DAG and backdoor framework while retaining the cautions around observational, proxy-defined exposures and well-defined interventions discussed by Hernan and Taubman [7, 9]. These cautions are especially relevant here because several variables named as “treatments” in the code are clinically closer to patient states or care-process proxies than to assigned therapeutic interventions.

The LLM has no estimator-time role in this chapter’s workflow. It is not called to inspect patient records, orient edges dynamically, select adjustment variables, or choose estimators during execution. The active graph scripts define the implemented nodes and edges, and the matching and CATE source code determines the implemented adjustment workflow from those encoded graphs and the columns available in each analysis dataframe.

Both graphs distinguish background variables, proxy states, observed measurements, care processes, missingness or measurement-intensity processes, and the mortality outcome. Only mapped background and eligible upstream proxy nodes can enter adjustment sets; measurement, care-process, missingness, and outcome nodes remain in the causal hypothesis but are excluded as adjustment candidates. MIMIC includes richer administrative and care-process structure, whereas PhysioNet uses a smaller background and observation vocabulary.

The PhysioNet graph contains 30 nodes and 45 directed edges. The implemented exposure list contains the ten non-chronic latent proxy states in the PhysioNet configuration: global severity, shock, respiratory failure, renal dysfunction, hepatic dysfunction, coagulation/hematologic dysfunction, inflammation/sepsis burden, neurologic dysfunction, cardiac injury/strain, and metabolic derangement. Chronic baseline risk is retained as an upstream baseline proxy and effect-modifier candidate rather than as an analyzed exposure.

The MIMIC graph contains 36 nodes and 57 directed edges. The implemented exposure list contains nine non-chronic latent proxy states: global severity, inflammation/sepsis, shock, respiratory failure, renal dysfunction, hepatic/coagulation dysfunction, neurologic dysfunction, cardiac strain, and metabolic derangement. Chronic burden is retained as an upstream baseline proxy and effect-modifier candidate. The MIMIC graph contains richer administrative and care-process structure than the PhysioNet graph, but the downstream adjustment implementation only uses graph nodes that can be mapped into the analysis dataframe.

Qualified clinical review of the graph arrows and a complete record of the human decisions applied to the LLM-assisted proposals remain unavailable. The graph artifacts and active source structure are recorded, but the exact producing source version, configuration, and command are not fully recovered.

The archived final run summaries and majority-vote logs establish the downstream proxy-label lineage at the stage boundary. In all twelve runs, the vote combines five enumerated inputs: one deterministic rule-derived proxy-label table and four thresholded prediction tables from GRU, GRU-D, STraTS, and TCN. Each run writes `majority_vote/latent_tags_majority_vote.csv` and passes that mixed rule-and-prediction aggregate to mortality prediction, matching, and CATE estimation. The logs preserve the external absolute filenames, but not the voter-file bytes, hashes, or checkpoint-to-export mappings; those parts of the lineage therefore remain incomplete.

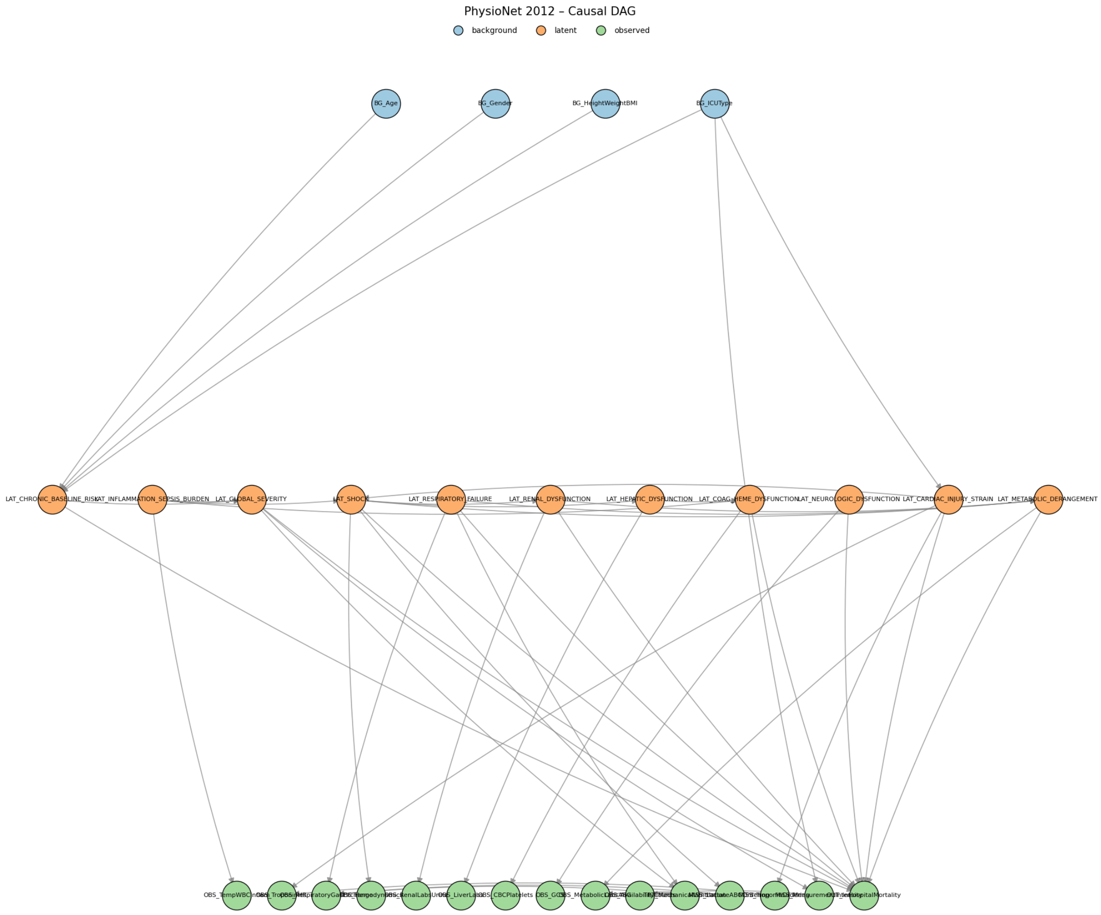

Figure 7.1: Project-specified PhysioNet 2012 DAG used by the causal pipeline. The active graph source reproduces the archived artifact; arrows encode assumed rather than learned or clinically validated relationships.

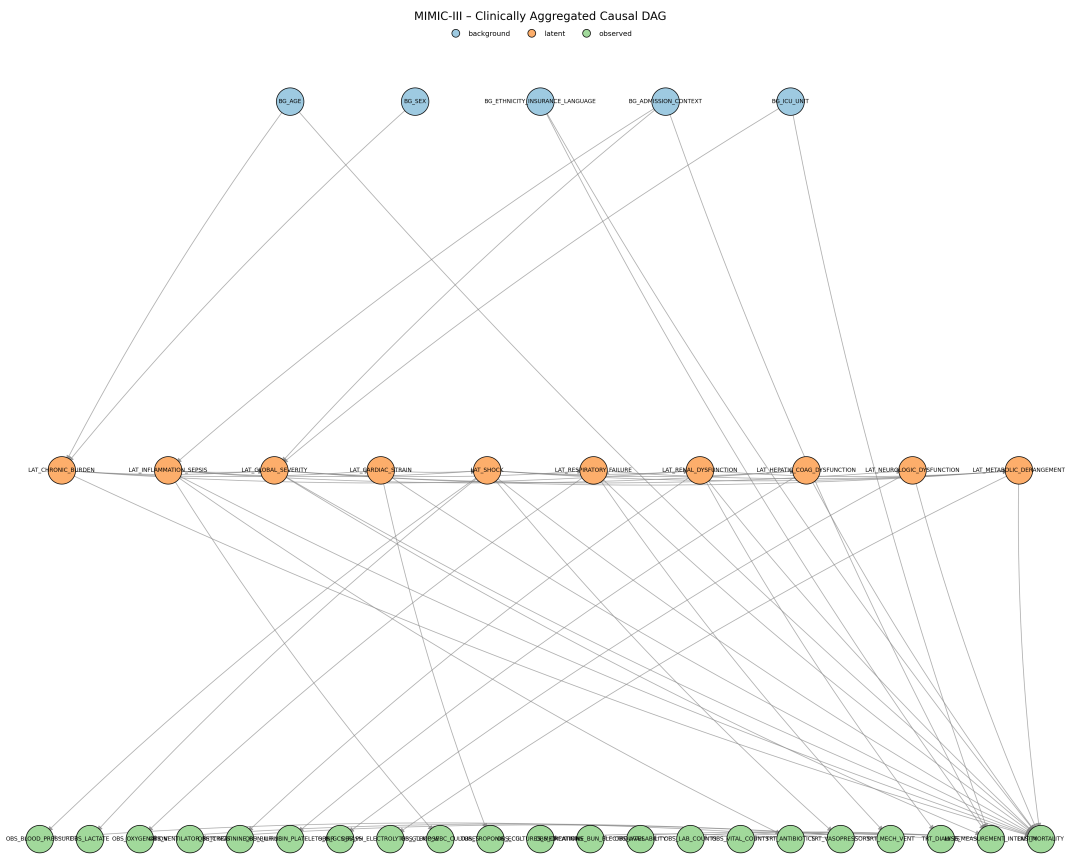

Figure 7.2: Project-specified MIMIC-III DAG used by the causal pipeline. The active graph source reproduces the archived artifact; arrows encode assumed rather than learned or clinically validated relationships.

7.2 Adjustment-Set Logic

Exposure-specific adjustment sets convert the project graphs into an executable analysis interface. They record what the encoded graph and available columns permit the pipeline to adjust for; they do not establish that the representation contains every confounder or that exchangeability holds.

For each dataset, treatment-like proxy state, and outcome pair, the causal pipeline derives an adjustment set from the project DAG and the columns available in the analysis dataframe. The implementation first maps dataframe variables to graph nodes and graph nodes back to dataframe columns. Background and latent graph nodes are eligible only if they can be represented by available columns. Observed measurement nodes, care-process nodes, missingness nodes, and the outcome node are excluded from candidate adjustment sets.

Let G denote the implemented project DAG, A the binary proxy-state exposure, and Y the in-hospital mortality outcome. The code constructs a graph $G_{\underline{A}}$ by removing outgoing edges from A. This is a project helper for finding adjustment candidates; it should not be confused with the usual intervention graph notation that removes incoming edges into A. The initial candidate pool is:

$$\mathcal{C}(A, Y) = \mathcal{A}_{\text{allowed}} \cap \mathrm{An}_G(A) \cap \mathrm{An}_{G_{\underline{A}}}(Y) \setminus \mathrm{De}_G(A) \setminus \{A, Y\},$$

where $\mathcal{A}_{\text{allowed}}$ is the set of available graph nodes whose node type is background or latent. The implementation then enumerates backdoor paths between A and Y in the undirected skeleton whose first directed edge points into A. A path is treated as blocked if an adjusted non-collider is present or if an unadjusted collider has no adjusted descendant. This is an encoded graphical test over the project DAG, not proof that all real-world confounding has been controlled.

The preferred helper is the safe expanded adjustment mode. It removes encoded colliders and their descendants from the full candidate pool, then tests whether the remaining mapped variables block every encoded backdoor path. If not, a deterministic greedy fallback removes sorted candidates only while path blocking is preserved. The returned set is inclusion-minimal under that order, not necessarily unique. The accompanying identifiability flag is therefore an availability diagnostic within the implemented graph, not proof of exchangeability in the clinical population.

Table 7.1: Causal assumptions required for interpreting the estimator outputs.

assumption	methodological requirement	repository support	unresolved limitation
Graph correctness for the target question	The DAG must correctly represent the relevant causal structure for the exposure contrast and mortality outcome.	Dataset-specific graphs were developed through LLM-assisted elicitation, project selection, and active source encoding.	The arrows were not learned from data, and the human clinical review record remains incomplete.
Sufficient measured adjustment	Available adjustment variables must block the relevant open backdoor paths for each proxy exposure.	The helper tests path blocking within the implemented DAG using mapped background and latent columns.	Graph variables may be unavailable, and unmeasured or misspecified confounding outside the project DAG remains possible.
Positivity or overlap	Comparable covariate strata must contain both exposed and unexposed records.	Matching outputs report match availability and match rates.	Dedicated treatment-specific overlap diagnostics and thresholds remain to be approved.
Consistency and well-defined exposure	Each binary proxy-state contrast must have a sufficiently stable meaning across records.	The pipeline applies one binary exposure column at a time.	Disease-state and severity proxies are not necessarily assignable interventions, so the causal estimand remains unresolved.
Proxy-state measurement	The binary exposure should measure the intended construct with sufficiently limited construct and classification error.	Final archived analyses use a deterministic aggregate of one rule-derived source and four predicted-label sources trained against the rule-derived targets.	No code path verifies construct validity or removes proxy-state measurement error; clinical and chart-review validation remains incomplete.

assumption	methodological requirement	repository support	unresolved limitation
Outcome measurement	The processed `in_hospital_mortality` field must be consistently and correctly constructed from the outcome artifact.	Both estimator paths merge this processed outcome column by record identifier and require it to be present.	Source checks establish schema availability, not endpoint validity or cross-dataset construction equivalence.
Temporal ordering and avoidance of post-exposure adjustment	Adjustment variables must be measured or conceptually occur before the relevant proxy-state contrast; post-exposure adjustment should be avoided.	Protection depends on the project DAG, source node classification, descendant filtering, and allowed-node filtering.	Repository graph ordering does not prove patient-record timing; actual variable timing and the correctness of the proxy-state construction window remain unverified.
Nuisance- and effect-model adequacy	LinearDML and CausalForestDML require sufficiently adequate nuisance and effect models, regularity conditions, and appropriate cross-fitting behavior for their intended interpretation.	The source configures EconML estimators, random-forest nuisance models, and estimator-provided fitting behavior.	Source configuration alone does not verify model adequacy, regularity conditions, calibration, or valid cross-fitting performance in these data.
No interference	One record's proxy-state exposure should not affect another record's outcome under the working unit-level interpretation.	Estimation treats records as separate units.	The repository does not empirically test interference or dependence across admissions, patients, units, or care processes.

assumption	methodological requirement	repository support	unresolved limitation
Sampling and target population	The analysis sample must support the stated target population, or reweighting or another transport argument must justify a sampled target.	The optional outcome-downsampling condition is deterministic given the configured seed and is applied consistently before treatment iteration.	Original and outcome-downsampled analyses correspond to different empirical populations; no weighting or target-population correction is documented.

7.3 Matching and Heterogeneous-Effect Estimators

The causal pipeline applies two families of estimators to the mixed-source majority-vote proxy-label exposures: a transparent matched-pair baseline and model-based heterogeneous-effect estimators. Matching is a secondary descriptive and support-oriented comparison, not a definitive treatment-effect estimator. Both families use the same broad analysis frame. Patient-level aggregated proxy labels are merged with the canonical in-hospital mortality outcome and baseline covariates. For a selected exposure $A_i \in \{0, 1\}$, outcome Y_i, adjustment variables W_i, and effect modifiers X_i, the target working quantity is a contrast in predicted or matched mortality outcome between $A_i = 1$ and $A_i = 0$, conditional on the source-defined covariate information. Because the exposures are proxy states and the data are observational, often without a simple well-specified intervention, these contrasts require cautious interpretation.

Measurement error can propagate through this entire chain. An unsupported design proposal can enter a selected rule or edge; a deterministic rule can misclassify the intended construct; a deep model can reproduce that error or add prediction error; aggregation can preserve correlated error; and the resulting exposure misclassification can alter adjustment, matching, and effect estimates. An LLM-assisted graph is not intrinsically more correct than a manually proposed graph, and predictive agreement with source-encoded labels does not repair graph or exposure-measurement assumptions.

An optional analysis condition, controlled by `DOWN_SAMPLE`, down-samples mortality outcome classes rather than exposure groups. All outcome-positive rows are retained. When the outcome-negative class is larger and the positive class is nonempty, outcome-negative rows are sampled without replacement until their count equals the number of outcome-positive rows; the configured random seed controls both this sample and the subsequent row shuffle. This occurs once before treatment-specific adjustment selection and estimation, including matching and DML/PFN fitting. It is not propensity trimming, treatment balancing, matching, or a correction for confounding, and it does not improve causal identification by itself. The procedure changes the analyzed population and observed outcome prevalence. The thesis therefore treats original cohorts as primary and outcome-downsampled cohorts as robustness analyses only, with no pooling between them.

Outcome-based downsampling can change nuisance-model estimation, the distribution of covariates and proxy exposures, patient-level fitted effects, the population over which `mean_cate` is averaged, and the availability and composition of matched pairs. It also changes the sample outcome rate and therefore affects `normalized_CATE`, whose denominator is that rate. Effect summaries from original and outcome-downsampled analyses should not be compared as though they estimated the same population quantity without an approved sampling and estimand argument. Outcome-based case-control sampling does not necessarily invalidate every estimator, but interpretation here requires estimator-specific methodological justification and an explicit target-population argument.

The matching estimator is a deterministic baseline. For each proxy exposure, the pipeline drops rows with missing exposure or outcome, coerces the exposure and outcome to binary variables, imputes numeric adjustment variables with medians, and converts non-binary numeric adjustment variables into binary columns using sample-median thresholds. Already-binary variables, including ICU-type indicators, remain binary. The matcher then greedily pairs exposed and unexposed patients by Hamming distance over these binary adjustment columns, increasing the allowed distance up to the configured maximum. By default, controls are not reused. The per-pair contrast is

$$\Delta_j = Y_{j,A=1} - Y_{j,A=0},$$

and the matching summary reports the mean and standard deviation of these matched-pair differences, the number of matched pairs, the match rate, the final allowed Hamming distance, and the observed adjustment columns. This output is useful as a transparent matched descriptive contrast. It should not be labeled as an average treatment effect unless the estimand, graph assumptions, overlap, and matching design are later approved for that interpretation.

For model-based heterogeneous effects, the pipeline includes EconML double-machine-learning estimators and an exploratory CausalPFN branch. The prespecified hierarchy treats CausalForestDML as primary, LinearDML as the secondary comparator, and CausalPFN as exploratory. Double machine learning estimates nuisance outcome and treatment models before estimating effects from residuals, reducing sensitivity to nuisance-model overfitting under appropriate assumptions [15, 34]. The LinearDML branch uses random-forest nuisance models with a linear final effect model. The CausalForestDML branch uses the same nuisance-model family and a causal-forest final stage, drawing on generalized random-forest and causal-forest ideas for heterogeneous treatment effects [16, 17]. This choice is consistent with broader machine-learning work on individualized treatment effects in electronic health records and critical care, while still requiring domain-specific validation [18, 19, 35, 36, 37].

The CATE pipeline saves patient-level CATE values, normalized CATE values, optional 95 percent effect-interval columns when the estimator provides them, feature-importance or coefficient diagnostics when available, model artifacts, and run-level summaries. In the current source, `mean_cate` is the arithmetic mean of patient-level CATE estimates over the model rows. It is an aggregate of the fitted model's conditional estimates and should not automatically be described as the study ATE. The normalized CATE column is computed by dividing CATE by the sample outcome rate when that rate is positive. It is an implementation-defined rescaling for comparison within this project; it is not a risk ratio, relative risk, or percent reduction.

The effect-modifier matrix X is selected from configured effect-modifier columns, typically baseline variables plus chronic and inflammation/sepsis proxy states when present. The adjustment matrix W is the observed confounder set selected from the DAG helper. The implementation can allow overlap between X and W, because configured effect modifiers are not automatically removed from the adjustment set. This is a source-supported behavior and a remaining methodological limitation.

Table 7.2: Implemented causal estimators and interpretation limits.

method	implementation summary	main outputs	interpretation boundary
Greedy Hamming matching	Binary adjustment matrix, median thresholding for non-binary covariates, median imputation, nearest available control within allowed Hamming distance, no default control reuse.	Matched pairs, pair effects, match rate, final distance, and per-treatment/global summaries.	Matched descriptive contrast; not automatically ATE or ATT.
LinearDML	EconML `LinearDML` with random-forest nuisance outcome and treatment models, binary treatment handling, configured W and X.	Patient-level CATE, optional intervals, coefficients when aligned, model artifact, global summaries.	Model-based heterogeneity estimate under project DAG, nuisance-model, overlap, and proxy-exposure assumptions.
CausalForest DML	EconML `CausalForestDML` with random-forest nuisance models and a causal-forest final stage.	Patient-level CATE, optional intervals, feature importance, model artifact, global summaries.	Flexible heterogeneity estimate; feature importance is a model diagnostic, not mechanism proof.

method	implementation summary	main outputs	interpretation boundary
CausalPFN	Exploratory branch using deduplicated confounder and effect-modifier features. It writes CATE CSVs and metadata but skips EconML-specific sensitivity and permutation stages.	Patient-level CATE and normalized CATE; interval columns are unavailable and stored as missing.	Exploratory only; its primary method source, implementation version, uncertainty, and comparable diagnostic support remain unresolved.

Across these interfaces, model-estimated CATE summaries remain conditional on proxy measurement, temporal ordering, graph adequacy, observed adjustment, empirical support, model specification, and the estimand definition. The hierarchy remains: CausalForestDML is primary, LinearDML is the secondary comparator, and CausalPFN is exploratory and lacks equivalent archived uncertainty, sensitivity, and permutation support. Sign, rank, or magnitude agreement among fitted estimators supports converging-evidence characterization under the fixed representation and population; it does not measure causal accuracy or establish identification.

Chapter 8

Converging-Evidence Evaluation, Robustness, and Sensitivity

This chapter is the methodological center for evaluating the two observational testbeds without a causal answer key. It defines the experiment families, population hierarchy, converging-evidence axes, diagnostic framework, result-admission policy, and reproducibility boundary underlying Chapter 9. The axes are deliberately non-equivalent: each investigates a different failure mode, and their findings cannot be added together as proof of clinical correctness, causal identification, or estimator accuracy. They supplement rather than replace the project DAG, exposure definition, adjustment assumptions, empirical support, and clinical review.

8.1 Converging-Evidence Axes

8.1.1 Construct plausibility

Construct plausibility asks whether a proposed representation has a recognizable clinical rationale and an inspectable operational definition. The evidence consists of literature-grounded representation design, the preserved rationales and source links, rule inspection, and the bounded informal ICU-clinician consultation reported by the finished study. There was no blinded formal clinician-rating study, chart adjudication, inter-rater reliability analysis, or clinical construct-validation study. This axis can identify an implausible or poorly specified construct, but it cannot validate a proxy as a diagnosis, establish that a graph is correct, or show that the encoded state is causally adequate.

8.1.2 Proxy recoverability and mortality-relevant predictive information

Proxy recoverability, detailed in Chapter 6, asks whether the deterministic rule-derived labels correspond to learnable patterns in the irregular source-derived representation. Performance against those labels measures reproduction of a representation-defined target, not clinical correctness. A separate mortality-prediction task asks whether proxy representations contain information associated with source-observed in-hospital mortality. Such prognostic information can characterize a resource, but it remains associational and does not establish causal relevance, diagnostic accuracy, intervention utility, correctness of the proxy, or correctness of the DAG.

8.1.3 Cross-estimator stability

Cross-estimator stability asks whether fitted effect patterns are unique to one estimator under a fixed representation and population. Three distinct dimensions are retained: sign agreement compares direction, rank agreement compares the ordering of exposures within a resource, and magnitude agreement compares numerical scale. These dimensions are not interchangeable, and estimators may also target or approximate different quantities. Agreement supplies evidence that a pattern is not confined to one fitted estimator; it does not establish estimator accuracy, estimand equivalence, identification, or truth.

8.1.4 Clinical-literature comparison

Clinical-literature comparison provides contextual corroboration by relating a CliniCause proxy and its model-estimated mortality contrast to the nearest usable published clinical construct. Where available, the comparison records the represented clinical construct, operational definition, study population, comparison group, mortality horizon, effect scale, adjustment method, reported mortality contrast, and major comparability limitations. Candidate comparisons are interpreted only at the level supported by the finished paper and supplement.

External studies can differ in clinical definition, severity, eligibility, data source, calendar period, treatment context, observation window, mortality horizon, matching or adjustment, estimand, empirical support, and residual confounding. The comparison is therefore a way to identify broad compatibility and discrepancies. It is not a meta-analysis, calibration against causal truth, external validation of a proxy, or proof that a CliniCause estimate is correct. Numerical synthesis is reserved for Chapter 9 and is not introduced in this methods chapter.

8.2 Populations and Experiment Families

8.2.1 Predictive experiments

The predictive comparison crosses PhysioNet 2012 and MIMIC-III with four learned models: STraTS, GRU, GRU-D, and TCN. Each model predicts dataset-specific deterministic rule-derived proxy labels, not mortality or the later aggregate. The workflow includes split-aware loading, multi-label training, validation-based checkpoint selection, test evaluation, and probability/binary prediction export; STraTS additionally supports self-supervised pretraining. These experiments operationalize the proxy-recoverability axis rather than clinical validation. Training summaries and prediction exports exist for all eight dataset–model combinations. Processed-pickle hashes, split identifiers, complete producing commands, checkpoint-to-export mappings, and archive-copy histories remain incomplete, so implementation availability, artifact availability, and complete provenance are not interchangeable claims.

8.2.2 Causal experiments and population hierarchy

The causal design crosses two datasets, three estimators, and two sampling conditions, yielding twelve archived run families. CausalForestDML is primary, LinearDML secondary, and CausalPFN exploratory. Original cohorts define the primary analyses. The robustness condition retains all mortality-positive rows and samples mortality-negative rows before exposure-specific matching and estimation; it therefore changes the empirical population and outcome prevalence and is never pooled with the original cohorts.

Every final run uses a deterministic five-source aggregate: one rule-derived table plus thresholded predictions from STraTS, GRU, GRU-D, and TCN. The runtime order is graph construction, mixed-source aggregation, mortality prediction, matching, CATE estimation, saved-CATE analysis, and permutations. CausalPFN completes CATE estimation but its EconML-specific saved-analysis and permutation stages are intentionally skipped, not failed. Within each run, all prespecified dataset-specific non-chronic proxy exposures are evaluated subject to column, graph-node, binary-value, adjustment, support, and artifact checks; the exposure list is a design commitment rather than a set of causal conclusions.

Table 8.1: Experiment families, populations, and evidential roles.

Family	Design	Primary output	Evidential role
Prediction	Two datasets × four learned models	Test metrics and binary/probability exports	Comparison of rule-label learnability; not clinical validation
Causal, original	Two datasets × three estimators	Matching and model-estimated CATE summaries	Primary forest, secondary linear, exploratory PFN analyses
Causal, downsampled	Same datasets and estimators after outcome-class sampling	Separate matching and CATE summaries	Robustness population only; no pooling or automatic common estimand

8.3 Executed Settings and Provenance Status

Table 8.2 separates what the checked archive supports from what only current source defines and what remains unresolved. *Artifact-supported* denotes a checked summary, export, run summary, or diagnostic record; *log-supported* denotes an argument recorded in archived logs without a complete immutable producing manifest; *source-defined* denotes implemented behavior that must not be mistaken for a recovered historical setting. *Unresolved* and *not applicable* retain their literal meanings. This classification is necessary because a current default, a historical path, and an archived output answer different reproducibility questions.

Table 8.2: Compact specification of executed or execution-supported settings and unresolved lineage.

Analysis component	Confirmed or execution-supported settings	Unresolved provenance or interpretation boundary
Prediction scope and artifacts	*Artifact-supported*: PhysioNet 2012 and MIMIC-III crossed with STraTS, GRU, GRU-D, and TCN; all eight dataset–model combinations have checked training summaries and prediction exports. The task is multi-label prediction of dataset-specific rule-derived proxy labels.	*Unresolved*: processed-pickle hashes, split identifiers, producing source revision, and archive-copy direction. Artifact availability does not establish an independently auditable out-of-sample membership.
Archived predictive run settings	*Artifact-supported*: checked training summaries identify run `1o10`, training fraction 0.5, validation and test evaluation, and checkpoint selection by AUPRC plus AUROC for each of the eight learned combinations. *Log-supported*: archived argument records give seed 2023 and `predict_split=all`.	*Unresolved*: the seed is not a split manifest; the raw shell command, immutable producing configuration, and checkpoint-to-export mapping are absent. `all` is the recorded export argument, not proof of a particular split membership.
Prediction export contract	*Artifact-supported*: each checked export has unique `ts_id` rows, one `<label>_prob` field and one binary `<label>` field per active target, valid probability ranges, and binary values. *Source-defined*: the binary export threshold is 0.5.	*Unresolved*: archived exports are hashable, as are candidate best checkpoints, but no immutable record proves which checkpoint produced each export.
STraTS pretraining	*Source-defined*: STraTS alone has a separate self-supervised pretraining path before supervised fine-tuning; GRU, GRU-D, and TCN do not use that path. Archived STraTS training and checkpoint artifacts exist.	*Unresolved*: pretraining-checkpoint, supervised-checkpoint, and final-export lineage is incomplete; source support does not prove the exact pretraining state used historically.
Causal run matrix and hierarchy	*Artifact-supported*: two datasets, CausalForestDML, LinearDML, and CausalPFN, and original/outcome-downsampled conditions yield twelve archived run families with stage records. Original cohorts are primary; outcome-downsampled populations are separate robustness analyses. CausalForestDML is primary, LinearDML secondary, and CausalPFN exploratory.	*Unresolved*: numbered final causal configuration files and processed-input hashes are missing. Recorded historical paths are not portable configurations, and current unnumbered configs are only candidates.

Analysis component	Confirmed or execution-supported settings	Unresolved provenance or interpretation boundary
Exposure, aggregation, and support	*Artifact-supported*: every archived final run records the same five-source mixed aggregate—one rule-derived table plus binary predictions from STraTS, GRU, GRU-D, and TCN. *Source-defined*: analysis is exposure specific; graph/column checks govern exposure admission, while matching supplies descriptive support and pair-availability diagnostics before CATE interpretation.	*Unresolved*: voter bytes and hashes are not co-archived per run, and matching does not establish positivity, balance, exchangeability, or a common estimand across sampling conditions.
Sensitivity evidence	*Artifact-supported*: CausalForestDML and LinearDML records distinguish fitting-time availability, saved-training direct values, later artifact analysis, labelled fallback reconstruction, benchmark status, partial rows, and failures. *Not applicable*: the archived CausalPFN branch has no equivalent EconML saved-analysis family.	*Unresolved*: evidence classes are not numerically interchangeable; incomplete or failed rows cannot be promoted to effects or sensitivity magnitudes, and producing environment lineage remains partial.
Permutation checks	*Artifact-supported*: validated CausalForestDML and LinearDML records contain 10 trials with seed 42 for both treatment-column and outcome permutations in each archived dataset–exposure–sampling combination. Treatment permutations alter the analyzed proxy column; outcome permutations alter mortality in a copied outcome object. *Not applicable*: CausalPFN permutations were intentionally skipped.	These checks diagnose behavior under disruption; they are not randomization tests or identification proof. Temporary trial artifacts were removed after aggregation, so only checked aggregate records remain.
Producing lineage	*Artifact-supported*: causal run summaries retain stage argv arrays, statuses, return codes, and limited interpreter paths; predictive logs retain argument records and limited device fields; result and reproducibility packages retain hashes and schemas.	*Unresolved*: missing processed-pickle hashes and split manifests; incomplete predictive producing commands and checkpoint mapping; missing numbered causal configs; incomplete source, package, hardware, and archive-copy lineage. Requirements are intended dependencies, not producing-environment proof.

8.4 Matching and Empirical Support

Matching provides a secondary descriptive comparison and indirect support evidence under its implemented binary representation. For each proxy exposure, the summaries record available pairs, match rate, final Hamming distance, a sufficient-pair flag, maximum-distance warnings, and explicit failures. Binary confounders are retained, while other numeric confounders may be median-binarized; greedy one-to-one matching then searches within an allowed distance. A larger accepted distance therefore indicates weaker exact similarity in this representation, not a calibrated distance in the original covariate space. Failure to obtain pairs is informative support evidence rather than a zero effect. Matched outcome differences remain descriptive and estimator-dependent; successful matching or a high match rate does not prove positivity, identification, or balance.

Dedicated propensity distributions, common-support plots, pre/post-match balance tables, standardized mean differences, and weighting effective sample sizes were not present in the approved archive. The available matching fields help identify unsupported comparisons but do not establish global or local positivity, covariate balance, exchangeability, or transportability [38].

8.5 Permutation or Disruption Checks

Permutation checks disrupt one relationship while retaining the remaining run structure. Treatment permutations shuffle only the currently analyzed proxy column; outcome permutations shuffle only the mortality column in a copied outcome object. The other proxy columns, time-series data, identifiers, graph, configuration, estimator family, and aggregation fields remain fixed as appropriate. Subprocess and schema failures are recorded rather than silently aggregated, and temporary trial artifacts are removed after summary extraction. The validated DML records contain ten trials with seed 42 for each dataset–exposure–sampling combination. These are negative-control-like sanity checks, not formal randomization tests, p-values, proof of calibration, or evidence that the DAG and exposure are valid [41]. CausalPFN permutations are intentionally absent.

A response to disrupted exposure or outcome labels tests whether the fitted pipeline distinguishes the observed structure from deliberately broken structure under the implemented procedure. Even a clear response is a sanity check on pipeline behavior, not causal validation or proof that the original estimate is correctly identified.

8.6 Omitted-Variable Sensitivity

The DML workflow distinguishes method availability, values called during fitting, saved-training direct values, later saved-artifact recomputation, explicitly labelled fallback reconstruction, benchmark-confounder comparisons, and rendered contours. These evidence types are not numerically interchangeable. A value must retain its source and status; unavailable, partial, failed, and not applicable are statuses rather than sensitivity magnitudes. Benchmarking against an observed covariate supplies an interpretable scale but does not bound every possible unmeasured confounder. Robustness-value and omitted-variable-bias methods motivate the analysis without validating the local artifacts or identification assumptions [39, 40, 34]. No sensitivity contour is admitted as main-text numerical evidence, and CausalPFN lacks an equivalent archived sensitivity family.

This axis asks how conclusions change under a specified model of unobserved confounding. It does not establish the actual amount of confounding, prove that no confounding exists below a threshold, validate the DAG, or determine whether an estimate is true or false.

8.7 Population Perturbation

The original-versus-outcome-downsampled design studies dependence on population composition and outcome prevalence. The original cohorts remain primary, and the sampled populations remain separate: downsampling changes the empirical population, nuisance-fitting problem, support, and averaging distribution. Stability or change across those populations is evidence about this specified perturbation, not a replacement for external validation and not evidence that the original cohort is unbiased.

8.8 Provenance, Evidence Admission, and Reproducibility Boundaries

Before a value enters Chapter 9, its producing run, dataset, model or estimator, sampling condition, source path, schema, status, and population must be identified. Failed artifacts are excluded; fallback diagnostics retain their labels; source fields are interpreted at their actual granularity; and the estimand and unresolved limitations remain explicit. Directory names, copied summaries, or non-null cells alone are insufficient. This policy preserves the original-cohort and estimator hierarchy and prevents numerical availability from being mistaken for scientific admissibility.

The result manifest and compact reproducibility package record available paths, hashes, schemas, commands and log arguments, seeds, environment fields, and artifact roles. They distinguish current source behavior, archived numerical artifacts, checked result rows, reconstructed summaries, unresolved producing lineage, and full clean-checkout reproducibility. These classes support numerical transcription and execution traceability, but not clean-checkout reproduction or scientific identification. Raw and processed data and numbered final causal configurations are absent; predictive split and checkpoint lineage, source versions, environments, and archive history are incomplete; and historical paths are not portable. Requirements and current defaults describe interfaces, not necessarily the producing environment. Provenance controls what may be claimed from a testbed; provenance quality does not itself establish clinical or causal validity.

8.9 Synthesis of the Non-Equivalent Axes

Each axis asks a different question and supplies a bounded form of evidence. Construct plausibility asks whether a representation and rationale are inspectable, but cannot establish clinical construct validity. Proxy recoverability asks whether rule-derived labels are learnable, but cannot establish diagnostic or causal correctness. Mortality prediction asks whether the representation contains prognostic information, but cannot establish causal relevance. Cross-estimator sign, rank, and magnitude stability asks whether a pattern depends on one fitted estimator, but cannot establish accuracy, common estimands, or truth. Clinical-literature comparison asks whether patterns are broadly compatible with heterogeneous external studies, but cannot provide a meta-analysis, external proxy validation, or calibration against causal truth.

Matching and empirical-support diagnostics ask whether usable comparisons are available under the implemented representation, but cannot prove positivity, balance, exchangeability, or identification. Permutation or disruption checks ask whether deliberately broken labels or outcomes change fitted behavior, but cannot validate the original causal analysis. Omitted-variable sensitivity asks how conclusions respond to a specified confounding model, but cannot reveal the actual confounding strength or validate the DAG. Population perturbation asks whether findings depend on the original-versus-downsampled population, but cannot establish external validity or remove selection bias. Provenance and evidence admission ask which artifacts and claims are traceable, but cannot establish clinical meaning or causal correctness. Together these axes characterize complementary strengths, discrepancies, and failure modes without producing an evaluation score or a causal answer key.

Chapter 9

Results

This chapter reports only results admitted through the validated result-source tables and prespecified evidence hierarchy. Original-cohort analyses are primary: CausalForestDML is the primary causal-model estimator, LinearDML is the secondary comparator, and CausalPFN is exploratory. All prespecified dataset-specific proxy-state exposures are retained; none was selected according to its observed result. In all twelve archived causal runs, the implemented majority-vote stage combines one rule-derived proxy-label source with predicted-label sources from GRU, GRU-D, STraTS, and TCN, and passes the resulting aggregate downstream. Detailed robustness outputs are summarized selectively, and outcome-downsampled analyses are treated only as a robustness perspective. Every model-estimated contrast remains conditional on the causal assumptions and proxy-exposure interpretation in Chapter 7.

9.1 Analysis-Population Counts

The original causal-analysis population contained 26,845 MIMIC-III records and 7,993 PhysioNet records (Table 9.1). MIMIC-III therefore contributed 18,852 more records and was approximately 3.36 times as large. These are causal-analysis population counts, not reconstructed raw cohort totals. Pipeline-stage artifacts have different contracts: prediction-export, proxy-label, majority-vote, and causal-analysis counts must not be interpreted as one universally identical cohort count.

Table 9.1: Original causal-analysis populations admitted to the primary analyses. Counts describe model rows in the checked original-cohort runs, not raw-source cohort totals.

Dataset	Pipeline population	Sampling	Records	Scope
MIMIC-III	Causal-analysis model rows	Original	26,845	Primary
PhysioNet 2012	Causal-analysis model rows	Original	7,993	Primary

9.2 Predictive Performance

The checked archive contains eight completed dataset–model test combinations spanning the four learned models STraTS, GRU, GRU-D, and TCN (Table 9.2). Every predictive metric compares model output with deterministic rule-derived proxy-label targets produced by the source-encoded decision trees. On MIMIC-III, STraTS had the smallest archived test loss and the highest archived test AUROC, AUPRC, and minRP. On PhysioNet, GRU-D led the same four archived metrics. Thus the leading model differed by dataset, although within each dataset the leader did not depend on which of these four metrics was used. Rankings among the remaining models did vary by metric; the table therefore does not support a universal model ordering or a claim of statistical superiority.

The numerical rows are test summaries and not clinical-validation values. Their summary and paired log values agreed within the validated tolerance, but the archived train/validation/test split manifest and checkpoint-to-export lineage remain incomplete. No confidence intervals or between-model significance tests were available in the validated sources. Strong agreement with a rule-derived target demonstrates learnability of that target, not validity of the clinical construct or correctness of the LLM-assisted rule design.

Table 9.2: Archived predictive test performance for the four learned models. AUROC, AUPRC, and minRP are macro-averaged across non-degenerate rule-derived proxy-label targets on the archived test split.

Dataset	Model	Test loss	Test AUROC	Test AUPRC	Test minRP
MIMIC-III	STraTS	0.348	0.905	0.869	0.795
MIMIC-III	GRU	0.386	0.881	0.840	0.770
MIMIC-III	GRU-D	0.404	0.884	0.841	0.773
MIMIC-III	TCN	0.414	0.867	0.823	0.758
PhysioNet 2012	STraTS	0.341	0.915	0.877	0.818
PhysioNet 2012	GRU	0.397	0.915	0.897	0.831
PhysioNet 2012	GRU-D	0.331	0.918	0.905	0.835
PhysioNet 2012	TCN	0.477	0.899	0.875	0.811

9.3 Primary CausalForestDML Results

Tables 9.3 and 9.4 report every prespecified original-cohort majority-vote proxy-state exposure from that mixed rule-and-prediction lineage. The reported quantity is the *mean model-estimated CATE over the analyzed sample*, on the model outcome scale. It is neither a risk ratio nor an unqualified population-average causal effect. Exposure prevalence is the proportion of analyzed records carrying the corresponding binary aggregated proxy-state tag. The adjustment-set codes refer to the observed variables archived for each exposure; identifiability with the available nodes remained unresolved for every validated row.

These results do not evaluate the LLM itself. The study contains no LLM ablation, clinician comparison, prompt-sensitivity experiment, alternative-LLM replication, or graph-correctness experiment. The numerical analyses therefore cannot establish that LLM assistance improved the rules or DAGs, nor can they separate design error from rule, prediction, aggregation, or estimator error.

For MIMIC-III, all nine archived means were positive. The largest mean was for `LAT_CARDIAC_STRAIN` (0.220), followed by `LAT_INFLAMMATION_SEPSIS` (0.161). The smallest were for `LAT_METABOLIC_DERANGEMENT` (0.020) and `LAT_SHOCK` (0.021). Figure 9.1 displays this source-exact ordering. It is descriptive and does not establish clinical importance or statistical significance.

Table 9.3: Primary original-cohort MIMIC-III CausalForestDML results (n = 26,845). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
`LAT_CARDIAC_STRAIN`	0.178	0.220	M1	Identifiability unresolved
`LAT_INFLAMMATION_SEPSIS`	0.472	0.161	M0	Identifiability unresolved
`LAT_HEPATIC_COAG_DYSFUNCTION`	0.241	0.098	M2	Identifiability unresolved
`LAT_RENAL_DYSFUNCTION`	0.363	0.091	M3	Identifiability unresolved
`LAT_GLOBAL_SEVERITY`	0.719	0.062	M4	Identifiability unresolved
`LAT_RESPIRATORY_FAILURE`	0.590	0.036	M5	Identifiability unresolved
`LAT_NEUROLOGIC_DYSFUNCTION`	0.434	0.034	M5	Identifiability unresolved
`LAT_SHOCK`	0.540	0.021	M6	Identifiability unresolved
`LAT_METABOLIC_DERANGEMENT`	0.373	0.020	M7	Identifiability unresolved

The MIMIC adjustment codes are: M0, no observed adjustment variables archived; M1, age, gender, and chronic burden; M2, inflammation/sepsis and shock; M3, chronic burden and shock; M4, age, gender, chronic burden, and inflammation/sepsis; M5, age, gender, chronic burden, global severity, and inflammation/sepsis; M6, age, gender, cardiac strain, chronic burden, and inflammation/sepsis; and M7, global severity, renal dysfunction, respiratory failure, and shock. These are observed adjustment-variable records, not proof that all confounding was controlled.

For PhysioNet, nine archived means were positive and the shock mean was negative. Renal dysfunction had the largest mean (0.120), followed by cardiac injury/strain (0.112) and global severity (0.108). Coagulation/hematologic dysfunction was small and positive (0.011), while

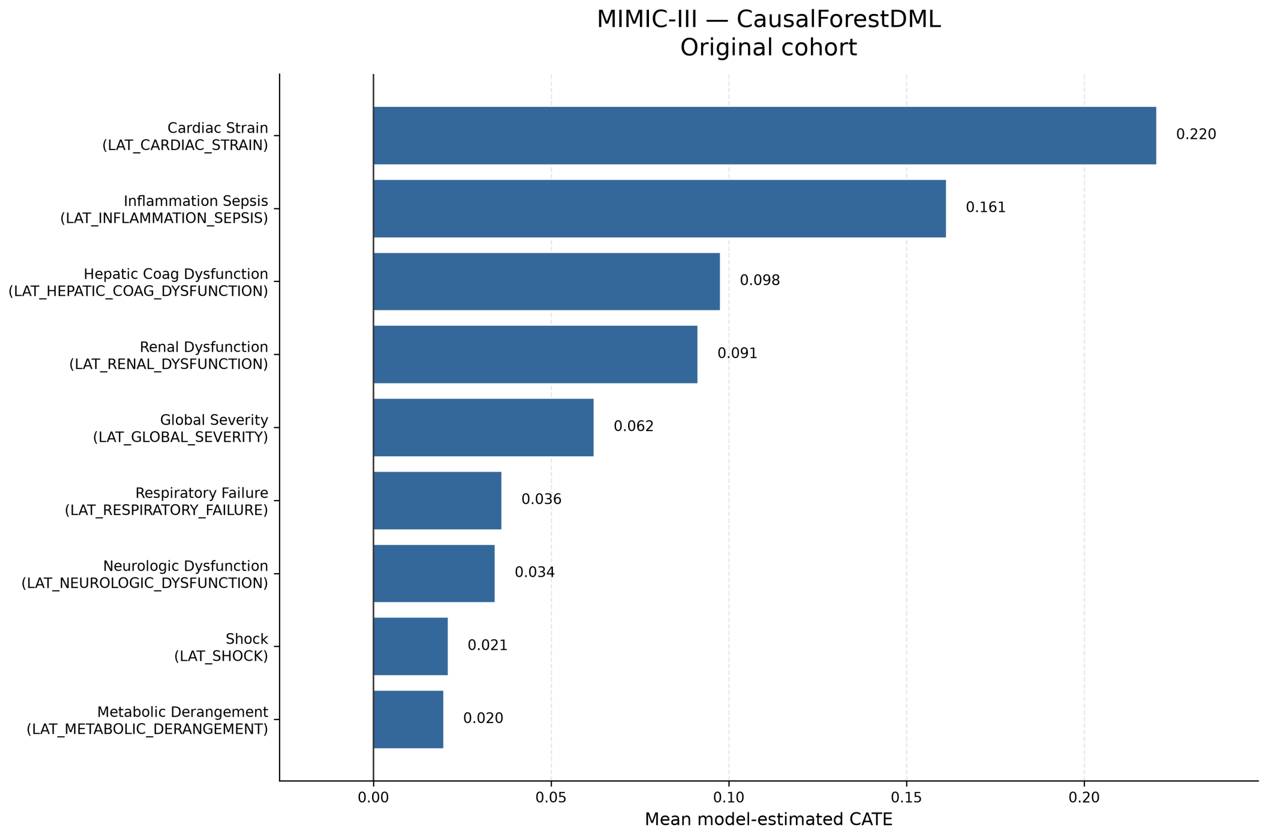

Figure 9.1: Original-cohort MIMIC-III mean model-estimated CATE values from CausalForest-DML. All nine prespecified proxy-state exposures are shown. The ordering is descriptive and does not establish statistical significance, clinical importance, or population-average causal effects.

shock was −0.014. Figure 9.2 retains that negative value and the source-exact ordering; the ranking is descriptive only.

Table 9.4: Primary original-cohort PhysioNet CausalForestDML results (n = 7,993). Prevalence and mean model-estimated CATE are proportions on their archived scales.

Proxy-state exposure	Prevalence	Mean CATE	Adjustment	Status
`LAT_RENAL_DYSFUNCTION`	0.230	0.120	P1	Identifiability unresolved
`LAT_CARDIAC_INJURY_STRAIN`	0.355	0.112	P2	Identifiability unresolved
`LAT_GLOBAL_SEVERITY`	0.777	0.108	P3	Identifiability unresolved
`LAT_HEPATIC_DYSFUNCTION`	0.142	0.091	P1	Identifiability unresolved
`LAT_NEUROLOGIC_DYSFUNCTION`	0.659	0.082	P0	Identifiability unresolved
`LAT_METABOLIC_DERANGEMENT`	0.558	0.078	P4	Identifiability unresolved
`LAT_INFLAMMATION_SEPSIS_BURDEN`	0.563	0.072	P0	Identifiability unresolved
`LAT_RESPIRATORY_FAILURE`	0.614	0.065	P0	Identifiability unresolved
`LAT_COAG_HEME_DYSFUNCTION`	0.395	0.011	P5	Identifiability unresolved
`LAT_SHOCK`	0.696	−0.014	P6	Identifiability unresolved

The PhysioNet adjustment codes are: P0, no observed adjustment variables archived; P1, ICU-type indicators 1–4, cardiac injury/strain, inflammation/sepsis burden, and shock; P2, ICU-type indicators 1–4; P3, age, gender, weight, ICU-type indicators 1–4, chronic baseline risk, and inflammation/sepsis burden; P4, renal dysfunction, respiratory failure, and shock; P5, inflammation/sepsis burden and shock; and P6, ICU-type indicators 1–4, cardiac injury/strain, and inflammation/sepsis burden. The same qualification about incomplete confounding control applies.

9.4 Matching and Empirical Support

Matching was identical across the three estimator run families within a dataset and sampling condition, so Table 9.5 reports the cross-run-consistent original-cohort rows once, as supporting evidence for the primary CausalForestDML analysis. The numerical contrast is a *descriptive matched-pair outcome difference*. Treated and control record counts were unavailable in the checked table and are left unreported rather than inferred.

Eight of nine MIMIC exposures and seven of ten PhysioNet exposures produced a matched-

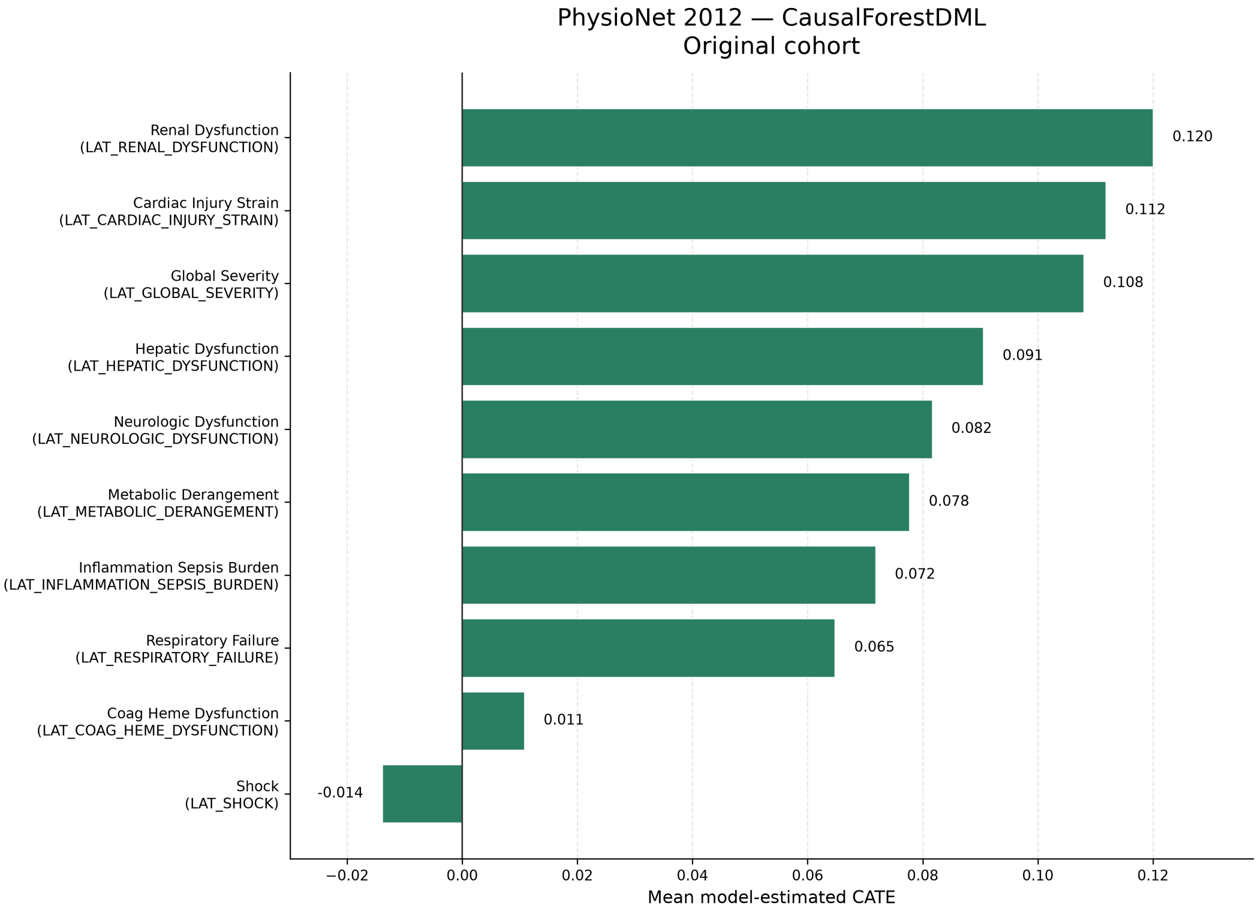

Figure 9.2: Original-cohort PhysioNet mean model-estimated CATE values from CausalForest-DML. All ten prespecified proxy-state exposures are shown. The ordering is descriptive; the negative shock estimate is retained. The figure does not establish statistical significance, clinical importance, or population-average causal effects.

pair summary. Matching failed for MIMIC inflammation/sepsis and for PhysioNet inflammation/sepsis burden, neurologic dysfunction, and respiratory failure because no binary matching columns remained after preprocessing. A failure is not a zero effect. Among successful rows, final allowed Hamming distance ranged from 0 to 2; increased allowed distance represents weaker exact similarity in the binary matching representation. Three successful rows were flagged as not reaching the configured sufficient-pair criterion: global severity in both datasets and PhysioNet shock.

Table 9.5: Original-cohort descriptive matching and empirical-support summaries. "Difference" is the descriptive matched-pair outcome difference; it is not a model-estimated CATE.

Dataset	Proxy-state exposure	Pairs	Rate	Distance	Difference	Support/status
MIMIC	`LAT_CARDIAC_STRAIN`	4,769	1.000	0	0.345	Sufficient
MIMIC	`LAT_GLOBAL_SEVERITY`	7,537	0.390	1	0.075	Weak: insufficient-pair flag
MIMIC	`LAT_HEPATIC_COAG_DYSFUNCTION`	6,478	1.000	0	0.133	Sufficient
MIMIC	`LAT_METABOLIC_DERANGEMENT`	5,061	0.506	0	0.037	Sufficient
MIMIC	`LAT_NEUROLOGIC_DYSFUNCTION`	6,275	0.539	0	0.027	Sufficient
MIMIC	`LAT_RENAL_DYSFUNCTION`	8,915	0.915	0	0.134	Sufficient
MIMIC	`LAT_RESPIRATORY_FAILURE`	10,998	0.694	2	0.106	Sufficient; distance 2
MIMIC	`LAT_SHOCK`	10,380	0.717	1	0.039	Sufficient
MIMIC	`LAT_INFLAMMATION_SEPSIS`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	2,840	1.000	0	0.134	Sufficient
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	1,763	0.558	0	0.0017	Sufficient
PhysioNet	`LAT_GLOBAL_SEVERITY`	1,782	0.287	1	0.095	Weak: insufficient-pair flag
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	1,133	1.000	0	0.137	Sufficient
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	3,378	0.757	2	0.073	Sufficient; distance 2
PhysioNet	`LAT_RENAL_DYSFUNCTION`	1,841	1.000	0	0.154	Sufficient
PhysioNet	`LAT_SHOCK`	2,433	0.438	1	0.010	Weak: insufficient-pair flag
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	–	–	–	–	Failed: no binary matching columns
PhysioNet	`LAT_RESPIRATORY_FAILURE`	–	–	–	–	Failed: no binary matching columns

For the 15 successful rows, matching and CausalForestDML shared direction in 14 cases. PhysioNet shock was the exception: its matched-pair difference was positive (0.010), while its model-estimated mean was negative (-0.014). This is a descriptive comparison, not independent causal validation. A high match rate does not establish positivity; the procedure is greedy and representation-dependent, and no dedicated propensity-overlap or balance figure was available.

9.5 LinearDML Comparison

LinearDML and CausalForestDML agreed in mean-effect direction for all 19 original-cohort dataset–exposure pairs (Table 9.6). This agreement does not imply equal magnitude or ranking. In MIMIC, CausalForestDML estimated a larger cardiac-strain mean by approximately 0.033, while LinearDML estimated a larger global-severity mean by approximately 0.018; renal and hepatic/coagulation dysfunction exchanged third and fourth positions. In PhysioNet, the largest absolute differences were for hepatic dysfunction (approximately 0.030) and renal dysfunction (approximately 0.029), and the top-ranked mean changed from renal dysfunction under CausalForestDML to cardiac injury/strain under LinearDML. LinearDML imposes a simpler final effect structure; similarity between these archived means does not show that either estimator is unbiased.

Table 9.6: Original-cohort comparison of mean model-estimated CATE from CausalForestDML and LinearDML.

Dataset	Proxy-state exposure	Forest	Linear	Same direction
MIMIC	`LAT_CARDIAC_STRAIN`	0.220	0.188	Yes
MIMIC	`LAT_INFLAMMATION_SEPSIS`	0.161	0.161	Yes
MIMIC	`LAT_HEPATIC_COAG_DYSFUNCTION`	0.098	0.083	Yes
MIMIC	`LAT_RENAL_DYSFUNCTION`	0.091	0.089	Yes
MIMIC	`LAT_GLOBAL_SEVERITY`	0.062	0.080	Yes
MIMIC	`LAT_RESPIRATORY_FAILURE`	0.036	0.039	Yes
MIMIC	`LAT_NEUROLOGIC_DYSFUNCTION`	0.034	0.032	Yes
MIMIC	`LAT_SHOCK`	0.021	0.021	Yes
MIMIC	`LAT_METABOLIC_DERANGEMENT`	0.020	0.018	Yes
PhysioNet	`LAT_RENAL_DYSFUNCTION`	0.120	0.091	Yes
PhysioNet	`LAT_CARDIAC_INJURY_STRAIN`	0.112	0.122	Yes
PhysioNet	`LAT_GLOBAL_SEVERITY`	0.108	0.107	Yes
PhysioNet	`LAT_HEPATIC_DYSFUNCTION`	0.091	0.060	Yes
PhysioNet	`LAT_NEUROLOGIC_DYSFUNCTION`	0.082	0.076	Yes
PhysioNet	`LAT_METABOLIC_DERANGEMENT`	0.078	0.080	Yes
PhysioNet	`LAT_INFLAMMATION_SEPSIS_BURDEN`	0.072	0.071	Yes
PhysioNet	`LAT_RESPIRATORY_FAILURE`	0.065	0.063	Yes
PhysioNet	`LAT_COAG_HEME_DYSFUNCTION`	0.011	0.004	Yes
PhysioNet	`LAT_SHOCK`	−0.014	−0.027	Yes

9.6 CausalPFN Exploratory Results

CausalPFN CATE execution completed for all 19 original-cohort comparisons. The three estimators agreed in mean-effect direction for 18 of the 19 dataset–exposure comparisons: MIMIC was concordant for 9 of 9 and PhysioNet for 9 of 10 (Table 9.7 and Figure 9.3). The sole exception was PhysioNet `LAT_SHOCK`: CausalForestDML was negative (−0.014), LinearDML was negative (−0.027), and CausalPFN was slightly positive (0.0041).

CausalPFN reproduced the prevailing direction in nearly every comparison, supporting its promise as a complementary estimator within this pipeline. This broad agreement is not complete agreement and does not establish estimator equivalence, superiority, causal validity, or interchangeable uncertainty quantification. Its later sensitivity and permutation stages were intentionally skipped rather than failed, so it lacks the archived robustness-diagnostic family available for the DML estimators. Its role remains exploratory, and the unresolved primary-citation limitation in Chapter 7 remains relevant.

Table 9.7: Original-cohort exploratory CausalPFN mean model-estimated CATE and direction agreement with both DML estimators.

Dataset	Proxy-state exposure	CausalPFN	Forest/Linear signs	All three agree
MIMIC	LAT_CARDIAC_STRAIN	0.259	+, +	Yes
MIMIC	LAT_INFLAMMATION_SEPSIS	0.149	+, +	Yes
MIMIC	LAT_HEPATIC_COAG_DYSFUNCTION	0.097	+, +	Yes
MIMIC	LAT_RENAL_DYSFUNCTION	0.087	+, +	Yes
MIMIC	LAT_GLOBAL_SEVERITY	0.073	+, +	Yes
MIMIC	LAT_RESPIRATORY_FAILURE	0.050	+, +	Yes
MIMIC	LAT_NEUROLOGIC_DYSFUNCTION	0.035	+, +	Yes
MIMIC	LAT_SHOCK	0.032	+, +	Yes
MIMIC	LAT_METABOLIC_DERANGEMENT	0.031	+, +	Yes
PhysioNet	LAT_CARDIAC_INJURY_STRAIN	0.119	+, +	Yes
PhysioNet	LAT_RENAL_DYSFUNCTION	0.111	+, +	Yes
PhysioNet	LAT_HEPATIC_DYSFUNCTION	0.107	+, +	Yes
PhysioNet	LAT_GLOBAL_SEVERITY	0.105	+, +	Yes
PhysioNet	LAT_METABOLIC_DERANGEMENT	0.091	+, +	Yes
PhysioNet	LAT_NEUROLOGIC_DYSFUNCTION	0.077	+, +	Yes
PhysioNet	LAT_INFLAMMATION_SEPSIS_BURDEN	0.067	+, +	Yes
PhysioNet	LAT_RESPIRATORY_FAILURE	0.063	+, +	Yes
PhysioNet	LAT_COAG_HEME_DYSFUNCTION	0.025	+, +	Yes
PhysioNet	LAT_SHOCK	0.0041	−, −	No

9.7 Cross-Dataset Comparison

MIMIC-III's original causal-analysis population was substantially larger: 26,845 versus 7,993 records, a difference of 18,852 and a ratio of approximately 3.36. The larger analysis population can affect estimate stability, feasible model flexibility, and the amount of empirical support available; it does not make the MIMIC estimates automatically correct. The checked outcome rates also differed (0.121 in MIMIC and 0.142 in PhysioNet), but these rates describe the analyzed original-cohort model rows rather than equivalent raw-source cohorts.

Some proxy names suggest nominal alignment, including global severity, shock, respiratory failure, renal dysfunction, neurologic dysfunction, and metabolic derangement. Others are only approximately corresponding project-defined states: MIMIC combines hepatic and coagulation dysfunction while PhysioNet separates them, and the inflammation/sepsis and cardiac constructs use dataset-specific names and rules. Consequently, similarly named states are not assumed to be construct-equivalent. Dataset era, measurement systems, variable availability, proxy rules, DAG structure, empirical support, and estimator behavior also differ. No effects are pooled, averaged across datasets, or interpreted as evidence of biological inconsistency; detailed clinical interpretation is deferred to Chapter 10.

9.8 Robustness and Sensitivity

9.8.1 Outcome-downsampled analyses

The outcome-downsampled analyses retained all outcome-positive records and sampled outcome-negative records, producing 6,486 MIMIC rows and 2,276 PhysioNet rows with sample outcome rates of 0.500. These are different empirical populations from the original cohorts, so their mean estimates are neither pooled with nor treated as estimates of the same population quantity.

Direction was preserved in 55 of 57 matched original-versus-downsampled estimator–exposure comparisons. CausalForestDML retained direction for all 19 comparisons. LinearDML changed direction only for PhysioNet `LAT_COAG_HEME_DYSFUNCTION`, from 0.0041 to -0.0174; CausalPFN changed direction only for PhysioNet `LAT_SHOCK`, from 0.0041 to -0.0413. All 57 paired means changed numerically, so magnitude changes were widespread and not confined to the two sign changes. This does not strengthen causal evidence because the sampled population changed.

Matching availability did not change: MIMIC again had eight successful summaries and one failure, while PhysioNet had seven successful summaries and three failures. Pair counts, match rates, distances, and descriptive differences nevertheless changed under downsampling. Detailed downsampled values remain supplementary.

9.8.2 Sensitivity, permutation, and support diagnostics

For both CausalForestDML and LinearDML, the validated sensitivity family was partial for eight of nine MIMIC exposures and seven of ten PhysioNet exposures in each sampling condition. It failed for MIMIC inflammation/sepsis and for the three PhysioNet exposures whose adjustment records were empty: inflammation/sepsis burden, neurologic dysfunction, and respiratory failure. Available rows combine saved-training-direct values with later recomputation and custom reconstruction from residual parameters; the planned real benchmark extraction is marked unimplemented by design. These source classifications prevent the values from being presented as a single estimator-native family. CausalPFN sensitivity was intentionally skipped and is not equivalent to a zero diagnostic.

Treatment- and outcome-permutation aggregate rows were available for every DML dataset–exposure–sampling combination, with ten archived trials and seed 42 in each row. The validated source table records these as sanity diagnostics and supplies no new p-values. They are not formal randomization tests and do not prove identification. CausalPFN permutations were intentionally skipped.

Table 9.8 summarizes availability rather than diagnostic magnitude. Matching remains indirect, greedy, binary-representation-dependent support evidence. No dedicated overlap or propensity figure, pre/post-match balance table, or empirical proof of positivity was available. The principal warnings are the four exposure-specific matching failures per sampling-condition pair, three successful original rows with insufficient-pair flags, and the absence of an equivalent PFN diagnostic family.

Table 9.8: Robustness and diagnostic availability. Counts are exposure-level statuses, not effect estimates.

Dataset	Estimator	Sampling	Matching	Sensitivity	Permutation
MIMIC	Forest	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Original	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Original	8 available; 1 failed	Intentional skip	Intentional skip
MIMIC	Forest	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	Linear	Downsampled	8 available; 1 failed	8 partial; 1 failed	Treatment/outcome available
MIMIC	PFN	Downsampled	8 available; 1 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Original	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Original	7 available; 3 failed	Intentional skip	Intentional skip
PhysioNet	Forest	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	Linear	Downsampled	7 available; 3 failed	7 partial; 3 failed	Treatment/outcome available
PhysioNet	PFN	Downsampled	7 available; 3 failed	Intentional skip	Intentional skip

9.8.3 Result provenance boundary

The checked hashes and source paths support arithmetic transcription of the admitted archive rows, but they do not remove the archive's reproducibility limitations. The `final-results/` tree is ignored or untracked in the main repository; exact numbered causal configuration files remain unavailable locally; some producing commits and archive-copy histories are incomplete; predictive split and checkpoint-to-export lineage is incomplete; and raw and processed data cannot be reproduced from the repository alone. These limitations qualify reproducibility claims, not the arithmetic correspondence between the checked rows and the tables above.

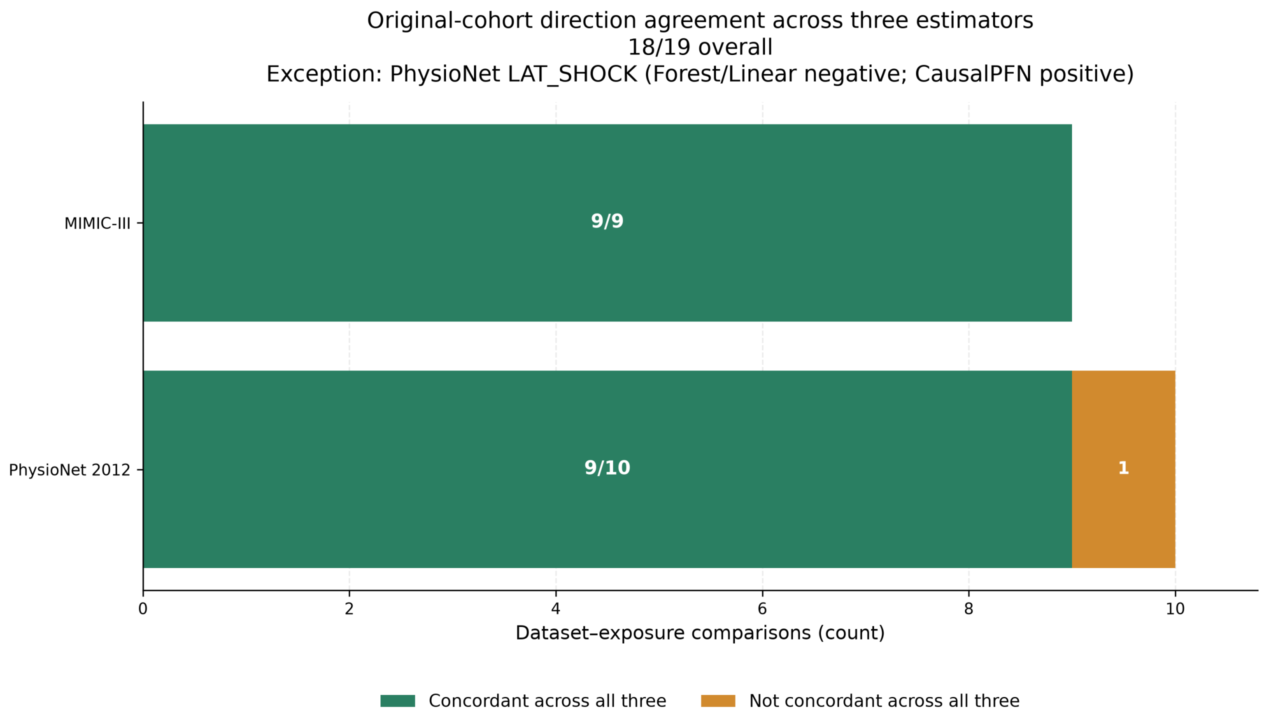

Figure 9.3: Directional concordance of the original-cohort mean model-estimated CATE across CausalForestDML, LinearDML, and exploratory CausalPFN. All three estimators shared the same direction for 18 of 19 dataset–exposure comparisons: 9 of 9 in MIMIC-III and 9 of 10 in PhysioNet. PhysioNet shock was the sole exception. Direction agreement does not imply equal magnitude, uncertainty, estimator equivalence, or causal validity.

Chapter 10

Discussion

This chapter interprets the evidence reported in Chapter 9 within the study design and assumptions established in Chapters 3–8. The central distinction is between what the thesis evidence demonstrates empirically—an implemented, evidence-tracked workflow and stable patterns across several analyses—and what it does not establish clinically or causally. The discussion therefore treats the reported quantities as prediction results, descriptive matched-pair comparisons, or adjusted model estimates under project assumptions, according to their source.

10.1 Answering the Research Questions

10.1.1 Main Research Question

The main research question asked: *How can LLM-assisted clinical and causal knowledge elicitation, deterministic proxy-label construction, and deep modeling of irregular ICU time series be integrated into an evidence-tracked framework for DAG-guided adjusted effect estimation of in-hospital mortality?*

The direct answer is that this thesis demonstrates an evidence-tracked methodological framework with a sequence of explicit interfaces. At design time, a structured LLM-assisted protocol generated proposals for dataset-specific proxy ontologies, decision rules, missingness considerations, and DAGs; project-selected designs were encoded in deterministic source. During execution, dataset-specific irregular measurements are converted into normalized data contracts; the source-coded rules create rule-derived proxy labels; irregular-time-series models predict those labels; and probabilities and binary predicted fields are exported and normalized. In every archived final causal run, the vote then combines one rule-derived table with four model-prediction tables. Source-coded, project-specified DAGs define exposure-specific adjustment logic, and matching and heterogeneous-effect estimators analyze the resulting mixed-source aggregate. Robustness and provenance checks qualify the interpretation. The framework was implemented and empirically exercised across MIMIC-III and PhysioNet 2012 without a runtime LLM call.

This is a feasibility and integration result, not a conclusive solution to the clinical causal question or an empirical evaluation of LLM quality. The proxy states are not chart-adjudicated diagnoses, many exposures describe illness states rather than well-defined interventions, and the LLM-assisted project DAGs are assumed rather than empirically established. The adjusted estimates remain conditional on construct measurement, consistency, exchangeability, temporal ordering, overlap, and model adequacy. The framework therefore supports auditable retrospective analysis and hypothesis generation, but it does not produce clinical treatment recommendations. In summary, the thesis demonstrates the feasibility of linking LLM-assisted knowledge construction, irregular-time-series representation, proxy-state prediction, and assumption-guided adjusted effect estimation in one auditable workflow, while showing that the strength of causal interpretation is constrained chiefly by construct validity, intervention definition, causal-graph validity, empirical support, and provenance.

10.1.2 Data Contracts and End-to-End Pipeline Integration

SRQ-1—data contracts. The pipeline requires at least two distinct processed-artifact contracts. The causal repository uses `[ts, oc, ts_ids]`, with a normalized `ts_id` linking the long event table to the outcome table. The STraTS and baseline workflow uses a split-aware artifact with events, outcomes, and explicit train, validation, and test identifier collections. The router and conversion logic bridge these contracts by normalizing identifiers and constructing the split-aware form when requested. Prediction exports add another interface: `ts_id`, paired probability and binary proxy-state columns, and subsequently binary-only voter files. The answer is therefore *implementation supported*: the repository defines and validates these boundaries, but exact processed-pickle hashes, producing commands, and final split lineage remain incomplete.

SRQ-4—prediction normalization and aggregation. Prediction outputs are normalized into paired probability and binary proxy-state fields, while downstream voter inputs retain the binary fields in an approved order. Majority voting aligns shared normalized identifiers, checks identical binary proxy-state schemas, and applies the implemented tie rule. In every archived final run, the logs enumerate one deterministic rule-derived source and four thresholded prediction sources—GRU, GRU-D, STraTS, and TCN. This supplies a deterministic mixed-source handoff into the causal repository. It is algorithmic aggregation rather than clinical consensus; the logs retain external filenames but do not co-archive the voter-file bytes or hashes, while checkpoint mappings and archive-copy provenance remain incomplete. Moreover, aggregation cannot automatically correct systematic error shared by models trained on the same rule-derived targets, especially when the originating rule source itself participates in the vote.

10.1.3 Proxy-State Construction, Prediction, and Aggregation

SRQ-2—LLM-assisted design selection, encoding, and traceability. The preserved generic and dataset-specific prompts, final run exports, active tagger and graph files, and generated proxy and DAG artifacts establish a traceable design sequence at artifact and schema level. The final prompt outputs align with the active eleven-state PhysioNet and ten-state MIMIC schemas, while deterministic source rules and stable `LAT_*` columns make implemented behavior inspectable. The answer remains *partially traceable and unresolved clinically*: complete accepted/rejected decision records and line-by-line prompt-output-to-code correspondence are unavailable; the two datasets use different ontologies and measurements; threshold rules may misclassify constructs; clinical review remains incomplete; and no chart-adjudicated reference labels were archived. These are LLM-assisted, project-selected proxy designs encoded as deterministic rules, not diagnoses, ground truth, or validated phenotypes. The rule-to-label transition is conceptually related to programmatic labeling, while the aggregation layer does not implement a learned Snorkel label model [12, 6].

SRQ-3—predictive modeling. The validated comparison included the four learned models STraTS, GRU, GRU-D, and TCN. STraTS led the four archived test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet. The leading architecture therefore differed by dataset. This is empirically supported evidence that representation choice is dataset dependent, not evidence for universal model superiority. Handling irregularity or missingness can be useful in several ways, and prior work distinguishes sparse-token representations from missingness-aware recurrent processing [1, 2, 3, 4]. The local comparison does not reproduce the original papers' benchmark tasks exactly.

Several mechanisms may contribute to the dataset-specific pattern: variable availability, measurement frequency, missingness, label prevalence, proxy definitions, preprocessing, dataset era, sample size, and ICU composition. The study did not isolate those mechanisms, so they remain possible explanations rather than findings. In addition, no uncertainty intervals or paired between-model tests were archived, checkpoint-to-export provenance is incomplete, and the prediction targets themselves lack clinical validation. Statistical superiority is therefore not established.

10.1.4 Adjusted Effect Estimation and Estimator Agreement

SRQ-5—DAG-guided adjustment. Dataset-specific, source-coded DAGs select exposure-specific observed adjustment sets from columns available in each analysis frame. LLM-assisted elicitation was a substantive design-time input, while the active graph code is the implementation authority. The graphs were neither learned from the data nor clinically or empirically validated, and the adjustment sets differ across exposures. Several validated rows contain empty observed adjustment sets, while treatment-specific identifiability remains unresolved throughout the result sources. The answer is thus *conditional on assumptions*: a computed set documents how the project applied its graph, but it does not prove exchangeability or the absence of unmeasured confounding [7].

SRQ-6—matching, CATE, sensitivity, and permutation evidence. The approved hierarchy uses original cohorts as primary, CausalForestDML as the primary estimator, LinearDML as secondary, CausalPFN as exploratory, and outcome-downsampled analyses as robustness evidence. In the primary forest results, all nine MIMIC original-cohort mean model-estimated CATE summaries were positive; nine of ten PhysioNet summaries were positive, with shock negative. This pattern supports the narrower conclusion that the project-defined proxy-state exposures contain systematic mortality-related variation after the implemented adjustment procedure, that patterns differ by construct and dataset, and that heterogeneous-effect estimators can be applied to the constructed analysis objects. It does not show that a proxy state caused death, that changing it would change mortality, or that a ranking identifies intervention priorities.

CausalForestDML and LinearDML agreed in direction for all 19 original-cohort dataset–exposure pairs. This complete directional concordance is useful model-form triangulation: the sign pattern was not unique to the nonlinear forest final stage. It does not imply equal magnitudes, orderings, estimands, uncertainty, or correctness, and it does not validate the DAG or rule out residual confounding. Indeed, Chapter 9 shows that magnitudes and within-dataset orderings can differ even when signs agree.

Within this pipeline, CausalPFN showed promise as a complementary estimator because it reproduced the prevailing direction across nearly all prespecified comparisons. It contributed to all-three directional concordance for 18 of 19 original-cohort dataset–exposure pairs despite using a different modeling approach. This positive result is bounded: magnitudes were not identical to DML magnitudes, the archived sensitivity and permutation stages were intentionally absent, and the approved literature corpus lacks a primary method citation. CausalPFN therefore remains an exploratory empirical contribution rather than an estimator with the same diagnostic support as the DML analyses.

PhysioNet shock is the central exception rather than a detail to be averaged away. CausalForestDML and LinearDML were negative, CausalPFN was slightly positive, and the descriptive matching summary was positive. This disagreement is a substantive caution about model, representation, and support sensitivity. It should not be resolved by a vote among estimators, and the negative DML estimates should not be described as a clinically protective effect.

Matching provides a transparent descriptive baseline and an empirical-support warning system. Most prespecified exposures produced summaries, and 14 of 15 successful original-cohort matching summaries shared the primary forest direction. Failures identified cases in which no usable binary matching columns remained; some successful rows carried insufficient-pair warnings; and accepted Hamming distances varied. Stable matching availability across original and downsampled populations supports the reproducibility of the availability pattern, but matching neither proves positivity nor supplies complete balance evidence. A failed comparison is missing support, not a zero estimate, and a successful descriptive matched-pair outcome difference is not automatically an ATT or independent causal confirmation.

Sensitivity and permutation evidence adds layers rather than closure. DML sensitivity outputs were saved-training direct, reconstructed, partial, or failed depending on the exposure, and reconstructed diagnostics are not identical to estimator-native calculations. Benchmark-confounder comparisons supply a scale, not proof that omitted-variable bias is absent. Treatment and outcome permutations provide disruption-based sanity checks for the DML estimators, but the archived procedure is not automatically a formal randomization test and does not establish identification. The corresponding CausalPFN stages were intentionally skipped. Finally, matching fields provide indirect support evidence, but dedicated propensity, common-support, and balance plots were not archived; positivity remains unestablished [38, 39, 40].

10.1.5 Cross-Dataset Interpretation and Robustness

The original causal-analysis populations contained 26,845 MIMIC records and 7,993 PhysioNet records, so the MIMIC analysis population was approximately 3.36 times larger. A larger analysis population can provide more information for nuisance-model fitting and modeled effect heterogeneity, and it may contribute to numerical stability. It does not demonstrate better data quality, make the MIMIC findings more valid, or erase differences in era, variables, measurement systems, missingness, proxy rules, DAGs, or patient composition. Predictive training and causal analysis also use distinct artifact contracts, so their populations should not be assumed identical merely because some row counts align.

Broad methodological replication across both datasets is a strength: the pipeline operated with the same overall sequence of representation, proxy prediction, aggregation, graph-guided adjustment, and estimator comparison. However, similarly named proxy states are not automatically construct-equivalent. MIMIC combines some domains that PhysioNet separates, and the rules and graphs are dataset specific. The deliberate no-pooling policy is therefore essential. Cross-dataset differences are evidence about portability of the workflow and sensitivity to context, not a pooled clinical effect or proof of biological inconsistency.

Outcome-based downsampling remained a contained robustness analysis. It retained outcome-positive records and sampled outcome-negative records, thereby changing the empirical population and outcome prevalence. Original and downsampled means do not automatically target the same population quantity. Nevertheless, direction was preserved in 55 of 57 matched comparisons across estimator, dataset, and exposure. This broad sign stability under a substantial sampling perturbation suggests that many sign patterns were not solely driven by the original mortality imbalance. Every paired mean changed numerically, and two PhysioNet comparisons changed sign. The procedure is not formal transport or reweighting; magnitude changes cannot be read as stronger or weaker original-population effects, and directional preservation does not validate causal identification.

10.1.6 Relationship to Prior Work and Methodological Implications

The contribution lies in connecting layers that are often evaluated separately. First, LLM-assisted elicitation converts unstructured clinical and causal reasoning into explicit proposals that can be selected, source-encoded, and audited. Second, a shared proxy-state interface links irregular-time-series prediction and observational effect-estimation workflows. Third, explicit separation of causal and split-aware predictive data contracts prevents silent identifier, split, and cohort conflation. Fourth, source-coded DAGs make adjustment choices inspectable even when their scientific correctness remains an assumption. Fifth, multi-estimator comparison exposes both broad directional agreement and important exceptions such as PhysioNet shock. Matching, sensitivity, and permutation diagnostics reveal support and provenance limitations that a single effect table would conceal, while result manifests and validated source tables materially improve numerical traceability in a complex workflow.

These implications align with the wider emphasis on explicit intervention definitions and target-trial thinking in observational research [8, 9]. They also address difficulties identified in observational ICU causal analysis and machine-learning HTE research: defining the target question, controlling bias, validating heterogeneity, handling covariate shift, and distinguishing predictive variation from actionable treatment-effect variation [18, 19, 35, 36, 37]. The literature supplies methodological context; it does not validate this project's proxy labels, graphs, estimates, or local execution. Accordingly, the framework should be described as an auditable research workflow rather than a generally validated clinical methodology.

10.1.7 Clinical Meaning and Appropriate Use

The proxy states summarize clinically inspired patterns and are analytically useful for organizing retrospective evidence. They may support hypothesis generation and help identify proxy-state contrasts that merit targeted clinical and causal investigation. High prediction performance against rule labels does not establish clinical accuracy, however, and a positive adjusted proxy-state contrast does not imply that clinicians should attempt to manipulate the proxy directly. The system did not undergo prospective validation, external clinical validation, or clinical impact analysis. Its causal quantities are not ready for bedside decision support, and no patient-level treatment recommendation follows from the results.

SRQ-7—principal limitations. The main limitations arise from proxy-label construct validity and measurement error; ill-defined or non-manipulable exposures; unvalidated DAG assumptions and possible unmeasured confounding; temporal ambiguity; incomplete empirical support; model dependence; outcome-class downsampling; dataset heterogeneity; incomplete final-run provenance; and the absence of clinical, fairness, and deployment validation. The answer is therefore *empirically supported with substantial unresolved boundaries*. These limitations determine the strength of every preceding answer and motivate the validity analysis below.

10.2 Limitations and Threats to Validity

10.2.1 Construct and Measurement Validity

The principal construct limitation is that clinically inspired rules approximate conditions but were not evaluated against chart-adjudicated diagnoses. Available evidence consists of active rule code, binary artifact schemas, and limited descriptive validation summaries. This makes the constructs transparent but does not establish their sensitivity, specificity, or equivalence across datasets. Prediction can add classification error, and the mixed-source majority vote can preserve shared systematic error among models trained on the same rules. Existing mitigations are deterministic definitions, source-code traceability, dataset-specific naming, and explicit proxy terminology; the residual risk remains high because clinical review and reference-standard validation are absent.

Binary exposure states also compress severity, duration, and timing. A positive tag may combine clinically distinct trajectories, while a negative tag can reflect missing measurement rather than absence of a condition. Proxy onset relative to covariates and mortality is not fully established, so exposure misclassification and temporal ambiguity can bias both prediction and adjusted analyses. The processed outcome, `in_hospital_mortality`, is consistently required by the software, but its producing-artifact provenance is incomplete, and no competing-risk or post-discharge mortality analysis was performed.

The observation process is itself informative. Measurement frequency may reflect clinical concern, resource use, or care pathways; models that encode masks or time gaps can exploit this signal without removing its causal implications. Preprocessing and missingness-aware representation are therefore not equivalent to adjustment for the decision to measure. This concern can affect proxy construction, predictive comparisons, and effect estimation simultaneously.

10.2.2 Causal Identification and Estimand Validity

These are the central constraints on causal interpretation. Illness-state proxies are not necessarily well-defined interventions, so consistency and the meaning of contrasting exposure states may be difficult to specify. No target-trial emulation was performed, and the pipeline does not implement a method for time-varying treatment–confounder feedback. Exchangeability depends on a project-specified DAG, correct temporal ordering, and measured variables; topology alone cannot establish patient-record timing. Some archived observed adjustment sets are empty, every checked identifiability field remains unresolved, and unmeasured or misspecified confounding remains possible.

Positivity was assumed rather than established. Matching availability, failures, pair counts, and distances provide indirect evidence, but the archive lacks dedicated propensity distributions, common-support plots, and pre/post-match balance summaries. Interference is also assumed, not tested, and repeated admissions or shared ICU processes may complicate unit-level independence.

Estimand language remains deliberately restricted. The mean model-estimated CATE over the analyzed sample is an aggregate of model outputs, not automatically a population ATE. The descriptive matched-pair outcome difference is not automatically an ATT. Normalized CATE was omitted because scaling by the sample outcome rate is unstable across populations and does not create a risk ratio or another standard clinical estimand. These boundaries mitigate overstatement but do not resolve the underlying intervention-definition and identification problems.

10.2.3 Statistical, Modeling, and Support Limitations

Many displayed main estimates lack uncertainty intervals, no formal multiple-comparison strategy was documented, and this thesis did not add significance tests. Absence of an interval cannot be interpreted as either significance or insignificance. Model selection, nuisance-model adequacy, regularization, hyperparameters, and cross-fitting behavior remain assumptions whose impact was not exhaustively tested. CausalForestDML, LinearDML, and CausalPFN do not provide interchangeable uncertainty, and directional agreement can coexist with material magnitude and ordering differences.

Support evidence is incomplete and representation dependent. Some matching runs failed, some successful matches required larger Hamming distances, and some carried insufficient-pair warnings. Sensitivity outputs were partial or failed for several exposure families; reconstructed or fallback diagnostics have different status from estimator-native outputs. Permutation aggregates are limited sanity checks rather than formal inferential tests. CausalPFN lacks the corresponding archived downstream diagnostic family. The multi-estimator and diagnostic hierarchy exposes these risks, but it cannot remove them.

10.2.4 External Validity and Generalizability

MIMIC-III and PhysioNet differ in size, era, variables, measurement practices, and patient composition. Their proxy ontologies and DAGs are dataset specific, and nominally similar states are not necessarily equivalent. Numerical estimates were therefore not pooled. Operation of the workflow on both sources is positive evidence of engineering and methodological portability, especially given the inclusion of several predictive and causal model families. It is not evidence that the estimates transport to other hospitals, countries, care practices, EHR systems, or time periods.

No prospective validation, independent external clinical validation, or formal transport analysis was performed. Subgroup performance and effect stability by age, gender, ICU type, or other characteristics were not evaluated as a fairness study. Consequently, the analysis does not establish clinical generalizability or subgroup equity.

10.2.5 Reproducibility and Provenance

The validated result manifest, source tables, source paths, and checksums provide strong numerical-transcription support for the values reported in Chapter 9. They also improve artifact traceability by distinguishing canonical sources, duplicates, and excluded artifacts. This is not the same as computational rerun reproducibility. The `final-results/` archive is ignored or untracked in the main repository; exact numbered causal configurations are unavailable; result-generating source versions and copy histories are incomplete; and the nested causal repository contains local modifications.

Predictive split provenance, checkpoint-to-export mapping, and copied-output lineage remain incomplete. Raw and processed datasets are external, exact processed-pickle hashes and producing commands are unavailable, and some paths are absolute to the producing machine. Final software, environment, and hardware records are also incomplete. Thus the available records support archived numerical transcription and materially improve artifact traceability, while a clean-checkout rerun is not currently demonstrated. Scientific reproducibility—replication of findings under independently reconstructed data, definitions, and analysis—is weaker still and requires new evidence rather than additional formatting of the archive.

10.2.6 LLM-Assisted Design Layer

The LLM-assisted design layer contributed structured proposals for proxy concepts, binary decision rules, missingness considerations, and dataset-specific DAGs. Potential advantages are the systematic elicitation of domain concepts, scalable generation of candidate proxy definitions and rules, parallel dataset-specific proposals, explicit prompt artifacts that improve design provenance, and conversion of otherwise unstructured reasoning into inspectable source-coded artifacts. These advantages concern organization, traceability, and design productivity; the thesis did not compare them with clinician-only construction or measure whether LLM assistance improved clinical or causal correctness.

The main design risks include hallucinated or unsupported clinical assertions; sensitivity to prompt wording, supplied context, model version, and hosted-model updates; unknown system, browsing, and decoding settings; incomplete reproducibility of hosted-model behavior; and possible dependence on hidden training data. Project selection can introduce anchoring and confirmation bias, especially when plausible generated explanations are mistaken for independent evidence. The prompt record does not establish complete accepted/rejected decisions, and only high-level rather than line-by-line prompt-output-to-code traceability is available.

Validation gaps remain substantial. There is no documented clinician-panel adjudication of every proxy definition or graph edge, no chart-level proxy validation, no comparison of LLM-assisted with clinician-only or hybrid construction, no replication with another LLM family, and no prompt-perturbation or model-version sensitivity experiment. Error can therefore propagate from an unsupported design proposal into a deterministic label, from that label into a prediction and aggregate, and from a misspecified exposure or graph into adjustment and effect estimation. Preserved prompts and explicit source-code authority make the chain inspectable but do not validate it.

Predictive agreement with LLM-assisted rule-derived labels validates neither the clinical construct nor the LLM-generated design. Sensitivity and permutation diagnostics do not validate the proxy ontology or DAG. No claim is made that an LLM provides authoritative clinical judgment, discovered the true causal graph, or can replace a clinician or causal expert.

10.2.7 Ethical and Clinical-Deployment Considerations

Misclassification can misrepresent a record's clinical condition, amplify measurement bias, and generate misleading exposure contrasts. Groups with different testing intensity or access to measurements may be affected disproportionately. Automation bias is therefore a direct risk: users could overread model probabilities as diagnoses, proxy-state rankings as clinical priorities, adjusted estimates as treatment recommendations, or estimator agreement as causal certainty.

No dedicated fairness audit was performed. The presence of age, gender, ICU type, or other variables in an adjustment set does not constitute fairness validation, and subgroup predictive performance and effect stability remain untested. The repository does not establish the project-specific institutional approval, consent or waiver procedure, data-use agreement, or governance determination. These facts must be supplied from authoritative institutional records and must not be inferred from dataset access, de-identification, or repository contents.

The deployment boundary is unambiguous: *the system is a retrospective research workflow and is not suitable for patient-level clinical deployment or treatment recommendation in its current form.* No prospective safety, workflow, impact, or human-factors evaluation was performed. This limitation is not cured by predictive accuracy, estimator agreement, or artifact traceability. Taken together, construct measurement, intervention definition, graph validity, temporal ordering, unmeasured confounding, support, uncertainty, transport, provenance, LLM-design reproducibility, fairness, and clinical validation remain the decisive threats to stronger interpretation.

Chapter 11

Conclusions and Future Work

11.1 Summary of the Study

This thesis developed and evaluated an evidence-tracked workflow connecting irregular ICU records to transparent retrospective analysis across PhysioNet 2012 and MIMIC-III. It makes the interfaces among LLM-assisted design proposals, deterministic proxy rules, four-model proxy prediction, and five-source mixed aggregation explicit. It then connects these interfaces to project-specified DAGs, matching, heterogeneous-effect estimation, and diagnostics.

The resulting framework is a retrospective research workflow. Its proxy states are rule-derived analytical constructs rather than verified clinical conditions, and its adjusted estimates remain conditional on the measurement, graph, intervention-definition, temporal-ordering, support, and modeling assumptions stated throughout the thesis. This distinction is central to the contribution: the workflow makes its handoffs, evidence sources, and unresolved requirements visible instead of presenting an apparently seamless output as a clinical fact.

11.2 Main Findings and Contributions

The strongest contribution is the integrated connection from LLM-assisted proxy and DAG design to source-encoded rules and graphs, irregular-time-series prediction, deterministic aggregation, and observational analysis. A shared proxy-state interface traces constructs from one rule-derived source through four predicted voters to dataset-specific exposure analyses, while separate causal and split-aware contracts keep identifiers, cohorts, and exports distinguishable. Checked sources and manifests improve numerical traceability without establishing clean-checkout reproduction.

The empirical predictive comparison covered the four learned models STraTS, GRU, GRU-D, and TCN. The leading model differed by dataset: STraTS led the archived proxy-label test metrics in MIMIC-III, whereas GRU-D led them in PhysioNet (Chapter 9, Section 9.2). This result supports a dataset-specific view of representation choice. It does not establish a universal model ordering, statistical superiority, or clinical validity of the rule-derived targets.

In the primary original cohorts, CausalForestDML means were positive for all nine MIMIC-

III exposures and nine of ten PhysioNet exposures; PhysioNet shock was negative. CausalForestDML and LinearDML shared direction in all 19 dataset–exposure comparisons, and exploratory CausalPFN agreed with both in 18 of 19. These patterns provide bounded triangulation, not proof of equal magnitude, estimator correctness, identification, or clinical actionability. Matching remains descriptive support, and CausalPFN lacks the full diagnostic and methodological provenance of the DML estimators (Chapter 9).

Together, the results support the methodological objective: these components can be connected while preserving dataset-specific definitions, estimator hierarchy, and evidence boundaries. They do not prove that the LLM proposals, proxy constructs, graphs, or effect estimates are clinically correct.

11.3 Limitations of the Conclusions

Construct validity is the first constraint: LLM-assisted, project-selected proxy rules lack chart adjudication; dataset ontologies and timing differ; prediction can add error; and mixed-source voting can preserve shared rule and measurement error. Clinical review, reference-standard evaluation, complete design-to-code decisions, and exact support for several rule families remain incomplete.

Identification and translation remain equally limited. Illness-state proxies are not necessarily well-defined interventions; the project DAGs are assumptions; temporal ordering, overlap, uncertainty, and unmeasured confounding remain unresolved; and diagnostics do not establish causal validity. The two datasets do not justify pooling or transport claims, and no prospective or external clinical validation, fairness or safety study, clinician-only comparison, alternative-LLM replication, or prompt-sensitivity analysis was performed. Ethics and governance wording and the data, configuration, split, checkpoint, source-version, environment, and archive records needed for full reproduction also remain incomplete.

11.4 Future Work

The first priority is clinical and construct validation. Future work should freeze the proxy rules and conduct blinded clinician review of every proxy definition and graph edge, with multiple reviewers and reported inter-reviewer agreement. Rule-derived and predicted states should be compared with chart-adjudicated reference labels, including dataset-specific assessment of timing, missingness, thresholds, and differences among rule-derived labels, individual predicted labels, and mixed-source aggregated proxy labels. A controlled comparison of LLM-assisted, clinician-only, and hybrid rule and DAG construction would test whether the design layer adds value rather than merely plausibility.

The second priority is a stronger observational and measurement design. A future study should specify a target-trial-aligned question with eligibility, time zero, intervention strategies, follow-up, outcome, estimand, and an explicit plan for time-varying confounding. It should add prespecified overlap and balance diagnostics, support-aware estimands, uncertainty and multiplicity procedures, and richer sensitivity analyses. Rule ablation and threshold-sensitivity experiments, formal label-noise modeling, and explicit measurement-error analyses should quantify how proxy construction, prediction, and aggregation affect downstream estimates. DAG review should proceed edge by edge and include alternative-graph and adjustment-set analyses rather than treating one encoded graph as fixed truth.

The third priority is reproducibility and external evaluation. Signed per-run manifests should link raw-data access records, processed-artifact hashes, split identifiers, checkpoints, commands, resolved configurations, software environments, source versions, and archived outputs. LLM design manifests should record complete prompts, system and decoding settings, model versions, follow-up interactions, reviewer decisions, and prompt-output-to-code mappings. Prompt perturbations, model-version sensitivity studies, and replication across multiple LLM families should be performed. Frozen definitions should then be evaluated on temporal and external cohorts and in prospective studies of clinically justified subgroups before any safety or impact study is considered. CausalPFN should remain exploratory until its primary source and implementation version are verified and an appropriate diagnostic and uncertainty plan is available. Authoritative ethics and data-governance wording should also be completed before prospective or deployment-related evaluation.

11.5 Closing Perspective

The thesis shows the value of treating integration itself as an object of evidence. LLM-assisted knowledge construction, irregular ICU records, deterministic proxy labels, deep predictive outputs, graph-guided adjustment, and model-estimated contrasts can be connected in a transparent workflow, but none of those handoffs removes the need for clinical review, careful observational design, diagnostic scrutiny, or reproducible provenance. An LLM cannot replace a clinician or causal expert on the evidence presented here. The appropriate conclusion is therefore measured: transparent integration can make retrospective evidence more inspectable and more useful for hypothesis generation, while the boundaries that prevent stronger clinical or causal claims remain visible and actionable for future research.

Appendix A

Reproducibility and Evidence Boundary

A.1 Current reproducibility interface

CliniCause now exposes a unified execution interface that can reconstruct causal-analysis resources from appropriately obtained raw inputs and execute the integrated predictive and causal workflow. The repository provides one canonical shell launcher, one project-wide dependency entry point, `requirements.txt`, dataset-isolated run directories, and explicit validation, cohort, artifact, receipt, and provenance contracts. This makes the current implementation and its reconstruction path substantially more reproducible without redistributing source ICU records.

A.2 Repository setup

On a fresh machine, clone the repository normally. The checkout contains the complete pipeline, including the `STraTS/` and `causal-irregular-time-series/` component directories:

```
git clone https://github.com/KenziVisor/CliniCause.git
cd CliniCause
```

The first component contains the temporal-model implementation; the second contains preprocessing, proxy construction, graph, and causal-analysis code. No separate repository initialization step is required.

A.3 Python 3.10 environment and dependencies

Create an isolated Conda environment, activate it, upgrade its packaging tool, and install the complete project through the root requirements file:

```
conda create -n clinicause python=3.10 -y
```

```
conda activate clinicause
python -m pip install --upgrade pip
python -m pip install -r requirements.txt
```

The root file declares the parent orchestration dependencies and recursively includes the independently maintained requirement files of both component directories. It is the single supported installation interface for the current checkout; it is not an environment lock or proof of the exact software environment used to produce every archived thesis result.

A.4 Source-data access

MIMIC-III and the PhysioNet/Computing in Cardiology Challenge 2012 records are not redistributed with CliniCause. Users must obtain their own copies from PhysioNet and follow the access and license terms shown on the respective project pages. MIMIC-III is a restricted resource: access requires PhysioNet credentialing, the currently specified human-subjects training, and acceptance of its data-use agreement. The Challenge 2012 files are currently presented by PhysioNet as open access under their specified license; users must still comply with that license and verify the current project-page terms. The pipeline neither automates nor bypasses these processes, and restricted source records must remain under the authorized user's control.

A.5 Default and external data locations

After obtaining and extracting the sources in a form consistent with their official distributions, place each raw dataset root at the launcher default:

```
data/physionet2012
data/mimiciii
```

These paths denote the raw dataset roots consumed directly by the two source-specific preprocessors; they are not locations for CliniCause-derived outputs. To keep protected data outside the repository, supply absolute paths:

```
export PHYSIONET_RAW_DATA_PATH=/absolute/path/to/physionet2012
export MIMIC_RAW_DATA_PATH=/absolute/path/to/mimiciii
```

The launcher accepts absolute paths. It resolves relative values, including the defaults, from the CliniCause repository root rather than from the caller's current directory.

A.6 Constructing the reusable datasets

From the repository root and the activated environment, run:

```
STAGES=dataset-extraction bash run_clinicause.sh
```

The default dataset selector is both. This preset executes the implemented preprocessing, deterministic tagging, tree generation, STraTS input preparation and model stages, prediction collection and normalization, then terminates the causal program at graph construction and deterministic majority vote. Its terminal resource artifacts are the dataset-specific causal graph and the aggregated latent-tag table, accompanied by a run summary. Thus the preset reconstructs the reusable resource through the implemented aggregation boundary; it does not run the later causal estimators.

By default, output is isolated below runs/<run-id>/, in separate physionet/ and mimic/ children, with coordinator metadata under the same run root. Dataset manifests, resolved configurations, stage logs, provenance sidecars, cohort fingerprints, artifact hashes, and per-stage receipts record and validate the current execution. A generated run identifier is used unless RUN_ID is set.

External raw-data paths can be applied to this command without copying the records into the checkout:

```
PHYSIONET_RAW_DATA_PATH=/absolute/path/to/physionet2012 \
MIMIC_RAW_DATA_PATH=/absolute/path/to/mimiciii \
STAGES=dataset-extraction \
bash run_clinicause.sh
```

A.7 Running the complete pipeline

The canonical complete run is:

```
STAGES=all bash run_clinicause.sh
```

This retains the full integrated downstream predictive and causal workflow rather than stopping at the dataset-construction boundary. It is computationally heavier, and the temporal-model stages may require suitable accelerator resources. Optional launcher controls include DATASET=physionet or DATASET=mimic for a single source, OUTPUT_ROOT, RUN_ID, the two raw-data path variables, STRATS_MAX_CONCURRENT, and PYTHON_BIN; the two commands above remain the canonical interfaces.

A.8 Reproducibility boundary and limitations

Three claims must remain distinct. First, the current implementation is reproducible at the level of repository composition, one dependency interface, explicit launcher controls, dataset-isolated outputs, and validated artifact contracts. Second, an authorized user can reconstruct the derived resources from appropriately obtained raw inputs without CliniCause redistributing those inputs. Third, neither achievement is automatically an exact historical reproduction of the archived thesis results. The available archive and compact reproducibility package provide checked result tables, manifests, checksums, and partial run lineage, but do not recover every producing revision, numbered configuration, split, checkpoint-to-export link, raw/processed manifest, or hardware and software detail. Runtime can also vary across systems. Finally, software reproducibility does not establish clinical construct validity or causal identification; those remain separate scientific and governance questions.

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אוניברסיטת בן-גוריון בנגב
Ben-Gurion University of the Negev

CliniCause: בניית סביבות מבחן תצפיתיות המעוגנות בידע קליני לניתוח סיבתי מתוך סדרות זמן לא־סדירות בטיפול נמרץ

מוגש לשם קבלת תואר "מגיסטר" בפקולטה להנדסה

קובי קנזי

בהנחיית: ד"ר דן וילנצ'יק